
JESD 21-C Section Title: Annex L: Serial Presence Detect (SPD) for DDR4 SDRAM Modules

Committee Document Reference Title: DDR4 SPD Document Release 4

Device Type Identifier: UDIMM Revision 1.1

RDIMM Revision 1.2

LRDIMM Revision 1.2

NVDIMM-N Revision 1.1

1 Scope

This annex describes the serial presence detect (SPD) values for all DDR4 modules. The SPD data provides critical information about all modules on the memory channel and is intended to be use by the system's BIOS in order to properly initialize and optimize the system memory channels. The storage capacity of the SPD EEPROMs is limited, so a number of techniques are employed to optimize the use of these bytes, including overlays and run length limited coding.

All unused entries will be coded as 0x00. All unused bits in defined bytes will be coded as 0 except where noted.

Timing parameters in the SPD represent the operation of the module including all DRAMs and support devices at the lowest supported supply voltage (see SPD byte 11), and are valid from $t_{CKAVGmin}$ to $t_{CKAVGmax}$ (see SPD bytes 18 and 19).

To allow for maximum flexibility as devices evolve, SPD fields described in this document may support device configuration and timing options that are not included in the JEDEC DDR4 SDRAM data sheet (JESD79-4). Please refer to DRAM supplier data sheets or JESD79-4 to determine the compatibility of components.

2 History

Computer main memory buses have traditionally been defined by the generation of memory attached to the bus, e.g., EDO, SDRAM, DDR1, etc. The bus interface protocol and characteristics have largely been defined by the memory type. Clock frequency, CAS latencies, refresh recovery times and similar parameters defined the timing of signals between memory controller and the memory, and parameters such as number of ranks installed and device widths allowed system software to determine the memory capacity and similar high level characteristics of each module.

Over time, the memory bus has been extended to include additional features for application specific requirements. Registered DIMMs, for example, increased total capacity by buffering the loading of the address bus signals, allowing more DRAM to be installed. Similarly, Load Reduced DIMMs buffered the data bus as well, allowing even more ranks of memory to be supported. As each new extension to the function of the memory bus was introduced, system software combined knowledge of those extensions with information programmed into the EEPROM in the SPD to determine how to use and optimize the new features. Using the RDIMM as an example, systems understood that an additional clock of latency needed to be added to the DRAM latency to accommodate propagation delay through the register.

In later generations, the DRAM to host interface is completely virtualized. A memory module may have no DRAM at all, yet may use the DRAM bus to communicate with the host by emulating the DRAM channel interface. These virtual interfaces must appear to the system as one of the base module types, i.e., UDIMM, RDIMM, or LRDIMM. Modules that incorporate at least one non-DRAM media type for the purpose of main memory data storage are called “hybrid”, they act like a DRAM but on the other side of the interface protocol are some other memory type(s).

3 SPD Architecture

The SPD contents architecture must support the many variations of module types while remaining efficient. A system of overlay information selected through the use of “key bytes”, or selectors for the type of information to load has been implemented. The following DDR4 module SPD address map describes where the individual lookup table entries will be held in the serial EEPROM.

Consistent with the definition of DDR4 generation SPD devices (EE1004 and TSE2004) which have four individual write protection blocks of 128 bytes in length each, the SPD contents are aligned with these blocks as shown in Table 1:

Table 1 — SPD Contents Architecture

Block	Range		Description
0	0~127	0x000~0x07F	Base Configuration and DRAM Parameters
1	128~191	0x080~0x0BF	Standard Module Parameters -- See annexes L.1.x for details
	192~255	0x0C0~0x0FF	Hybrid Module Parameters -- See annexes L.2.x for details
2	256~319	0x100~0x13F	Hybrid Module Extended Function Parameters - See Annexes L.3.x for details
	320~383	0x140~0x17F	Manufacturing Information
3	384~511	0x180~0x1FF	End User Programmable

Operating parameters for the different module types are defined in the following subsections and will reside in the appropriate address ranges of the EEPROM address map depending on the module type. Please see Overlay Schema for further detail.

1. Section 9 - Standard Module Parameter - Overlay Bytes 128-191
 - Section 9.1 - UDIMMs
 - Section 9.2 - RDIMM
 - Section 9.3 - LRDIMM
2. Section 10 - Hybrid Module Parameters - Overlay Bytes 192 - 255
 - Section 10.1 NVDIMM
3. Section 11 - Hybrid Module Extended Function Parameters - Overlay Bytes 256-319
 - Section 11.1 - Energy Backed Byte Addressable NVDIMM
 - Section 11.2 - Energy Backed Block Addressable NVDIMM
 - Section 11.3 - Non-Energy Backed Byte Addressable NVDIMM

4 Overlay Schema

The following Schema exemplify the manner in which the base configuration information along with the appropriate subsections are to be overlaid onto the appropriate address spaces in order to provide a complete definition of the module.

4.1 UDIMM Overlay Schema

Table 2 shows the UDIMM overlay schema.

Key Byte 2 contains value 0x0C or 0x0E

Key Byte 3 contains any of the following values:

- 0x02, UDIMM
- 0x03, SO-DIMM
- 0x06, Mini-UDIMM
- 0x09, 72b-SO-UDIMM
- 0x0C, 16b-SO-DIMM
- 0x0D, 32b-SO-DIMM

4 Overlay Schema (Cont'd)

4.1 UDIMM Overlay Schema (Cont'd)

Table 2 — UDIMM Overlay Schema

Block	Range	Description
0	0~127	Base Configuration and DRAM Parameters
1	128~191	Insert Section 9.1: Unbuffered Memory Module Types
	192~255	Unused
2	256~319	Reserved
	320~383	Module Supplier's Data
3	384~511	End User Programmable

4.2 RDIMM Overlay Schema

Table 3 shows the RDIMM overlay schema.

Key Byte 2 contains value 0x0C or 0x0E

Key Byte 3 contains any of the following values:

- 0x01, RDIMM
- 0x05, Mini-RDIMM
- 0x08, 72b-SO-RDIMM

Table 3 — RDIMM Overlay Schema

Block	Range	Description
0	0~127	Base Configuration and DRAM Parameters
1	128~191	Insert Section 9.2: RDIMM Memory Module Types
	192~255	Unused
2	256~319	Reserved
	320~383	Module Supplier's Data
3	384~511	End User Programmable

4.3 LRDIMM Overlay Schema

Table 4 shows the LRDIMM overlay schema.

Key Byte 2 contains value 0x0C or 0x0E

Key Byte 3 contains any of the following values:

- 0x04, LRDIMM

Table 4 — LRDIMM Overlay Schema

Block	Range	Description
0	0~127	Base Configuration and DRAM Parameters
1	128~191	Insert Section 9.3: LRDIMM Memory Module Types
	192~255	Unused
2	256~319	Reserved
	320~383	Module Supplier's Data
3	384~511	End User Programmable

4.4 Hybrid DIMM Overlay Schema

Table 5 shows the Hybrid DIMM overlay schema.

Key Byte 2 contains value 0x0C or 0x0E

Key Byte 3 contains any of the following values:

- 0x9M, NVDIMM - Non-volatile DIMM; M defines base architecture

Table 5 — Hybrid DIMM Overlay Schema

Block	Range	Description
0	0~127	Base Configuration and DRAM Parameters
1	128~191	Insert Base Module Type Section 9.x
	192~255	Insert Hybrid Memory Parameter Section 9.x
2	256~319	Insert Extended Function Parameter Block Section 9.x
	320~383	Module Supplier's Data
3	384~511	End User Programmable

After programming the SPD contents, suppliers of JEDEC compliant modules must set the write protect bits for SPD device blocks 0 and 1 and must not set the write protect bit for block 3. Write protection of block 2 is required for modules using the Extended Function Parameter Block, such as NVDIMMs. See the EE1004-v/TSE2004av Device Specification for details on the SWPn command protocol.

5 Parsing the SPD

Table 6 provides relevant SPD bytes for parsing.

Table 6 — Some Relevant SPD Bytes for Parsing

SPD Byte(s)	Definition
1	SPD revision for this memory module
2	DRAM interface type presented or emulated
3	Memory module interface type
0~127	Base configuration and DRAM parameters
128~191	Module specific parameters
192~255	Hybrid memory parameters
204~219	NVDIMM function interface descriptors
256~319	Extended function parameter block

The system BIOS will acquire information from the SPD in order to properly configure the systems memory controller. It is assumed the BIOS will parse the SPD data in the order listed below.

Step 1: Parse Byte 2 - Verify the installed DRAM type is supported.

The first step in parsing the SPD is to verify that the DRAM type installed is supported by looking at DRAM device type byte 2. While it is usually not possible to physically plug in the wrong memory type, for example a DDR3 module size or key location should prevent insertion into a DDR4 system, there are cases where byte 2 is used to prevent accidental use of an incompatible memory type such as DDR4E modules in a DDR4-only system.

5 Parsing the SPD (Cont'd)

Step 2: Parse Byte 1 - Verify SPD compatibility. See Section 6 - SPD Revision Progression

The SPD revision byte 1 “encoding” nibble may be used to force legacy systems to reject newer modules. This would typically only occur if a critical error were found in SPD encoding that would require a “fix”. In this case, as in the case of an unsupported DRAM type, system initialization must be halted immediately.

The SPD revision stored in Byte 1 applies to all information for the module, including base information, module specific information, and hybrid information. Each SPD revision exactly defines how many bytes are valid in all other SPD blocks. The number of supported bytes (or bits) may increase from one SPD revision to another within each block as indicated by the “additions” nibble of the SPD revision. For example, an SPD revision 1.3 has more bytes or bits defined than SPD revision 1.2.

This progression of SPD contents is important for the BIOS to DIMM compatibility model. An older system may have a BIOS that only understands SPD revision 1.2 encoding, so if a module is installed that contains revision 1.3 information, the system can accurately interpret all of the historical revision 1.2 information retained in the module that is the subset of the revision 1.3 specification. Similarly, if a module with SPD revision 1.1 information is installed, that same BIOS can interpret all information that was current at the time that SPD 1.1 was defined. Therefore, BIOSes must maintain a knowledge of the active information for each historical SPD revision in order to support older modules.

Step 3: Parse Byte 3 - Determine module type and appropriate overlays.

Key byte 3 for module type determines the subsequent use of overlay information. Byte 3 bits 3~0 define the host to module interface style: unbuffered, registered, or load reduced. Standard module types are defined in Annexes L.1.x.

Key byte 3 also determines the presence of hybrid module types. Byte 3, bit 7 asserts if the module is hybrid, and Byte 3, bits 6~4 define the type of hybrid module. A hybrid module may appear to the system as a superset of any of the base module types. For example, an NVDIMM may present a UDIMM, RDIMM, or LRDIMM compatible interface to the system. Because of this unique second overlay for hybrid information on top of the base module type, the SPD revision for Hybrid modules must include all relevant information for any of the base module types if applicable. Refer to SPD revision progression in Section 6 for more information.

If Byte 3, bits 7~4 indicate the presence of a Hybrid module type, the BIOS must parse two more overlays. Annex L.2, Hybrid Module Parameters (Bytes 192~255) and Section 11 Hybrid Modules Extended Function Parameters (Bytes 256~319). As with the module specific parameters, both of the Hybrid Module parameter sets may increase from SPD revision to revision.

Step 4: Parse Bytes 0-127 - Make base configuration settings to memory interface based on these bytes.

All module types are required to read and interpret this block of data to set up the DRAM type, maximum operating frequency, the number of row, column, bank bits, write recovery time, etc. While these bytes primarily describe the timing of the DRAMs, timing represents the capabilities of the module and it may be necessary to downgrade the timing based on other factors including layout or support components on the module, such as registers.

Step 5: Parse Bytes 128-191 - Configure Standard module memory interface.

These bytes will be referenced and used by the system as needed based on the Standard module type that was determined after Byte 3 was parsed as indicated in Step 2. Bytes 128~191 are coded differently for each standard module type as defined in Section 9.

Step 6: Parse Bytes 192-255 (If Hybrid) - Read Hybrid Module parameters settings.

These bytes will be referenced and used by the system as needed based on the Hybrid module type that was determined after Byte 3 was parsed as indicated in Step 2. Function Format Interface Descriptors defined in bytes 204~205 through 218~219 will be read and utilized as indicated in Step 7. Bytes 192~255 are coded differently for each Hybrid module type and defined in Section 10.

Step 7: Parse Bytes 256-319 (If Hybrid Extended Functions are specified).

These bytes will be referenced and used by the system as needed based on the Function Format interface descriptors read previously during Step 6 from SPD bytes 204~205 through 218~219. Support up to 8 functions per Hybrid module can be defined within Bytes 256~319 and are coded based on Section 11.

For NVDIMM hybrid modules, parsing the two data blocks of hybrid memory parameters in SPD bytes 192~255 and the extended function parameters in SPD bytes 256~319 is intimately related. Each function descriptor contains a 4-bit field in bits 13~10 that creates an address pointer into the extended function parameter block. The formula is: $256 + 4 * \text{function descriptor bits 13~10}$.

The extended function parameter blocks are run length limited, meaning that when one function parameter set stops the next set starts

5 Parsing the SPD (Cont'd)

(with a granularity of 4 bytes). For example, imagine an NVDIMM with two functions, the first function is a byte addressable energy backed memory which requires 14 bytes of extended parameters and the second function is block addressed requiring 23 bytes of extended parameters. The resulting encoding for that NVDIMM are shown in Table 7:

Table 7 — Example with Imaginary SPD Revision 1.2

SPD Bytes	Coding (binary)	Definition
204~205	1000 0000 0000 0001	Implemented; byte addressable energy backed JEDEC byte addressable energy backed protocol Offset of extended function parameters = $256 + 4 * 0 = 256$
206~207	1001 0000 0100 0001	Implemented; block addressed JEDEC block addressed protocol Offset of extended function parameters = $256 + 4 * 4 = 272$
256~269	<valid>	Extended parameters for byte addressed energy backed function
272~294	<valid>	Extended parameters for block addressed function

The example in Table 7 assumes a certain SPD revision in order for the BIOS to know how many bytes are defined in the extended function parameter fields. If the above example were for SPD revision 1.2, then SPD revision 1.3 is released with 18 bytes for the first function and 26 bytes for the second function, the coding would change as shown in Table 8:

Table 8 — Example with Imaginary SPD Revision 1.3

SPD Bytes	Coding (binary)	Definition
204~205	1000 0000 0000 0001	Implemented; byte addressable energy backed JEDEC byte addressable energy backed protocol Offset of extended function parameters = $256 + 4 * 0 = 256$
206~207	1001 0100 0100 0001	Implemented; block addressed JEDEC block addressed protocol Offset of extended function parameters = $256 + 4 * 5 = 276$
256~273	<valid>	Extended parameters for byte addressed energy backed function
276~301	<valid>	Extended parameters for block addressed function

When an NVDIMM with SPD revision 1.3 is plugged into a legacy system written for SPD 1.2 or less, the BIOS would interpret the first 14 bytes of extended parameters for the byte addressed energy backed function (256~269) and the first 23 bytes of the block addressed function (276~294). The rest of the parameter bytes are ignored.

A newer system that supports SPD revisions up through 1.3 would interpret all bytes on either NVDIMM type in the correct SPD byte addresses, knowing that an NVDIMM with SPD revision 1.2 would have parameters in 256~269 and 272~294, and an NVDIMM with SPD revision 1.3 would have parameters in bytes 256~273 and 276~301.

6 SPD Revision Progression

The use of indexing for functional overlays creates a set of dependencies that affect how SPD revisions must be defined. All module types are dependent on the definition of the general DRAM section bytes 0~127, therefore any change in the general DRAM section requires an increase in the SPD revision for all module types UDIMM, RDIMM, LRDIMM and Hybrid DIMM.

A change in the indexed overlay information for specific module types does not require that the SPD revision for the other module types change. For example, adding information to an RDIMM 1.1 SPD specification would require increasing the RDIMM SPD specification to 1.2, however it would not affect UDIMM, LRDIMM or Hybrid DIMM SPD revisions.

Since hybrid modules may be implemented emulating any module type interface, then any change in the general DRAM section 0~127 or any change in any module type specific information 128~191 does require an increase in the NVDIMM SPD revision level. Since all NVDIMM types share a single overlay 192~255 and 256~319, a change in any NVDIMM SPD revision changes the revision for all NVDIMM types even if nothing changes for a specific NVDIMM type. Table 9 shows a hypothetical sequence starting from all module types at revision 1.0, then showing how SPD revisions would change based on certain actions.

6 SPD Revision Progression (Cont'd)

Table 9 — Hypothetical SPD Revision Progression Showing Revision Relationships

Event	UDIMM	RDIMM	LRDIMM	NVDIMM
Initial SPD release	1.0	1.0	1.0	1.0
Addition in RDIMM Annex	1.0	1.1	1.0	1.1
Addition in LRDIMM Annex	1.0	1.1	1.1	1.2
Addition in LRDIMM Annex	1.0	1.1	1.2	1.3
Addition in General Section	1.1	1.2	1.3	1.4
Addition in UDIMM Annex	1.2	1.2	1.3	1.5
Addition in NVDIMM Annex	1.2	1.2	1.3	1.6
Addition in RDIMM and UDIMM Annexes	1.3	1.3	1.3	1.7
Encoding change in LRDIMM Annex	1.3	1.3	2.3	2.7
Addition in LRDIMM Annex	1.3	1.3	2.4	2.8
Encoding change in General Section	2.3	2.3	3.4	3.8
Addition in RDIMM Annex	2.3	2.4	3.4	3.9

7 Address Map

This section provides the SPD address map for all DDR4 modules. It describes where the individual lookup table entries will be held in the serial EEPROM. Consistent with the definition of DDR4 generation SPD devices which have four individual write protection blocks of 128 bytes in length each, the SPD contents are aligned with these blocks as shown in Table 10:

Table 10 — Address Map

Block	Range		Description
0	0~127	0x000~0x07F	Base Configuration and DRAM Parameters
1	128~191	0x080~0x0BF	Module Specific Parameters -- See annexes for details
	192~255	0x0C0~0x0FF	Hybrid Memory Parameters -- See annexes for details
2	256~319	0x100~0x13F	Extended Function Parameter Block
	320~383	0x140~0x17F	Manufacturing Information
3	384~511	0x180~0x1FF	End User Programmable

After programming the SPD contents, suppliers of JEDEC compliant modules must set the write protect bits for SPD device blocks 0 and 1 and must not set the write protect bit for block 3. Write protection of block 2 is required for modules using the Extended Function Parameter Block, such as NVDIMMs. See the EE1004-v/TSE2004av Device Specification for details on the SWPn command protocol.

7.1 Block 0: Base Configuration and DRAM Parameters

Table 11 details the location of each byte in this block.

Table 11 — Block 0: Base Configuration and DRAM Parameters

Byte Number	Function Described		Notes
0	0x000	Number of Serial PD Bytes Written / SPD Device Size	1, 2
1	0x001	SPD Revision	
2	0x002	Key Byte / DRAM Device Type	
3	0x003	Key Byte / Module Type	
NOTE 1 Number of SPD bytes written will typically be programmed as 384 bytes.			
NOTE 2 Size of SPD device will typically be programmed as 512 bytes.			
NOTE 3 From DDR4 SDRAM datasheet.			

7 Address Map (Cont'd)

7.1 Block 0: Base Configuration and DRAM Parameters (Cont'd)

Table 11 — Block 0: Base Configuration and DRAM Parameters (Cont'd)

Byte Number		Function Described	Notes
4	0x004	SDRAM Density and Banks	3
5	0x005	SDRAM Addressing	3
6	0x006	Primary SDRAM Package Type	3
7	0x007	SDRAM Optional Features	3
8	0x008	SDRAM Thermal and Refresh Options	3
9	0x009	Other SDRAM Optional Features	3
10	0x00A	Secondary SDRAM Package Type	3
11	0x00B	Module Nominal Voltage, VDD	3
12	0x00C	Module Organization	
13	0x00D	Module Memory Bus Width	
14	0x00E	Module Thermal Sensor	
15	0x00F	Extended module type	
16	0x010	Reserved -- must be coded as 0x00	
17	0x011	Timebases	
18	0x012	SDRAM Minimum Cycle Time ($t_{CKAVGmin}$)	3
19	0x013	SDRAM Maximum Cycle Time ($t_{CKAVGmax}$)	3
20	0x014	CAS Latencies Supported, First Byte	3
21	0x015	CAS Latencies Supported, Second Byte	3
22	0x016	CAS Latencies Supported, Third Byte	3
23	0x017	CAS Latencies Supported, Fourth Byte	3
24	0x018	Minimum CAS Latency Time (t_{AAmin})	3
25	0x019	Minimum RAS to CAS Delay Time (t_{RCDmin})	3
26	0x01A	Minimum Row Precharge Delay Time (t_{RPmin})	3
27	0x01B	Upper Nibbles for t_{RASmin} and t_{RCmin}	3
28	0x01C	Minimum Active to Precharge Delay Time (t_{RASmin}), Least Significant Byte	3
29	0x01D	Minimum Active to Active/Refresh Delay Time (t_{RCmin}), Least Significant Byte	3
30	0x01E	Minimum Refresh Recovery Delay Time ($t_{RFC1min}$), LSB	3
31	0x01F	Minimum Refresh Recovery Delay Time ($t_{RFC1min}$), MSB	3
32	0x020	Minimum Refresh Recovery Delay Time ($t_{RFC2min}$), LSB	3
33	0x021	Minimum Refresh Recovery Delay Time ($t_{RFC2min}$), MSB	3
34	0x022	Minimum Refresh Recovery Delay Time ($t_{RFC4min}$), LSB	3
35	0x023	Minimum Refresh Recovery Delay Time ($t_{RFC4min}$), MSB	3
36	0x024	Minimum Four Activate Window Time (t_{FAWmin}), Most Significant Nibble	3
37	0x025	Minimum Four Activate Window Time (t_{FAWmin}), Least Significant Byte	3
38	0x026	Minimum Activate to Activate Delay Time (t_{RRD_smin}), different bank group	3
39	0x027	Minimum Activate to Activate Delay Time (t_{RRD_Lmin}), same bank group	3
40	0x28	Minimum CAS to CAS Delay Time (t_{CCD_Lmin}), same bank group	3
41	0x29	Upper Nibble for t_{WRmin}	3
NOTE 1 Number of SPD bytes written will typically be programmed as 384 bytes. NOTE 2 Size of SPD device will typically be programmed as 512 bytes. NOTE 3 From DDR4 SDRAM datasheet.			

7 Address Map (Cont'd)

7.1 Block 0: Base Configuration and DRAM Parameters (Cont'd)

Table 11 — Block 0: Base Configuration and DRAM Parameters (Cont'd)

Byte Number		Function Described	Notes
42	0x2A	Minimum Write Recovery Time (t_{WRmin})	3
43	0x2B	Upper nibbles for t_{WTRmin}	3
44	0x02C	Minimum Write to Read Time (t_{WTR_Smin}), different bank group	3
45	0x02D	Minimum Write to Read Time (t_{WTR_Lmin}), same bank group	3
46~59	0x02E~0x03B	Reserved -- must be coded as 0x00	
60~77	0x03C~0x04D	Connector to SDRAM Bit Mapping	
78~116	0x04E~0x074	Reserved -- must be coded as 0x00	
117	0x75	Fine Offset for Minimum CAS to CAS Delay Time (t_{CCD_Lmin}), same bank group	
118	0x76	Fine Offset for Minimum Activate to Activate Delay Time (t_{RRD_Lmin}), same bank group	3
119	0x77	Fine Offset for Minimum Activate to Activate Delay Time (t_{RRD_Smin}), different bank group	3
120	0x078	Fine Offset for Minimum Activate to Activate/Refresh Delay Time (t_{RCmin})	3
121	0x079	Fine Offset for Minimum Row Precharge Delay Time (t_{RPmin})	3
122	0x07A	Fine Offset for Minimum RAS to CAS Delay Time (t_{RCDmin})	3
123	0x07B	Fine Offset for Minimum CAS Latency Time (t_{AAmin})	3
124	0x07C	Fine Offset for SDRAM Maximum Cycle Time ($t_{CKAVGmax}$)	3
125	0x07D	Fine Offset for SDRAM Minimum Cycle Time ($t_{CKAVGmin}$)	3
126	0x07E	CRC for Base Configuration Section, Least Significant Byte	
127	0x07F	CRC for Base Configuration Section, Most Significant Byte	
NOTE 1 Number of SPD bytes written will typically be programmed as 384 bytes.			
NOTE 2 Size of SPD device will typically be programmed as 512 bytes.			
NOTE 3 From DDR4 SDRAM datasheet.			

7.2 Block 1, lower half: Module Specific Parameters

Bytes 128~191 (0x080~0x0BF) Parameters in this block are specific to the module type as selected by the contents of SPD Key Byte 3 bits 3~0. Refer to the appropriate subsection for detailed byte descriptions.

7.3 Block 1, upper half: Hybrid Memory Parameters

Bytes 192~255 (0x0C0~0x0FF) Parameters in this block are specific to the hybrid memory type as selected by the contents of SPD Key Byte 3 bits 7~4. Refer to the appropriate subsection for detailed byte descriptions. The CRC field in bytes 254~255 covers all of block 1, bytes 128~255.

7.4 Block 2, lower half: Extended Function Parameters

Bytes 256~319 (0x100~0x13F). Parameters in this block are used to define the extended functions available on Hybrid modules. Values in this block are based on indices in the NVDIMM annex, and the size of each function parameter block is unique to that function type.

7.5 Block 2, upper half: Manufacturing Information

Bytes 320~383 (0x140~0x17F) Table 12 details the location of each byte in this block.

Table 12 — Block 2, upper half: Manufacturing Information

Byte Number	Function Described	Notes
320	0x140	Module Manufacturer's ID Code, Least Significant Byte
321	0x141	Module Manufacturer's ID Code, Most Significant Byte
322	0x142	Module Manufacturing Location
323~324	0x143~0x144	Module Manufacturing Date
325~328	0x145~0x148	Module Serial Number
329~348	0x149~0x15C	Module Part Number
349	0x15D	Module Revision Code
350	0x15E	DRAM Manufacturer's ID Code, Least Significant Byte
351	0x15F	DRAM Manufacturer's ID Code, Most Significant Byte
352	0x160	DRAM Stepping
353~381	0x161~0x17D	Module Manufacturer's Specific Data
382~383	0x17E~0x17F	Reserved; must be coded as 0x00

7.6 Block 3: End User Programmable

Bytes 384~511 (0x180~0x1FF) Bytes in this block are reserved for use by end users.

8 Details of Each Byte

8.1 General Configuration Section: Bytes 0~127 (0x000~0x07F)

This section defines parameters that are common to all DDR4 module types.

8.1.1 Byte 0 (0x000): Number of Bytes Used / Number of Bytes in SPD Device

The least significant nibble of this byte, shown in Table 13, describes the total number of bytes used by the module manufacturer for the SPD data and any (optional) specific supplier information. The byte count includes the fields for all required and optional data. Bits 6~4 describe the total size of the serial memory used to hold the Serial Presence Detect data.

Table 13 — Byte 0 (0x000): Number of Bytes Used / Number of Bytes in SPD Device

Bit 7	Bits 6~4	Bits 3~0
Reserved	SPD Bytes Total	SPD Bytes Used
Reserved; must be coded as 0	Bit [6, 5, 4] : 000 = Undefined 001 = 256 010 = 512 All others reserved	Bit [3, 2, 1, 0] : 0000 = Undefined 0001 = 128 0010 = 256 0011 = 384 0100 = 512 All others reserved
NOTE Typical programming of bits 3~0 will be 0011 (384 bytes).		

8.1.2 Byte 1 (0x001): SPD Revision

This byte, shown in Table 14, describes the compatibility level of the encoding of the bytes contained in the SPD EEPROM, and the current collection of valid defined bytes. Software should examine the upper nibble (Encoding Level) to determine if it can correctly interpret the contents of the module SPD. The lower nibble (Additions Level) can optionally be used to determine which additional bytes or attribute bits have been defined; however, since any undefined additional byte must be encoded as 0x00 or undefined

8.1 General Configuration Section: Bytes 0~127 (0x000~0x07F) (Cont'd)

8.1.2 Byte 1 (0x001): SPD Revision (Cont'd)

attribute bit must be defined as 0, software can safely detect additional bytes and use safe defaults if a zero encoding is read for these bytes.

Table 14 — Byte 1 (0x001): SPD Revision

Production Status	SPD Revision	Encoding Level				Additions Level				Hex
		Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	
Pre-production	Revision 0.0	0	0	0	0	0	0	0	0	00
	Revision 0.1	0	0	0	0	0	0	0	1	01
	...	.	.	.	.	.	.	.	.	.
	Revision 0.9	0	0	0	0	1	0	0	1	09
Production	Revision 1.0	0	0	0	1	0	0	0	0	10
	Revision 1.1	0	0	0	1	0	0	0	1	11
	...	.	.	.	.	.	.	.	.	...
Undefined	Undefined	1	1	1	1	1	1	1	1	FF

The Additions Level is never reduced even after an increment of the Encoding Level. For example, if the current SPD revision level were 1.2 and a change in Encoding Level were approved, the next revision level would be 2.2. If additions to revision 2.2 were approved, the next revision would be 2.3. Changes in the Encoding Level are extremely rare, however, since they can create incompatibilities with older systems.

The exceptions to the above rule are the SPD revision levels used during development prior to the Revision 1.0 release. Revisions 0.0 through 0.9 are used to indicate sequential pre-production SPD revision levels, however the first production release will be Revision 1.0.

This document defines the SPD contents for multiple families of DDR4 memory modules, with a separate section for each family that defines the bytes in SPD locations 128~255 (0x080~0x0FF). These module families and their respective sections are:

- Section 9.1: Unbuffered Memory Modules
- Section 9.2: Registered Memory Modules
- Section 9.3: Load Reduced Memory Modules
- Section 10.1: Non-Volatile Memory Modules

The SPD revision level for each module family type is independent. This allows changes to be made to the Registered DIMM annex, for example, without necessarily changing the revision of Unbuffered DIMMs. In this context, the SPD revision value corresponds to all SPD bytes for that DIMM type. It also means that over time, the revisions for each module type may vary. Note that changes to a DIMM specific annex does not affect the revisions of other module types, but changes in the General Section of the SPD affect all DIMM types. The example in Table 15 suggests a possible historical progression:

Table 15 — Hypothetical Historic Progression of SPD Revisions by DIMM Type

Event	UDIMM	RDIMM	LRDIMM
Initial SPD release	1.0	1.0	1.0
Addition in RDIMM Annex	1.0	1.1	1.0
Addition in LRDIMM Annex	1.0	1.1	1.1
Addition in LRDIMM Annex	1.0	1.1	1.2
Addition in General Section	1.1	1.2	1.3
Addition in UDIMM Annex	1.2	1.2	1.3

Table 15 — Hypothetical Historic Progression of SPD Revisions by DIMM Type (Cont'd)

Event	UDIMM	RDIMM	LRDIMM
Encoding change in LRDIMM Annex	1.2	1.2	2.3
Addition in LRDIMM Annex	1.2	1.2	2.4
Encoding change in General Section	2.2	2.2	3.4
Addition in RDIMM Annex	2.2	2.3	3.4

8.1.3 Byte 2 (0x002): Key Byte / DRAM Device Type

This byte, shown in Table 16, is the key byte used by the system BIOS to determine how to interpret all other bytes in the SPD EEPROM. The BIOS must check this byte first to ensure that the EEPROM data is interpreted correctly. Any DRAM or Module type that requires significant changes to the SPD format (beyond defining previously undefined bytes or bits) also requires a new entry in the key byte table below.

Table 16 — Byte 2 (0x002): Key Byte / DRAM Device Type

Line #	Hex	SDRAM / Module Type Corresponding to Key Byte
0	00	Reserved
1	01	Fast Page Mode
2	02	EDO
3	03	Pipelined Nibble
4	04	SDRAM
5	05	ROM
6	06	DDR SGRAM
7	07	DDR SDRAM
8	08	DDR2 SDRAM
9	09	DDR2 SDRAM FB-DIMM
10	0A	DDR2 SDRAM FB-DIMM PROBE
11	0B	DDR3 SDRAM
12	0C	DDR4 SDRAM
13	0D	Reserved
14	0E	DDR4E SDRAM
15	0F	LPDDR3 SDRAM
16	10	LPDDR4 SDRAM
17	11	LPDDR4X SDRAM
-	-	-
253	FD	Reserved
254	FE	Reserved
255	FF	Reserved

The DRAM Device Type byte defines an interface compatibility family more than it identifies a specific memory device. For example, where an unbuffered memory module may expose the devices directly to the edge connector contacts, a load reduced memory module presents the interfaces of registering clock drivers and data buffers to the edge connector contacts. There may be memory modules with completely different devices, such as non-volatile memories, however these may define themselves as a base memory type such as DDR4 SDRAM for interface compatibility with the memory controller. The memory controller must be aware of its capabilities when parsing the SPD.

8 Details of Each Byte (Cont'd)**8.1 General Configuration Section: Bytes 0~127 (0x000~0x07F) (Cont'd)****8.1.4 Byte 3 (0x003): Key Byte / Module Type**

This byte, shown in Table 17, is a Key Byte used to index the module specific section of the SPD from bytes 128~191 and bytes 192~255. Byte 3 bits 3~0 identifies the SDRAM memory module type.

Bits 7~4 identifies the architecture of the secondary memory type, if any, present on the module in addition to the base DRAM. These secondary hybrid architectures may apply to any Base Module Type. The index in bits 7~4 are used to index the hybrid memory architecture specific section of the SPD from bytes 192~255.

System software supporting hybrid memory modules should parse the DRAM attributes per the Base Module Type and combine the indexed information per the Hybrid Memory Type to determine the combined memory module capabilities.

Some modules may have no base memory, but will have only a secondary memory type. For example, a Flash-only memory module. These are classified as “hybrid” for the purposes of interpreting the SPD. Where base memory parameters apply to this class of hybrid module, these will be documented with those bytes in the base section. Extended parameters are encoded in SPD bytes 192~255.

Table 17 — Byte 3 (0x003): Key Byte / Module Type

Bit 7	Bits 6~4	Bits 3~0
Hybrid	Hybrid Media	Base Module Type
0 = Not hybrid (Module is DRAM only) 1 = Hybrid module (See bits 6~4 for hybrid type)	Bits [6, 5, 4]: 000 = Not hybrid 001 = NVDIMM Hybrid All other codes reserved	Bits [3, 2, 1, 0]: 0000 = Extended DIMM type, see byte 15 (0x00F) 0001 = RDIMM 0010 = UDIMM 0011 = SO-DIMM 0100 = LRDIMM 0101 = Mini-RDIMM 0110 = Mini-UDIMM 0111 = Reserved 1000 = 72b-SO-RDIMM 1001 = 72b-SO-UDIMM 1010 = Reserved 1011 = Reserved 1100 = 16b-SO-DIMM 1101 = 32b-SO-DIMM 1110 = Reserved 1111 = Reserved
Base Module Type Definitions: RDIMM: Registered Dual In-Line Memory Module UDIMM: Unbuffered Dual In-Line Memory Module SO-DIMM: Unbuffered Small Outline Dual In-Line Memory Module, 64-bit data bus LRDIMM: Load Reduced Dual In-Line Memory Module Mini-RDIMM: Mini Registered Dual In-Line Memory Module Mini-UDIMM: Mini Unbuffered Dual In-Line Memory Module 72b-SO-RDIMM: Small Outline Registered Dual In-Line Memory Module, 72-bit data bus 72b-SO-UDIMM: Small Outline Unbuffered Dual In-Line Memory Module, 72-bit data bus 16b-SO-DIMM: Small Outline Unbuffered Dual In-Line Memory Module, 16-bit data bus 32b-SO-DIMM: Small Outline Unbuffered Dual In-Line Memory Module, 32-bit data bus Hybrid Memory Type Definitions: NVDIMM: Non-Volatile Dual In-Line Memory Module, Hybrid module with a DRAM-style interface with one or more non-DRAM components for data storage		

Examples:

0x01 = RDIMM, no hybrid memory present

0x91 = RDIMM, NVDIMM hybrid memory present

0x94 = LRDIMM, NVDIMM hybrid memory present

8 Details of Each Byte (Cont'd)

8.1 General Configuration Section: Bytes 0~127 (0x000~0x07F) (Cont'd)

8.1.5 Byte 4 (0x004): SDRAM Density and Banks

This byte, shown in Table 18, defines the total density of the DDR4 SDRAM, in bits, and the number of internal banks and bank groups into which the memory array is divided. For multi-die stacks (DDP, QDP, or 3DS), this represents the capacity of each DRAM die in the stack. These values come from the DDR4 SDRAM data sheet.

Table 18 — Byte 4 (0x004): SDRAM Density and Banks

Bits 7~6	Bits 5~4	Bits 3~0
Bank Group Bits	Bank Address Bits ¹	Total SDRAM capacity per die, in megabits
Bits [7, 6]: 00 = 0 (no bank groups) 01 = 1 (2 bank groups) 10 = 2 (4 bank groups) 11 = reserved	Bit [5, 4]: 00 = 2 (4 banks) 01 = 3 (8 banks) All others reserved	Bit [3, 2, 1, 0]: 0000 = 256 Mb 0001 = 512 Mb 0010 = 1 Gb 0011 = 2 Gb 0100 = 4 Gb 0101 = 8 Gb 0110 = 16 Gb 0111 = 32 Gb 1000 = 12 Gb 1001 = 24 Gb All others reserved
NOTE 1 Bank Address Bits determine the number of banks in each Bank Group.		

8.1.6 Byte 5 (0x005): SDRAM Addressing

This byte, shown in Table 19, describes the row addressing and the column addressing in the SDRAM device. Bits 2~0 encode the number of column address bits, and bits 5~3 encode the number of row address bits. These values come from the DDR4 SDRAM data sheet.

Table 19 — Byte 5 (0x005): SDRAM Addressing

Bits 7~6	Bits 5~3	Bits 2~0
Reserved	Row Address Bits	Column Address Bits
Reserved; must be coded as 00	Bit [5, 4, 3]: 000 = 12 001 = 13 010 = 14 011 = 15 100 = 16 101 = 17 110 = 18 All others reserved	Bit [2, 1, 0]: 000 = 9 001 = 10 010 = 11 011 = 12 All others reserved

8 Details of Each Byte (Cont'd)

8.1 General Configuration Section: Bytes 0~127 (0x000~0x07F) (Cont'd)

8.1.7 Byte 6 (0x006): Primary SDRAM Package Type

This byte, shown in Table 20, describes the type of SDRAM devices on the module. For modules having asymmetrical assembly of multiple SDRAM package types, this byte defines the primary set of SDRAMs; Byte 10 defines the secondary set.

Table 20 — Byte 6 (0x006): Primary SDRAM Package Type

Bit 7	Bits 6~4	Bits 3~2	Bit 1~0
Primary SDRAM Package Type	Die Count	Reserved	Signal Loading ²
0 = Monolithic DRAM Device 1 = Non-Monolithic Device ¹	000 = Single die 001 = 2 die 010 = 3 die 011 = 4 die 100 = 5 die 101 = 6 die 110 = 7 die 111 = 8 die	00	00 = Not specified ³ 01 = Multi load stack 10 = Single load stack (3DS) 11 = Reserved
NOTE 1 This includes Dual Die, Quad Die, Multi-Die, 3DS, or physically stacked devices - anything that is outside the standard monolithic device. NOTE 2 Refers to loading on signals at the SDRAM balls. Loading on certain signals (CKE, ODT, etc.) per specification of device stacking as defined in JESD79-4. NOTE 3 Monolithic DRAM device coded as 00 in bits 1~0.			

Terminology: See Table 21.

Table 21 — Package Type Terminology

SDRAM Package Type	Abbreviation	Description	# Electrical Loads ...			
			... On data, mask, and strobe signals	... On address and command signals	... On control signals except CKE	... On CKE signals
Monolithic	SDP	Single die package	1	1	1	1
Multi-load stack	DDP	Dual Die Package	2	2	1	1
	QDP	Quad Die Package	4	4	1	2
Single load stack	2H 3DS	Two SDRAM die single load stack	1	1	1	1
	3H 3DS	Three SDRAM die single load stack	1	1	1	1
	4H 3DS	Four SDRAM die single load stack	1	1	1	1
	5H 3DS	Five SDRAM die single load stack	1	1	1	1
	6H 3DS	Six SDRAM die single load stack	1	1	1	1
	7H 3DS	Seven SDRAM die single load stack	1	1	1	1
	8H 3DS	Eight SDRAM die single load stack	1	1	1	1

8 Details of Each Byte (Cont'd)

8.1 General Configuration Section: Bytes 0~127 (0x000~0x07F) (Cont'd)

8.1.8 Byte 7 (0x007): SDRAM Optional Features

This byte, shown in Table 22, defines support for certain SDRAM features. This value comes from the DDR4 SDRAM data sheet.

Table 22 — Byte 7 (0x007): SDRAM Optional Features

Bits 7~6	Bits 5~4	Bits 3~0
Reserved	Maximum Activate Window (tMAW)	Maximum Activate Count (MAC)
Reserved; must be coded as 00	Bits [5, 4]: 00 = 8192 * tREFI 01 = 4096 * tREFI 10 = 2048 * tREFI 11 = Reserved	Bits [3, 2, 1, 0]: 0000 = Untested MAC ¹ 0001 = 700 K 0010 = 600 K 0011 = 500 K 0100 = 400 K 0101 = 300 K 0110 = 200 K 0111 = Reserved 1000 = Unlimited MAC ² All others reserved
NOTE 1 Untested MAC means the device is not tested for tMAW and/or MAC; no particular value should be assumed. NOTE 2 Unlimited MAC means there is no restriction to the number of activates to a given row in a refresh period providing DRAM timing requirements such as tRCmin and refresh requirements are not violated.		

8.1.9 Byte 8 (0x008): SDRAM Thermal and Refresh Options

This byte, shown in Table 23, describes the module's supported operating temperature ranges and refresh options. These values come from the DDR4 SDRAM data sheet. Please refer to the DDR4 SDRAM data sheet (JESD79-4 or supplier data sheet) for a complete description of these options.

Table 23 — Byte 8 (0x008): SDRAM Thermal and Refresh Options

Bits 7~0
Reserved
Reserved; must be coded as 0x00

8.1.10 Byte 9 (0x009): Other SDRAM Optional Features

This byte, shown in Table 24, defines support for certain SDRAM features. This value comes from the DDR4 SDRAM data sheet.

Table 24 — Byte 9 (0x009): Other SDRAM Optional Features

Bits 7~6	Bit 5	Bits 4~0
Post Package Repair (PPR)	Soft PPR	Reserved
00: PPR not supported 01: Post package repair supported, one row per bank group 10: Reserved 11: Reserved	0 = Soft PPR not supported 1 = Soft PPR supported	Reserved; must be coded as 00000
NOTE If PPR is supported, Hard PPR is always supported by the device. Optional additional support for Soft PPR is indicated in bit 5.		

8 Details of Each Byte (Cont'd)**8.1 General Configuration Section: Bytes 0~127 (0x000~0x07F) (Cont'd)****8.1.11 Byte 10 (0x00A): Secondary SDRAM Package Type**

For modules having asymmetrical assembly of multiple SDRAM package types, this byte, shown in Table 25, defines the secondary set of SDRAMs. For modules with symmetrical assembly, this byte must be coded as 0x00.

Table 25 — Byte 10 (0x00A): Secondary SDRAM Package Type

Bit 7	Bits 6~4	Bits 3~2	Bits 1~0
SDRAM Package Type	Die Count	DRAM Density Ratio ¹	Signal Loading
0 = Monolithic DRAM Device 1 = Non-Monolithic Device ¹	000 = Single die 001 = 2 die 010 = 3 die 011 = 4 die 100 = 5 die 101 = 6 die 110 = 7 die 111 = 8 die	00 = Rank 1 and 3 device densities are the same as rank 0 and 2 densities 01 = Rank 1 and 3 are one standard device density smaller than rank 0 and 2 10 = Rank 1 and 3 are two standard device densities smaller than rank 0 and 2 11 = Reserved	00 = Not specified 01 = Multi load stack 10 = Single load stack (3DS) 11 = Reserved
NOTE 1 When DRAM Density Ratio (bits 3~2) is non-zero, all SPD settings except timing parameters are specified for the higher density devices in rank 0 (e.g., density, row, column, etc.). SPD timing parameters are specified for the slowest device timings of any rank (e.g., tCK, tAA, tRFC, etc.). NOTE 2 See Byte 6 for packaging notes.			

DRAM Density Ratio Settings (Table 26):**Table 26 — DRAM Density Ratio Settings**

SPD Byte 4, Bits 3~0	Ranks 0, 2 Device Density	SPD Byte 10, Bits 3~2	Ranks 1, 3 Device Density	SPD Byte 10, Bits 3~2	Ranks 1, 3 Device Density
0000	256 Mb	01	Not Defined	10	Not Defined
0001	512 Mb	01	256 Mb	10	Not Defined
0010	1 Gb	01	512 Mb	10	256 Mb
0011	2 Gb	01	1 Gb	10	512 Mb
0100	4 Gb	01	2 Gb	10	1 Gb
0101	8 Gb	01	4 Gb	10	2 Gb
0110	16 Gb	01	12 Gb	10	8 Gb
0111	32 Gb	01	24 Gb	10	16 Gb
1000	12 Gb	01	8 Gb	10	4 Gb
1001	24 Gb	01	16 Gb	10	12 Gb

8.1.12 Byte 11 (0x00B): Module Nominal Voltage, VDD

This byte, shown in Table 27, describes the Voltage Level for DRAM and other components on the module such as the register or memory buffer if applicable. Note that SPDs or thermal sensor components are on the VDDSPD supply and are not affected by this byte.

'Operable' is defined as the VDD voltage at which module operation is allowed using the performance values programmed in the SPD.

'Endurant' is defined as the VDD voltage at which the module may be powered without adversely affecting the life expectancy or

reliability. Operation is not supported at this voltage.

Table 27 — Byte 11 (0x00B): Module Nominal Voltage, VDD

Byte 11: Module Nominal Voltage, VDD		
Reserved	DRAM VDD 1.2 V	
Bits 7~2	Bit 1	Bit 0
Reserved; must be coded as 000000	0 = not endurant 1 = endurant	0 = not operable 1 = operable

Examples:

A value on bits 1~0 of 11 defines DRAM support of a nominal operable voltage of 1.2 V only.

8.1.13 Byte 12 (0x00C): Module Organization

This byte, shown in Table 28, describes the organization of the SDRAM module. Bits 2~0 encode the device width of the SDRAM devices. Bits 5~3 encode the number of package ranks on the module.

Table 28 — Byte 12 (0x00C): Module Organization

Bit 7	Bit 6	Bits 5~3	Bits 2~0
Reserved	Rank Mix	Number of Package Ranks per DIMM	SDRAM Device Width
Reserved; must be coded as 0	0 = Symmetrical 1 = Asymmetrical	Bit [5, 4, 3] : 000 = 1 Package Rank 001 = 2 Package Ranks 010 = 3 Package Ranks 011 = 4 Package Ranks 100 = 5 Package Ranks 101 = 6 Package Ranks 110 = 7 Package Ranks 111 = 8 Package Ranks	Bit [2, 1, 0] : 000 = 4 bits 001 = 8 bits 010 = 16 bits 011 = 32 bits All others reserved

“Package ranks per DIMM” refers to the collections of devices on the module sharing common chip select signals (across the data width of the DIMM), either from the edge connector for unbuffered modules or from the outputs of a registering clock driver for RDIMMs and LRDIMMs.

“Logical rank” refers the individually addressable die in a 3DS stack and has no meaning for monolithic or multi-load stacked SDRAMs, however, for the purposes of calculating the capacity of the module, one should treat monolithic and multi-load stack SDRAMs as having one logical rank per package rank.

8.1.13.1 Symmetrical Configurations

Byte 12 Bit 6 = 0 defines a memory module as being “symmetrical” where all DRAM devices on the module are the same type as described in Byte 6.

Symmetrical configurations are shown in Figure 1 and Table 29.

8.1.13 Byte 12 (0x00C): Module Organization (Cont'd)

8.1.13.1 Symmetrical Configurations (Cont'd)

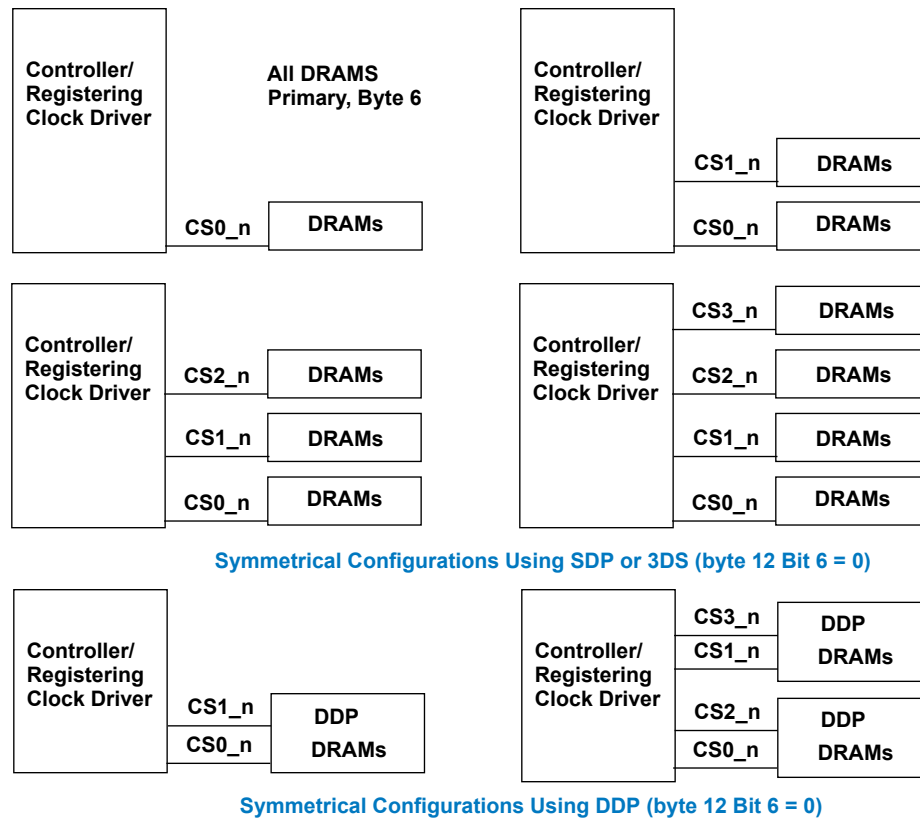


Figure 1 — Module Organization: Symmetrical Configurations

Table 29 — Rank Matrix for Symmetrical Modules (Byte 12 Bit 6 = 0)

SDRAM Package Type(s) (Byte 6)	# Package Ranks per DIMM	# Logical Ranks per Package Rank (for calculation only)	# Logical Ranks per DIMM
SDP	1	1	1
	2		2
	3		3
	4		4
DDP	2	1	2
	4		4
2H 3DS	1	2	2
	2		4
3H 3DS	1	3	3
	2		6
4H 3DS	1	4	4
	2		8
5H 3DS	1	5	5
	2		10
6H 3DS	1	6	6
	2		12

Table 29 — Rank Matrix for Symmetrical Modules (Byte 12 Bit 6 = 0) (Cont'd)

SDRAM Package Type(s) (Byte 6)	# Package Ranks per DIMM	# Logical Ranks per Package Rank (for calculation only)	# Logical Ranks per DIMM
7H 3DS	1	7	7
	2		14
8H 3DS	1	8	8
	2		16
Logical Ranks per DIMM = # Package Ranks per DIMM * # Logical Ranks per Package Rank Byte 10 coded as 0x00 for all symmetrical configurations			

8.1.13.2 Asymmetrical Configurations

Byte 12 Bit 6 = 1 describes support for modules having a asymmetrical DRAM capacities, for example a 3 package rank module comprised of a combination of SDP and DDP devices, or 3DS-based designs having different numbers of logical ranks in each package rank. When Byte 12 Bit 6 = 1, Byte 6 indicates the package type for the first collection of devices on the module and Byte 10 indicates the package type for the second collection of devices on the module.

Asymmetrical configurations are shown in Figure 2 and Table 30.

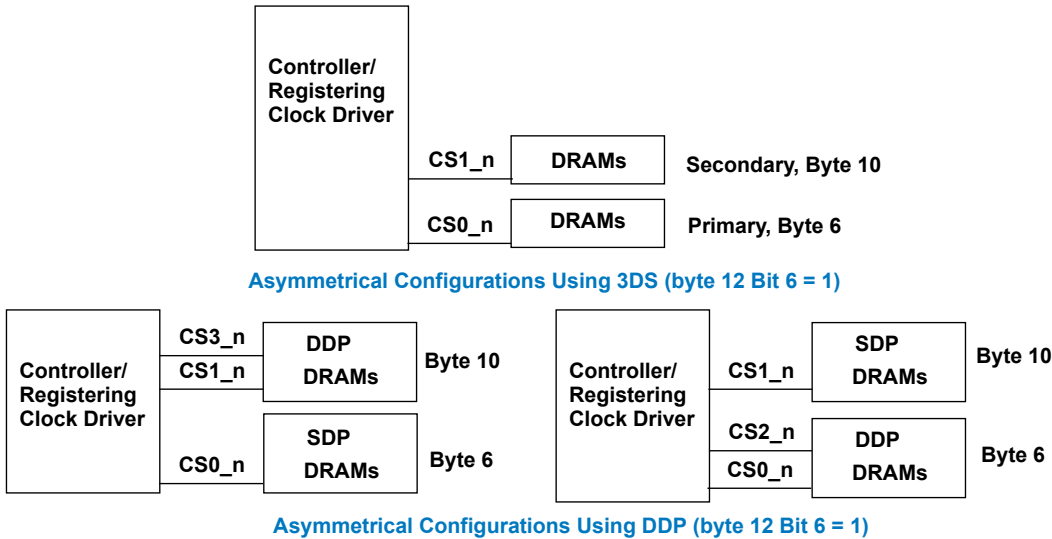


Figure 2 — Module Organization: Asymmetrical Configurations

8.1.13 Byte 12 (0x00C): Module Organization (Cont'd)
8.1.13.2 Asymmetrical Configurations (Cont'd)

Table 30 — Rank Mix for Asymmetrical Modules (Byte 12 Bit 6 = 1)

First SDRAM Package Type(s) (Byte 6) and Chip Select Alignment		Second SDRAM Package Type(s) (Byte 10) and Chip Select Alignment		# Package Ranks per DIMM	# Logical Ranks per DIMM
Non-3DS Asymmetrical Configurations					
SDP	CS0_n	DDP	CS1_n, CS3_n	3	3
DDP	CS0_n, CS2_n	SDP	CS1_n	3	3
3DS Asymmetrical Configurations					
2H 3DS	CS0_n	3H 3DS	CS1_n	2	5
2H 3DS	CS0_n	4H 3DS	CS1_n	2	6
...		...		...	...
7H 3DS	CS0_n	8H 3DS	CS1_n	2	15
Logical Ranks per DIMM = # Logical Ranks in Primary SDRAM type + # Logical Ranks in Secondary SDRAM type					

Examples: see Table 31

Table 31 — Module Configuration Examples

Byte 12, Bit 6	Byte 6	Byte 10	Byte 12, Bits 5~3, 2~0	Logical Ranks per DIMM	Module Configuration
Symmetry	Primary	Secondary	Ranks, Width		
0 = Sym.	0x00	0x00	000, 001	1	1 Package Rank x8, Monolithic
0 = Sym.	0x00	0x00	001, 001	2	2 Package Ranks x8, Monolithic
0 = Sym.	0x91	0x00	001, 000	2	2 Package Ranks x4, DDP
0 = Sym.	0xB2	0x00	000, 000	4	1 Package Rank x4, 4H 3DS
0 = Sym.	0xB2	0x00	001, 000	8	2 Package Ranks x4, 4H 3DS
1 = Asym.	0x00	0x91	010, 000	3	3 Package Ranks x4, SDP + DDP
1 = Asym.	0xF2	0xE2	001, 000	15	2 Package Ranks x4, 8H 3DS + 7H 3DS

8.1.14 Byte 13 (0x00D): Module Memory Bus Width

This byte, shown in Table 32, describes the width of the SDRAM memory bus on the module. Bits 2~0 encode the primary bus width. Bits 4~3 encode the bus extensions such as parity or ECC.

Table 32 — Byte 13 (0x00D): Module Memory Bus Width

Bits 7~5	Bits 4~3	Bits 2~0
Reserved	Bus width extension, in bits	Primary bus width, in bits
Reserved; must be coded as 000	Bit [4, 3] : 000 = 0 bits (no extension) 001 = 8 bits All others reserved	Bit [2, 1, 0] : 000 = 8 bits 001 = 16 bits 010 = 32 bits 011 = 64 bits All others reserved

Examples:

- 64 bit primary bus, no parity or ECC (64 bits total width): xxx 000 011
- 64 bit primary bus, with 8 bit ECC (72 bits total width): xxx 001 011

8.1.14.1 Calculating Module DRAM Capacity

The total memory capacity of the DRAM on the module may be calculated from SPD values. For example, to calculate the total capacity, in megabytes or gigabytes, of a typical module:

$$\text{Total} = \text{SDRAM Capacity} \div 8 * \text{Primary Bus Width} \div \text{SDRAM Width} * \text{Logical Ranks per DIMM}$$

where:

- SDRAM Capacity = SPD byte 4 bits 3~0
- Primary Bus Width = SPD byte 13 bits 2~0
- SDRAM Width = SPD byte 12 bits 2~0
- Logical Ranks per DIMM =

$$\text{for SDP, DDP, QDP:} = \text{SPD byte 12 bits 5~3}$$

$$\text{for 3DS:} = \text{SPD byte 12 bits 5~3 times SPD byte 6 bits 6~4}$$

Examples:

Two package ranks per DIMM using monolithic SDRAMs, 2 Gb per die, with x4 organization on a module with a 64 bit primary bus:

- $$\text{Total} = 2 \text{ Gb} \div 8 * 64 \div 4 * 2 * 1 = 8 \text{ GB}$$

Two package ranks per DIMM using DDP SDRAMs, 4 Gb per die, with x4 organization on a module with a 64 bit primary bus:

- $$\text{Total} = 4 \text{ Gb} \div 8 * 64 \div 4 * 2 = 16 \text{ GB}$$

Two package ranks per DIMM using 4H 3DS SDRAMs, 2 Gb per die, having a x8 organization on a module with a 64 bit primary bus:

- $$\text{Total} = 2 \text{ Gb} \div 8 * 64 \div 8 * 2 * 4 = 16 \text{ GB}$$

Commonly, parity or ECC are not counted in total module capacity, though they can also be included by adding the bus width extension in SPD byte 13 bits 4~3 to the primary bus width in the previous examples.

8.1.15 Byte 14 (0x00E): Module Thermal Sensor

This byte, shown in Table 33, describes the module’s supported thermal options.

Table 33 — Byte 14 (0x00E): Module Thermal Sensor

Bit 7	Bits 6~0
Thermal Sensor ¹	Reserved
0 = Thermal sensor not incorporated onto this assembly 1 = Thermal sensor incorporated onto this assembly	0 = Undefined All others reserved
NOTE 1 Thermal sensor compliant with TSE2004av specifications.	

8 Details of Each Byte (Cont'd)

8.1 General Configuration Section: Bytes 0~127 (0x000~0x07F) (Cont'd)

8.1.16 Byte 15 (0x00F): Extended Module Type

This byte, shown in Table 34, extends the module type field of byte 3. Used when byte 3 bits 3~0 = 0000.

Table 34 — Byte 15 (0x00F): Extended Module Type

Bit 7~4	Bits 3~0
Reserved	Extended Base Module Type
Reserved; must be coded as 0000	Bits [3, 2, 1, 0]: 0000 = Reserved; must be coded as 0000 ... 1111 = Reserved; must be coded as 0000

8.1.17 Byte 16 (0x010):

Reserved, must be coded as 0x00

8.1.18 Byte 17 (0x011): Timebases

This byte, shown in Table 35, defines a value in picoseconds that represents the fundamental timebase for fine grain and medium grain timing calculations. These values are used as a multiplier for formulating subsequent timing parameters.

Table 35 — Byte 17 (0x011): Timebases

Bits 7~4	Bits 3~2	Bits 1~0
Reserved	Medium Timebase (MTB)	Fine Timebase (FTB)
Reserved; must be coded as 0000	Bits [3, 2]: 00 = 125 ps All others reserved	Bits [0, 1]: 00 = 1 ps All others reserved

8.1.18.1 Relating the MTB and FTB

When a timing value tXX cannot be expressed by an integer number of MTB units, the SPD must be encoded using both the MTB and FTB. The Fine Offsets are encoded using a two's complement value which, when multiplied by the FTB yields a positive or negative correction factor. Typically, for safety and for legacy compatibility, the MTB portion is rounded UP and the FTB correction is a negative value. The general algorithm for programming SPD values is:

```

Temp_val = tXX / MTB           // Calculate as real number
Remainder = Temp_val modulo 1   // Determine if integer # MTBs
Fine_Correction = 1 - Remainder // If needed, what correction
if (Remainder == 0) then        // Integer # MTBs?
    tXX(MTB) = Temp_val         // Convert to integer
    tXX(FTB) = 0                // No correction needed
else                             // Needs correction
    tXX(MTB) = ceiling (Temp_val) // Round up for safety in legacy systems
    tXX(FTB) = Fine_Correction * MTB / FTB // Correction is negative offset
endif

```

To recalculate the value of tXX from the SPD values, a general formula BIOSes may use is:

$$tXX = tXX(MTB) * MTB + tXX(FTB) * FTB$$

Example (see Table 36):

Table 36 — $t_{CKAVGmin}$ SPD Calculations Using MTB and FTB

Speed Bin	t _{CKAVGmin} Value Decimal	SPD byte 18 Decimal (Hexadecimal)	SPD byte 125 Decimal (Hexadecimal)
DDR4-1866	1.071 ns	9 (0x09)	-54 (0xCA)
	=	(9 * 0.125) + (-54 * 0.001)	
NOTE Examples assume MTB of 0.125 ns and FTB of 0.001 ns			

Timing parameters using both MTB and FTB are shown in Table 37:

Table 37 — Timing parameters using both MTB and FTB

Parameter	MTB Byte(s)	FTB Byte
$t_{CKAVGmin}$	18 (0x012)	125 (0x07D)
$t_{CKAVGmax}$	19 (0x013)	124 (0x07C)
t_{AAmin}	24 (0x018)	123 (0x07B)
t_{RCDmin}	25 (0x019)	122 (0x07A)
t_{RPmin}	26 (0x01A)	121 (0x079)
t_{RCmin}	27, 29 (0x01B, 0x01D)	120 (0x078)
t_{RRD_Smin}	39 (0x027)	119 (0x077)
t_{RRD_Lmin}	40 (0x028)	118 (0x076)

The encoding of two's complement fine timebase offsets are shown in Table 38:

Table 38 — eEncoding of Two's Complement Fine Timebase Offsets

Coding		Value (Dec)	Value (Hex)	FTB Timebase
Bit 7	Bits 6~0			1 ps
0	1111111	+127	7F	+127 ps
0	1111110	+126	7E	+126 ps
...	...	...	...	
0	0000001	+1	01	+1 ps
0	0000000	0	00	0
1	1111111	-1	FF	-1 ps
1	1111110	-2	FE	-2 ps
...	...	...	...	
1	0000000	-128	80	-128 ps

8.1.18.2 Rounding Algorithms

Software algorithms for calculation of timing parameters are subject to rounding errors from many sources. For example, a system may use a memory clock with a nominal frequency of 933.33... MHz, or a clock period of 1.0714... ns. Similarly, a system with a memory clock frequency of 1066.66... MHz yields mathematically a clock period of 0.9375... ns. In most cases, it is impossible to express all digits after the decimal point exactly, and rounding must be done because the SPD establishes a minimum granularity for timing parameters of 1 ps.

Rules for rounding must be defined to allow optimization of memory module performance without violating device parameters. In some cases, rounding errors cause an unnecessary clock of delay, affecting system performance. These rounding algorithms rely on results that are within correction factors on device testing and specification to avoid losing performance due to rounding errors.

8.1.18 Byte 17 (0x011): Timebases (Cont'd)

8.1.18.2 Rounding Algorithms (Cont'd)

These rules are:

- Clock periods such as $t_{CKAVGmin}$ are defined to 1 ps of accuracy; for example, 0.9375... ns is defined as 937 ps and 1.0714... ns is defined as 1071 ps.
- Using real math, parameters like t_{AAmin} , t_{RCDmin} , etc. which are programmed in systems in numbers of clocks (nCK) but expressed in the SPD in units of time (in ns) are divided by the clock period (in ns) yielding a ratio of clock units (nCK), a correction factor of 2.5% is subtracted, then the result is set to the next higher integer number of clocks:

$$nCK = \text{ceiling} \left[\left(\frac{\text{parameter_in_ns}}{\text{application_t}_{CK_in_ns}} \right) - 0.025 \right]$$

- Alternatively, programmers may prefer to use integer math instead of real math by expressing timing in ps, scaling the desired parameter value by 1000, dividing by the application clock period, adding an inverse correction factor of 97.4%, dividing the result by 1000, then truncating down to the next lower integer value:

$$nCK = \text{truncate} \left[\frac{\left(\frac{\text{parameter_in_ps} \times 1000}{\text{application_t}_{CK_in_ps}} \right) + 974}{1000} \right]$$

- Either algorithm should yield identical results. In case of any conflict between the two methodologies, the integer method shall prevail.

Examples

Example 1, using REAL math to convert t_{AAmin} from ns to nCK (see Table 39):

```
// This algorithm subtracts 2.5% correction factor and rounds up to next integer value
real MTB, FTB, TaaMin, Correction, ClockPeriod, TempNck;
int TaaInNck;

MTB = 0.125; // From SPD[17], in ns
FTB = 0.001; // From SPD[17], in ns
TaaMin = (SPD[18] * MTB) + (SPD[123] * FTB); // Calculate tAAmin in ns (FTB is negative offset)
Correction = 0.025; // 2.5%, per rounding algorithm
ClockPeriod = ApplicationTck; // Frequency (clock period) is application dependent
TempNck = TaaMin / ClockPeriod; // Initial calculation of nCK
TempNck = TempNck - Correction; // Subtract correction factor from nCK
TaaInNck = (int)ceiling(TempNck); // Ceiling to next higher integer value
```

Table 39 — Example 1, using REAL math to convert t_{RCDmin} from ns to nCK

DDR4-2666W Device Operating at Standard Application Data Rates (Full & Downbinned)						
Timing Parameter: $t_{AAmin} = 15.0$ ns						
Application Speed Grade	Device t_{AA}	Application t_{CK}	Device $t_{AA} \div$ Application t_{CK}	2.5% Correction	$t_{AA} / t_{CK} -$ Correction	Ceiling Result
	ns	ns	ratio (real)	(real)	ratio (real)	nCK (integer)
2666	15.000	0.750	20.0	0.025	19.975	20
2400	15.000	0.833	18.0072	0.025	17.9822	18
2133	15.000	0.937	16.00854	0.025	15.9835	16
1866	15.000	1.071	14.0056	0.025	13.9806	14

Table 39 — Example 1, using REAL math to convert t_{RCDmin} from ns to nCK (Cont'd)

DDR4-2666W Device Operating at Standard Application Data Rates (Full & Downbinned) Timing Parameter: $t_{AAmin} = 15.0$ ns						
Application Speed Grade	Device t_{AA}	Application t_{CK}	Device $t_{AA} \div$ Application t_{CK}	2.5% Correction	$t_{AA} / t_{CK} -$ Correction	Ceiling Result
	ns	ns	ratio (real)	(real)	ratio (real)	nCK (integer)
1600	15.000	1.250	12.0	0.025	11.975	12
NOTE Roundup values for bins 2400, 2133, and 1866 would have lost one clock of performance without the application of the rounding algorithm. For example, a DDR4-2666W device running at DDR4-2400 data rates would have been required to set t_{AA} to 19 clocks without correction, but with correction t_{AA} may be safely programmed to 18 clocks.						

Example 2, using INT math to convert t_{AAmin} from ns to nCK (see Table 40):

```
// This algorithm uses adds 97.4% of a clock and truncates down to the next lower integer value
int MTB, FTB, TaaMin, ClockPeriod, TempNck, TaaInNck;

MTB = 125; // From SPD[17], in ps
FTB = 1; // From SPD[17], in ps
TaaMin = (SPD[18] * MTB) + (SPD[123] * FTB); // Calculate tAAmin in ps (FTB is negative offset)
ClockPeriod = ApplicationTckInPs; // Clock period is application specific
TempNck = (TaaMin * 1000) / ApplicationTckInPs; // Preliminary nCK calculation, scaled by 1000
TempNck = TempNck + 974; // Apply inverse of 2.5% correction factor
TaaInNck = (int)(TempNck / 1000); // Truncate to next lower integer
```

Table 40 — Example 2, using INT math to convert t_{RCDmin} from ns to nCK

DDR4-2666W Device Operating at Standard Application Data Rates (Full & Downbinned) Timing Parameter: $t_{AAmin} = 15.0$ ns (15000 ps)					
Application Speed Grade	Device t_{AA}	Application t_{CK}	(Device $t_{AA} * 1000$) Application t_{CK}	Add Inverse Correction	Truncate Corrected nCK / 1000
	ps	ps	Scaled nCK	Scaled nCK	nCK (integer)
2666	15000	750	20000	20974	20
2400	15000	833	18007	18981	18
2133	15000	937	16008	16982	16
1866	15000	1071	14005	14979	14
1600	15000	1250	12000	12974	12

8.1.19 Byte 18 (0x012): SDRAM Minimum Cycle Time ($t_{CKAVGmin}$)

This byte, shown in Table 41, defines the minimum cycle time for the SDRAM module, in medium timebase (MTB) units. This number applies to all applicable components on the module. This byte applies to SDRAM and support components as well as the overall capability of the DIMM. This value comes from the DDR4 SDRAM and support component data sheets.

Table 41 — Byte 18 (0x012): SDRAM Minimum Cycle Time ($t_{CKAVGmin}$)

Bits 7~0
Minimum SDRAM Cycle Time ($t_{CKAVGmin}$) MTB Units
Values defined from 1 to 255

If $t_{CKAVGmin}$ cannot be divided evenly by the MTB, this byte must be rounded up to the next larger integer and the Fine Offset for $t_{CKAVGmin}$ (SPD byte 125) used for correction to get the actual value.

8.1 General Configuration Section: Bytes 0~127 (0x000~0x07F) (Cont'd)

8.1.19 Byte 18 (0x012): SDRAM Minimum Cycle Time ($t_{CKAVGmin}$) (Cont'd)

Examples (see Table 42):

Table 42 — Examples of SDRAM Minimum Cycle Time

$t_{CKAVGmin}$ (MTB units)		MTB (ns)	$t_{CKAVGmin}$ Offset (FTB units) ¹		FTB (ns)	$t_{CKAVGmin}$ Result (ns)	Use
10	0x0A	0.125	0	0	0.001	1.25	DDR4-1600 (800 MHz clock)
9	0x09	0.125	-54	0xCA	0.001	1.071	DDR4-1866 (933 MHz clock)
8	0x08	0.125	-63	0xC1	0.001	0.937	DDR4-2133 (1067 MHz clock)
7	0x07	0.125	-42	0xD6	0.001	0.833	DDR4-2400 (1200 MHz clock)
6	0x06	0.125	0	0	0.001	0.750	DDR4-2666 (1333 MHz clock)
6	0x06	0.125	-68	0xBC	0.001	0.682	DDR4-2933 (1466 MHz clock)
5	0x05	0.125	0	0	0.001	0.625	DDR4-3200 (1600 MHz clock)

NOTE 1 1: See SPD byte 125.

8.1.20 Byte 19 (0x013): SDRAM Maximum Cycle Time ($t_{CKAVGmax}$)

This word defines the maximum cycle time for the SDRAM module, in medium timebase (MTB) units. This number applies to all applicable components on the module. This byte, shown in Table 43, applies to SDRAM and support components as well as the overall capability of the DIMM. This value comes from the DDR4 SDRAM and support component data sheets.

Table 43 — Byte 19 (0x013): SDRAM Maximum Cycle Time ($t_{CKAVGmax}$)

Bits 7~0	
Minimum SDRAM Cycle Time ($t_{CKAVGmax}$) MTB Units	
Values defined from 1 to 255	

If $t_{CKAVGmax}$ cannot be divided evenly by the MTB, this byte must be rounded up to the next larger integer and the Fine Offset for $t_{CKAVGmax}$ (SPD byte 124) used for correction to get the actual value.

Examples: See Table 44.

Table 44 — Examples of SDRAM Maximum Cycle Time

$t_{CKAVGmax}$ (MTB units)		MTB (ns)	$t_{CKAVGmax}$ Offset (FTB units) ¹		FTB (ns)	$t_{CKAVGmax}$ Result (ns)	Use
13	0x0D	0.125	-25	0xE7	0.001	1.600	DDR4-1600 (800 MHz clock)
13	0x0D	0.125	-25	0xE7	0.001	1.600	DDR4-1866 (933 MHz clock)
13	0x0D	0.125	-25	0xE7	0.001	1.600	DDR4-2133 (1067 MHz clock)
13	0x0D	0.125	-25	0xE7	0.001	1.600	DDR4-2400 (1200 MHz clock)
13	0x0D	0.125	-25	0xE7	0.001	1.600	DDR4-2666 (1333 MHz clock)
13	0x0D	0.125	-25	0xE7	0.001	1.600	DDR4-2933 (1466 MHz clock)
13	0x0D	0.125	-25	0xE7	0.001	1.600	DDR4-3200 (1600 MHz clock)
12	0x0C	0.125	0	0	0.001	1.500	DDR4-1600 3DS (800 MHz clock)
12	0x0C	0.125	0	0	0.001	1.500	DDR4-1866 3DS (933 MHz clock)
12	0x0C	0.125	0	0	0.001	1.500	DDR4-2133 3DS (1067 MHz clock)
12	0x0C	0.125	0	0	0.001	1.500	DDR4-2400 3DS (1200 MHz clock)
12	0x0C	0.125	0	0	0.001	1.500	DDR4-2666 3DS (1333 MHz clock)
12	0x0C	0.125	0	0	0.001	1.500	DDR4-2933 3DS (1466 MHz clock)
12	0x0C	0.125	0	0	0.001	1.500	DDR4-3200 3DS (1600 MHz clock)

NOTE 1 1: See SPD byte 124.

8 Details of Each Byte (Cont'd)

8.1 General Configuration Section: Bytes 0~127 (0x000~0x07F) (Cont'd)

8.1.21 Byte 20 (0x014): CAS Latencies Supported, First Byte
Byte 21 (0x015): CAS Latencies Supported, Second Byte
Byte 22 (0x016): CAS Latencies Supported, Third Byte
Byte 23 (0x017): CAS Latencies Supported, Fourth Byte

These bytes, shown in Tables 45 and 46, define which CAS Latency (CL) values are supported with one bit per possible CAS Latency. A 1 in a bit position means that CL is supported, a 0 in that bit position means it is not supported. Bit 7 of byte 23 selects the range of CL values, 7~36 or 23~52. Byte 23 bit 6 is reserved for future use and must be coded as 0. These values come from the DDR4 SDRAM data sheet, JESD79-4.

CAS Latency Mask may cover either non-DBI mode or both DBI and non-DBI modes of operation; this is vendor specific.

Table 45 — CAS Latencies, Low Range (Byte 23 Bit 7 = 0)

Byte 20	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
	CL = 14	CL = 13	CL = 12	CL = 11	CL = 10	CL = 9	CL = 8	CL = 7
Byte 21	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
	CL = 22	CL = 21	CL = 20	CL = 19	CL = 18	CL = 17	CL = 16	CL = 15
Byte 22	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
	CL = 30	CL = 29	CL = 28	CL = 27	CL = 26	CL = 25	CL = 24	CL = 23
Byte 23	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
	0 = Low CL range	Rsvd	CL = 36	CL = 35	CL = 34	CL = 33	CL = 32	CL = 31

NOTE 1 Byte 23 bit 7 selects the possible CAS Latency range.

NOTE 2 Byte 23 bit 6 is reserved and must be coded as 0.

NOTE 3 For each other bit position, 0 means this CAS Latency is not supported, 1 means this CAS Latency is supported.

Table 46 — CAS Latencies, High Range (Byte 23 Bit 7 = 1)

Byte 20	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
	CL = 30	CL = 29	CL = 28	CL = 27	CL = 26	CL = 25	CL = 24	CL = 23
Byte 21	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
	CL = 38	CL = 37	CL = 36	CL = 35	CL = 34	CL = 33	CL = 32	CL = 31
Byte 22	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
	CL = 46	CL = 45	CL = 44	CL = 43	CL = 42	CL = 41	CL = 40	CL = 39
Byte 23	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
	1 = High CL range	Rsvd	CL = 52	CL = 51	CL = 50	CL = 49	CL = 48	CL = 47

NOTE 1 Byte 23 bit 7 selects the possible CAS Latency range.

NOTE 2 Byte 23 bit 6 is reserved and must be coded as 0.

NOTE 3 For each other bit position, 0 means this CAS Latency is not supported, 1 means this CAS Latency is supported.

Example 1:

Byte 23 bit 7 = 0 (Low CL Range)

Byte 20 = 0xB4 (= 1011 0100) -- first byte

Byte 21 = 0x05 (= 0000 0101) -- second byte

Byte 22 = 0x00 (= 0000 0000) -- third byte

Byte 23 = 0x00 (= 0000 0000) -- fourth byte

CAS Latencies	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7
CL Mask	0	0	0	0	0	1	0	1	1	0	1	1	0	1	0	0
CAS Latencies	R	X	36	35	34	33	32	31	30	29	28	27	26	25	24	23

8.1 General Configuration Section: Bytes 0~127 (0x000~0x07F) (Cont'd)**8.1.21 Bytes 20, 21, 22, and 23: CAS Latencies Supported (Cont'd)**

CL Mask	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
----------------	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

Results: Actual CAS Latencies supported = 9, 11, 12, 14, 15, 17

Example 2:

Byte 23 bit 7 = 1 (High CL Range)

Byte 20 = 0x00 (= 0000 0000) -- first byte

Byte 21 = 0x78 (= 0111 1000) -- second byte

Byte 22 = 0x01 (= 0000 0001) -- third byte

Byte 23 = 0x80 (= 1000 0000) -- fourth byte

CAS Latencies	38	37	36	35	34	33	32	31	30	29	28	27	26	25	24	23
CL Mask	0	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0
CAS Latencies	R	X	52	51	50	49	48	47	46	45	44	43	42	41	40	39
CL Mask	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Results: Actual CAS Latencies supported = 34, 35, 36, 37, 39

8.1.22 Byte 24 (0x018): Minimum CAS Latency Time (t_{AAmin})

This word, shown in Table 47, defines the minimum CAS Latency in medium timebase (MTB) units. This value comes from the DDR4 SDRAM data sheet.

Table 47 — Byte 24 (0x018): Minimum CAS Latency Time (t_{AAmin})

Bits 7~0
Minimum SDRAM CAS Latency Time (t_{AAmin})
MTB Units
Values defined from 1 to 255

If t_{AAmin} cannot be divided evenly by the MTB, this byte must be rounded up to the next larger integer and the Fine Offset for t_{AAmin} (SPD byte 123) used for correction to get the actual value.

Examples: See Table 48.**Table 48 — Examples of Minimum CAS Latency Time**

t _{AA} min (MTB units)		MTB (ns)	t _{AA} min Offset (FTB units) ¹		FTB (ns)	t _{AA} min Result (ns)	Use	Notes
100	0x64	0.125	0	0	0.001	12.50	DDR4-1600J	
110	0x6E	0.125	0	0	0.001	13.75	DDR4-1600K	
108	0x6C	0.125	0	0	0.001	13.50	DDR4-1600K-D	2
120	0x78	0.125	0	0	0.001	15.00	DDR4-1600L	
103	0x67	0.125	-26	0xE6	0.001	12.85	DDR4-1866L	
112	0x70	0.125	-81	0xAF	0.001	13.92	DDR4-1866M	
108	0x6C	0.125	0	0	0.001	13.50	DDR4-1866M-D	2
120	0x78	0.125	0	0	0.001	15.00	DDR4-1866N	
106	0x6A	0.125	-120	0x88	0.001	13.13	DDR4-2133N	
113	0x71	0.125	-65	0xBF	0.001	14.06	DDR4-2133P	
108	0x6C	0.125	0	0	0.001	13.50	DDR4-2133P-D	2
120	0x78	0.125	0	0	0.001	15.00	DDR4-2133R	
100	0x64	0.125	0	0	0.001	12.50	DDR4-2400P	
107	0x6B	0.125	-55	0xC9	0.001	13.32	DDR4-2400R	
114	0x72	0.125	-90	0xA6	0.001	14.16	DDR4-2400T	

8.1 General Configuration Section: Bytes 0~127 (0x000~0x07F) (Cont'd)

8.1.22 Byte 24 (0x018): Minimum CAS Latency Time (t_{AAmin}) (Cont'd)

Table 48 — Examples of Minimum CAS Latency Time (Cont'd)

t_{AAmin} (MTB units)		MTB (ns)	t_{AAmin} Offset (FTB units) ¹		FTB (ns)	t_{AAmin} Result (ns)	Use	Notes
110	0x6E	0.125	0	0	0.001	13.75	DDR4-2400T-D	2
120	0x78	0.125	0	0	0.001	15.00	DDR4-2400U	
102	0x66	0.125	0	0	0.001	12.75	DDR4-2666T	
108	0x6C	0.125	0	0	0.001	13.50	DDR4-2666U	
114	0x72	0.125	0	0	0.001	14.25	DDR4-2666V	
110	0x6E	0.125	0	0	0.001	13.75	DDR4-2666V-D	2
120	0x78	0.125	0	0	0.001	15.00	DDR4-2666W	
104	0x68	0.125	-40	0xD8	0.001	12.96	DDR4-2933V	
110	0x6E	0.125	-110	0x92	0.001	13.64	DDR4-2933W	
115	0x73	0.125	-55	0xC9	0.001	14.32	DDR4-2933Y	
110	0x6E	0.125	0	0	0.001	13.75	DDR4-2933Y-D	2
120	0x78	0.125	0	0	0.001	15.00	DDR4-2933AA	
100	0x64	0.125	0	0	0.001	12.50	DDR4-3200W	
110	0x6E	0.125	0	0	0.001	13.75	DDR4-3200AA	
120	0x78	0.125	0	0	0.001	15.00	DDR4-3200AB	
120	0x78	0.125	0	0	0.001	15.00	DDR4-1600J-3DS2B	
130	0x82	0.125	0	0	0.001	16.25	DDR4-1600K-3DS2B	
140	0x8C	0.125	0	0	0.001	17.50	DDR4-1600L-3DS2B	
120	0x78	0.125	0	0	0.001	15.00	DDR4-1866L-3DS2B	
120	0x78	0.125	0	0	0.001	15.00	DDR4-1866L-D-3DS2B	2
129	0x81	0.125	-55	0xC9	0.001	16.07	DDR4-1866M-3DS2B	
129	0x81	0.125	-55	0xC9	0.001	16.07	DDR4-1866M-D-3DS2B	2
138	0x8A	0.125	-110	0x92	0.001	17.14	DDR4-1866N-3DS2B	
128	0x80	0.125	-51	0xCD	0.001	15.95	DDR4-2133P-3DS2A	
128	0x80	0.125	-51	0xCD	0.001	15.95	DDR4-2133P-D-3DS2A	2
136	0x88	0.125	-121	0x87	0.001	16.88	DDR4-2133R-3DS3A	
151	0x97	0.125	-115	0x8D	0.001	18.76	DDR4-2133R-3DS4A	
138	0x8A	0.125	-110	0x92	0.001	17.14	DDR4-2133R-D-3DS4A	2
120	0x78	0.125	0	0	0.001	15.00	DDR4-2400P-3DS3B	
127	0x7F	0.125	-45	0xD3	0.001	15.83	DDR4-2400T-3DS2A	
134	0x86	0.125	-80	0xB0	0.001	16.67	DDR4-2400U-3DS2A	
147	0x93	0.125	-46	0xD2	0.001	18.33	DDR4-2400U-3DS4A	
138	0x8A	0.125	-110	0x92	0.001	17.14	DDR4-2400U-D-3DS4A	2
120	0x78	0.125	0	0	0.001	15.00	DDR4-2666T-3DS3A	
132	0x84	0.125	0	0	0.001	16.50	DDR4-2666V-3DS3A	
144	0x90	0.125	0	0	0.001	18.00	DDR4-2666W-3DS4A	
138	0x8A	0.125	-110	0x92	0.001	17.14	DDR4-2666W-D-3DS4A	2
		0.125			0.001	tbd	DDR4-2933tbd-3DStbd	
		0.125			0.001	tbd	DDR4-2933tbd-3DStbd	
		0.125			0.001	tbd	DDR4-2933tbd-3DStbd	
120	0x78	0.125	0	0	0.001	15.00	DDR4-3200W-3DS4A	
130	0x82	0.125	0	0	0.001	16.25	DDR4-3200AA-3DS4A	
140	0x8C	0.125	0	0	0.001	17.50	DDR4-3200AC-3DS4A	
138	0x8A	0.125	-110	0x92	0.001	17.14	DDR4-3200AC-D-3DS4A	2
NOTE 1 1: See SPD byte 123								
NOTE 2 2: Refer to device data sheet for downbin support details.								

8 Details of Each Byte (Cont'd)

8.1 General Configuration Section: Bytes 0~127 (0x000~0x07F) (Cont'd)

8.1.23 Byte 25 (0x019): Minimum RAS to CAS Delay Time (t_{RCDmin})

This word, shown in Table 49, defines the minimum SDRAM RAS to CAS Delay Time in medium timebase (MTB) units. This value comes from the DDR4 SDRAM data sheet.

Table 49 — Byte 25 (0x019): Minimum RAS to CAS Delay Time (t_{RCDmin})

Bits 7~0
Byte 25: Minimum SDRAM RAS to CAS Delay Time (t_{RCDmin}) MTB Units
Values defined from 1 to 255

If t_{RCDmin} cannot be divided evenly by the MTB, this byte must be rounded up to the next larger integer and the Fine Offset for t_{RCDmin} (SPD byte 122) used for correction to get the actual value

Examples: See Table 50.

Table 50 — Examples of Minimum RAS to CAS Delay Time

t _{RCD} min (MTB units)		MTB (ns)	t _{RCD} min Offset (FTB units) ¹		FTB (ns)	t _{RCD} min Result (ns)	Use	Notes
100	0x64	0.125	0	0	0.001	12.50	DDR4-1600J	
110	0x6E	0.125	0	0	0.001	13.75	DDR4-1600K	
108	0x6C	0.125	0	0	0.001	13.50	DDR4-1600K-D	2
120	0x78	0.125	0	0	0.001	15.00	DDR4-1600L	
103	0x67	0.125	-26	0xE6	0.001	12.85	DDR4-1866L	
112	0x70	0.125	-81	0xAF	0.001	13.92	DDR4-1866M	
108	0x6C	0.125	0	0	0.001	13.50	DDR4-1866M-D	2
120	0x78	0.125	0	0	0.001	15.00	DDR4-1866N	
106	0x6A	0.125	-120	0x88	0.001	13.13	DDR4-2133N	
113	0x71	0.125	-65	0xBF	0.001	14.06	DDR4-2133P	
108	0x6C	0.125	0	0	0.001	13.50	DDR4-2133P-D	2
120	0x78	0.125	0	0	0.001	15.00	DDR4-2133R	
100	0x64	0.125	0	0	0.001	12.50	DDR4-2400P	
107	0x6B	0.125	-55	0xC9	0.001	13.32	DDR4-2400R	
114	0x72	0.125	-90	0xA6	0.001	14.16	DDR4-2400T	
110	0x6E	0.125	0	0	0.001	13.75	DDR4-2400T-D	2
120	0x78	0.125	0	0	0.001	15.00	DDR4-2400U	
102	0x66	0.125	0	0	0.001	12.75	DDR4-2666T	
108	0x6C	0.125	0	0	0.001	13.50	DDR4-2666U	
114	0x72	0.125	0	0	0.001	14.25	DDR4-2666V	
110	0x6E	0.125	0	0	0.001	13.75	DDR4-2666V-D	2
120	0x78	0.125	0	0	0.001	15.00	DDR4-2666W	
104	0x68	0.125	-40	0xD8	0.001	12.96	DDR4-2933V	
110	0x6E	0.125	-110	0x92	0.001	13.64	DDR4-2933W	
115	0x73	0.125	-55	0xC9	0.001	14.32	DDR4-2933Y	
110	0x6E	0.125	0	0	0.001	13.75	DDR4-2933Y-D	2
120	0x78	0.125	0	0	0.001	15.00	DDR4-2933AA	
100	0x64	0.125	0	0	0.001	12.50	DDR4-3200W	
110	0x6E	0.125	0	0	0.001	13.75	DDR4-3200AA	
120	0x78	0.125	0	0	0.001	15.00	DDR4-3200AB	
110	0x78	0.125	0	0	0.001	13.75	DDR4-1600J-3DS2B	
120	0x78	0.125	0	0	0.001	15.00	DDR4-1600K-3DS2B	
130	0x82	0.125	0	0	0.001	16.25	DDR4-1600L-3DS2B	
112	0x70	0.125	-81	0xAF	0.001	13.92	DDR4-1866L-3DS2B	
110	0x6E	0.125	0	0	0.001	13.75	DDR4-1866L-D-3DS2B	2

8.1 General Configuration Section: Bytes 0~127 (0x000~0x07F) (Cont'd)
8.1.23 Byte 25 (0x019): Minimum RAS to CAS Delay Time (t_{RCDmin}) (Cont'd)

Table 50 — Examples of Minimum RAS to CAS Delay Time (Cont'd)

t_{RCDmin} (MTB units)		MTB (ns)	t_{RCDmin} Offset (FTB units) ¹		FTB (ns)	t_{RCDmin} Result (ns)	Use	Notes
120	0x78	0.125	0	0	0.001	15.00	DDR4-1866M-3DS2B	
120	0x78	0.125	0	0	0.001	15.00	DDR4-1866M-D-3DS2B	2
129	0x81	0.125	-55	0xC9	0.001	16.07	DDR4-1866N-3DS2B	
113	0x71	0.125	-65	0xBF	0.001	14.06	DDR4-2133P-3DS2A	
113	0x71	0.125	-65	0xBF	0.001	14.06	DDR4-2133P-D-3DS2A	2
113	0x71	0.125	-65	0xBF	0.001	14.06	DDR4-2133R-3DS3A	
120	0x78	0.125	0	0	0.001	15.00	DDR4-2133R-3DS4A	
120	0x78	0.125	0	0	0.001	15.00	DDR4-2133R-D-3DS4A	2
107	0x6B	0.125	-45	0xD3	0.001	13.33	DDR4-2400P-3DS3B	
114	0x72	0.125	-90	0xA6	0.001	14.16	DDR4-2400T-3DS2A	
120	0x78	0.125	0	0	0.001	15.00	DDR4-2400U-3DS2A	
120	0x78	0.125	0	0	0.001	15.00	DDR4-2400U-3DS4A	
120	0x78	0.125	0	0	0.001	15.00	DDR4-2400U-D-3DS4A	2
102	0x66	0.125	0	0	0.001	12.75	DDR4-2666T-3DS3A	
114	0x72	0.125	0	0	0.001	14.25	DDR4-2666V-3DS3A	
120	0x78	0.125	0	0	0.001	15.00	DDR4-2666W-3DS4A	
120	0x78	0.125	0	0	0.001	15.00	DDR4-2666W-D-3DS4A	2
		0.125			0.001	tbd	DDR4-2933tbd-3DS2b	
		0.125			0.001	tbd	DDR4-2933tbd-3DS2b	
		0.125			0.001	tbd	DDR4-2933tbd-3DS2b	
100	0x64	0.125	0	0	0.001	12.50	DDR4-3200W-3DS4A	
107	0x6B	0.125	-105	0x97	0.001	13.27	DDR4-3200AA-3DS4A	
120	0x78	0.125	0	0	0.001	15.00	DDR4-3200AC-3DS4A	
120	0x78	0.125	0	0	0.001	15.00	DDR4-3200AC-D-3DS4A	2
NOTE 1 See SPD byte 122								
NOTE 2 Refer to device data sheet for downbin support details.								

8.1.24 Byte 26 (0x01A): Minimum Row Precharge Delay Time (t_{RPmin})

This word, shown in Table 51, defines the minimum SDRAM Row Precharge Delay Time in medium timebase (MTB) units. This value comes from the DDR4 SDRAM data sheet.

Table 51 — Byte 26 (0x01A): Minimum Row Precharge Delay Time (t_{RPmin})

Bits 7~0	
Minimum Row Precharge Time (t_{RPmin})	
MTB Units	
Values defined from 1 to 255	

If t_{RPmin} cannot be divided evenly by the MTB, this byte must be rounded up to the next larger integer and the Fine Offset for t_{RPmin} (SPD byte 121) used for correction to get the actual value.

Examples: See Table 52.

Table 52 — Examples of Minimum Row Precharge Delay Time

t_{RPmin} (MTB units)		MTB (ns)	t_{RPmin} Offset (FTB units) ¹		FTB (ns)	t_{RPmin} Result (ns)	Use	Notes
100	0x64	0.125	0	0	0.001	12.50	DDR4-1600J	
110	0x6E	0.125	0	0	0.001	13.75	DDR4-1600K	
108	0x6C	0.125	0	0	0.001	13.50	DDR4-1600K-D	2

8.1 General Configuration Section: Bytes 0~127 (0x000~0x07F) (Cont'd)
8.1.24 Byte 26 (0x01A): Minimum Row Precharge Delay Time (t_{RPmin}) (Cont'd)

Table 52 — Examples of Minimum Row Precharge Delay Time (Cont'd)

t_{RPmin} (MTB units)		MTB (ns)	t_{RPmin} Offset (FTB units) ¹		FTB (ns)	t_{RPmin} Result (ns)	Use	Notes
120	0x78	0.125	0	0	0.001	15.00	DDR4-1600L	
103	0x67	0.125	-26	0xE6	0.001	12.85	DDR4-1866L	
112	0x70	0.125	-81	0xAF	0.001	13.92	DDR4-1866M	
108	0x6C	0.125	0	0	0.001	13.50	DDR4-1866M-D	2
120	0x78	0.125	0	0	0.001	15.00	DDR4-1866N	
106	0x6A	0.125	-120	0x88	0.001	13.13	DDR4-2133N	
113	0x71	0.125	-65	0xBF	0.001	14.06	DDR4-2133P	
108	0x6C	0.125	0	0	0.001	13.50	DDR4-2133P-D	2
120	0x78	0.125	0	0	0.001	15.00	DDR4-2133R	
100	0x64	0.125	0	0	0.001	12.50	DDR4-2400P	
107	0x6B	0.125	-55	0xC9	0.001	13.32	DDR4-2400R	
114	0x72	0.125	-90	0xA6	0.001	14.16	DDR4-2400T	
110	0x6E	0.125	0	0	0.001	13.75	DDR4-2400T-D	2
120	0x78	0.125	0	0	0.001	15.00	DDR4-2400U	
102	0x66	0.125	0	0	0.001	12.75	DDR4-2666T	
108	0x6C	0.125	0	0	0.001	13.50	DDR4-2666U	
114	0x72	0.125	0	0	0.001	14.25	DDR4-2666V	
110	0x6E	0.125	0	0	0.001	13.75	DDR4-2666V-D	2
120	0x78	0.125	0	0	0.001	15.00	DDR4-2666W	
104	0x68	0.125	-40	0xD8	0.001	12.96	DDR4-2933V	
110	0x6E	0.125	-110	0x92	0.001	13.64	DDR4-2933W	
115	0x73	0.125	-55	0xC9	0.001	14.32	DDR4-2933Y	
110	0x6E	0.125	0	0	0.001	13.75	DDR4-2933Y-D	2
120	0x78	0.125	0	0	0.001	15.00	DDR4-2933AA	
100	0x64	0.125	0	0	0.001	12.50	DDR4-3200W	
110	0x6E	0.125	0	0	0.001	13.75	DDR4-3200AA	
120	0x78	0.125	0	0	0.001	15.00	DDR4-3200AB	
100	0x64	0.125	0	0	0.001	12.50	DDR4-1600J-3DS2B	
110	0x6E	0.125	0	0	0.001	13.75	DDR4-1600K-3DS2B	
120	0x78	0.125	0	0	0.001	15.00	DDR4-1600L-3DS2B	
103	0x67	0.125	-26	0xE6	0.001	12.85	DDR4-1866L-3DS2B	
103	0x67	0.125	-26	0xE6	0.001	12.85	DDR4-1866L-D-3DS2B	2
112	0x70	0.125	-81	0xAF	0.001	13.92	DDR4-1866M-3DS2B	
110	0x6E	0.125	0	0	0.001	13.75	DDR4-1866M-D-3DS2B	2
120	0x78	0.125	0	0	0.001	15.00	DDR4-1866N-3DS2B	
113	0x71	0.125	-65	0xBF	0.001	14.06	DDR4-2133P-3DS2A	
110	0x6E	0.125	0	0	0.001	13.75	DDR4-2133P-D-3DS2A	2
113	0x71	0.125	-65	0xBF	0.001	14.06	DDR4-2133R-3DS3A	
120	0x78	0.125	0	0	0.001	15.00	DDR4-2133R-3DS4A	
120	0x78	0.125	0	0	0.001	15.00	DDR4-2133R-D-3DS4A	2
100	0x64	0.125	0	0	0.001	12.50	DDR4-2400P-3DS3B	
114	0x72	0.125	-90	0xA6	0.001	14.16	DDR4-2400T-3DS2A	
120	0x78	0.125	0	0	0.001	15.00	DDR4-2400U-3DS2A	
120	0x78	0.125	0	0	0.001	15.00	DDR4-2400U-3DS4A	
120	0x78	0.125	0	0	0.001	15.00	DDR4-2400U-D-3DS4A	2
102	0x66	0.125	0	0	0.001	12.75	DDR4-2666T-3DS3A	
114	0x72	0.125	0	0	0.001	14.25	DDR4-2666V-3DS3A	
120	0x78	0.125	0	0	0.001	15.00	DDR4-2666W-3DS4A	
120	0x78	0.125	0	0	0.001	15.00	DDR4-2666W-D-3DS4A	2
		0.125			0.001	tbd	DDR4-2933tbd-3DStbd	
		0.125			0.001	tbd	DDR4-2933tbd-3DStbd	
		0.125			0.001	tbd	DDR4-2933tbd-3DStbd	

8.1 General Configuration Section: Bytes 0~127 (0x000~0x07F) (Cont'd)
8.1.24 Byte 26 (0x01A): Minimum Row Precharge Delay Time (t_{RPmin}) (Cont'd)

Table 52 — Examples of Minimum Row Precharge Delay Time (Cont'd)

t_{RPmin} (MTB units)		MTB (ns)	t_{RPmin} Offset (FTB units) ¹		FTB (ns)	t_{RPmin} Result (ns)	Use	Notes
100	0x64	0.125	0	0	0.001	12.50	DDR4-3200W-3DS4A	
107	0x6B	0.125	-105	0x97	0.001	13.27	DDR4-3200AA-3DS4A	
120	0x78	0.125	0	0	0.001	15.00	DDR4-3200AC-3DS4A	
120	0x78	0.125	0	0	0.001	15.00	DDR4-3200AC-D-3DS4A	2
NOTE 1 See SPD byte 121								
NOTE 2 Device supports downbinning in lower frequency applications; see supplier data sheet								

8.1.25 Byte 27 (0x01B): Upper Nibbles for t_{RASmin} and t_{RCmin}

This byte, shown in Table 53, defines the most significant nibbles for the values of t_{RASmin} (byte 28) and t_{RCmin} (byte 29). These values come from the DDR4 SDRAM data sheet.

Table 53 — Byte 27 (0x01B): Upper Nibbles for t_{RASmin} and t_{RCmin}

Bits 7~4	Bits 3~0
t_{RCmin} Most Significant Nibble	t_{RASmin} Most Significant Nibble
See Byte 28 description	See Byte 29 description

8.1.26 Byte 28 (0x01C): Minimum Active to Precharge Delay Time (t_{RASmin}), Least Significant Byte

The lower nibble of Byte 27 and the contents of Byte 28, shown in Table 54, combined create a 12-bit value which defines the minimum SDRAM Active to Precharge Delay Time in medium timebase (MTB) units. The most significant bit is Bit 3 of Byte 27, and the least significant bit is Bit 0 of Byte 28. This value comes from the DDR4 SDRAM data sheet.

Table 54 — Byte 28 (0x01C): Minimum Active to Precharge Delay Time (t_{RASmin}), Least Significant Byte

Byte 27 Bits 3~0, Byte 28 Bits 7~0
Minimum Active to Precharge Time (t_{RASmin}) MTB Units
Values defined from 1 to 4095

Examples: See Table 55.

Table 55 — Examples of Minimum Active to Precharge Delay Time

t_{RASmin} (MTB units)		MTB (ns)	t_{RASmin} Result (ns)	Use
280	0x118	0.125	35	DDR4-1600
272	0x110	0.125	34	DDR4-1866
264	0x108	0.125	33	DDR4-2133
256	0x100	0.125	32	DDR4-2400
256	0x100	0.125	32	DDR4-2666
256	0x100	0.125	32	DDR4-2933
256	0x100	0.125	32	DDR4-3200
280	0x118	0.125	35	DDR4-1600-3DS
272	0x110	0.125	34	DDR4-1866-3DS
264	0x108	0.125	33	DDR4-2133-3DS
256	0x100	0.125	32	DDR4-2400-3DS
256	0x100	0.125	32	DDR4-2666-3DS
		0.125	tbd	DDR4-2933-3DS
256	0x100	0.125	32	DDR4-3200-3DS

8 Details of Each Byte (Cont'd)

8.1 General Configuration Section: Bytes 0~127 (0x000~0x07F) (Cont'd)

8.1.27 Byte 29 (0x01D): Minimum Active to Active/Refresh Delay Time (t_{RCmin}), Least Significant Byte

The upper nibble of Byte 27 and the contents of Byte 29, shown in Table 56, combined create a 12-bit value which defines the minimum SDRAM Active to Active/Refresh Delay Time in medium timebase (MTB) units. The most significant bit is Bit 7 of Byte 27, and the least significant bit is Bit 0 of Byte 29. This value comes from the DDR4 SDRAM data sheet.

Table 56 — Byte 29 (0x01D): Minimum Active to Active/Refresh Delay Time (t_{RCmin}), Least Significant Byte

Byte 27 Bits 7~4, Byte 29 Bits 7~0
Minimum Active to Active/Refresh Time (t_{RCmin}) MTB Units
Values defined from 1 to 4095

If t_{RCmin} cannot be divided evenly by the MTB, this byte must be rounded up to the next larger integer and the Fine Offset for t_{RCmin} (SPD byte 120) used for correction to get the actual value.

By DDR4 device design, t_{RCmin} must always be greater than or equal to $t_{RASmin} + t_{RPmin}$. When converting time in ns to nCK units using the appropriate rounding algorithms for these three parameters, systems must account for the overall design requirement and choose the larger of t_{RCmin} or ($t_{RASmin} + t_{RPmin}$) as a minimum setting.

Examples: See Table 57.

Table 57 — Examples of Minimum Active to Active/Refresh Delay Time

t _{RC} min (MTB units)		MTB (ns)	t _{RC} min Offset (FTB units) ¹		FTB (ns)	t _{RC} min Result (ns)	Use	Notes
380	0x17C	0.125	0	0	0.001	47.50	DDR4-1600J	
390	0x186	0.125	0	0	0.001	48.75	DDR4-1600K	
388	0x184	0.125	0	0	0.001	48.50	DDR4-1600K-D	2
400	0x190	0.125	0	0	0.001	50.00	DDR4-1600L	
375	0x177	0.125	-25	0xE7	0.001	46.85	DDR4-1866L	
384	0x180	0.125	-80	0xB0	0.001	47.92	DDR4-1866M	
380	0x17C	0.125	0	0	0.001	47.50	DDR4-1866M-D	2
392	0x188	0.125	0	0	0.001	49.00	DDR4-1866N	
370	0x172	0.125	-120	0x88	0.001	46.13	DDR4-2133N	
377	0x179	0.125	-65	0xBF	0.001	47.06	DDR4-2133P	
372	0x174	0.125	0	0	0.001	46.50	DDR4-2133P-D	2
384	0x180	0.125	0	0	0.001	48.00	DDR4-2133R	
356	0x164	0.125	0	0	0.001	44.50	DDR4-2400P	
363	0x16B	0.125	-55	0xC9	0.001	45.32	DDR4-2400R	
370	0x172	0.125	-91	0xA5	0.001	46.16	DDR4-2400T	
366	0x16E	0.125	0	0	0.001	45.75	DDR4-2400T-D	2
376	0x178	0.125	0	0	0.001	47.00	DDR4-2400U	
358	0x166	0.125	0	0	0.001	44.75	DDR4-2666T	
364	0x16C	0.125	0	0	0.001	45.50	DDR4-2666U	
370	0x172	0.125	0	0	0.001	46.25	DDR4-2666V	
366	0x16E	0.125	0	0	0.001	45.75	DDR4-2666V-D	2
376	0x178	0.125	0	0	0.001	47.00	DDR4-2666W	
360	0x168	0.125	-40	0xD8	0.001	44.96	DDR4-2933V	
366	0x16E	0.125	-110	0x92	0.001	45.64	DDR4-2933W	
371	0x173	0.125	-55	0xC9	0.001	46.32	DDR4-2933Y	
366	0x16E	0.125	0	0	0.001	45.75	DDR4-2933Y-D	2
376	0x178	0.125	0	0	0.001	47.00	DDR4-2933AA	
356	0x164	0.125	0	0	0.001	44.50	DDR4-3200W	
366	0x16E	0.125	0	0	0.001	45.75	DDR4-3200AA	
376	0x178	0.125	0	0	0.001	47.00	DDR4-3200AB	

8.1 General Configuration Section: Bytes 0~127 (0x000~0x07F) (Cont'd)

8.1.27 Byte 29 (0x01D): Minimum Active to Active/Refresh Delay Time (t_{RCmin}), Least Significant Byte (Cont'd)

Table 57 — Examples of Minimum Active to Active/Refresh Delay Time (Cont'd)

t_{RCmin} (MTB units)		MTB (ns)	t_{RCmin} Offset (FTB units) ¹		FTB (ns)	t_{RCmin} Result (ns)	Use	Notes
380	0x17C	0.125	0	0	0.001	47.50	DDR4-1600J-3DS2B	
390	0x186	0.125	0	0	0.001	48.75	DDR4-1600K-3DS2B	
400	0x190	0.125	0	0	0.001	50.00	DDR4-1600L-3DS2B	
375	0x177	0.125	-25	0xE7	0.001	46.85	DDR4-1866L-3DS2B	
375	0x177	0.125	-25	0xE7	0.001	46.85	DDR4-1866L-D-3DS2B	2
384	0x180	0.125	-80	0xB0	0.001	47.92	DDR4-1866M-3DS2B	
384	0x180	0.125	-80	0xB0	0.001	47.92	DDR4-1866M-D-3DS2B	2
392	0x188	0.125	0	0	0.001	49.00	DDR4-1866N-3DS2B	
377	0x179	0.125	-65	0xBF	0.001	47.06	DDR4-2133P-3DS2A	
377	0x179	0.125	-65	0xBF	0.001	47.06	DDR4-2133P-D-3DS2A	2
377	0x179	0.125	-65	0xBF	0.001	47.06	DDR4-2133R-3DS3A	
384	0x180	0.125	0	0	0.001	48.00	DDR4-2133R-3DS4A	
384	0x180	0.125	0	0	0.001	48.00	DDR4-2133R-D-3DS4A	2
356	0x164	0.125	0	0	0.001	44.50	DDR4-2400P-3DS3B	
370	0x172	0.125	-91	0xA5	0.001	46.16	DDR4-2400T-3DS2A	
376	0x178	0.125	0	0	0.001	47.00	DDR4-2400U-3DS2A	
376	0x178	0.125	0	0	0.001	47.00	DDR4-2400U-3DS4A	
376	0x178	0.125	0	0	0.001	47.00	DDR4-2400U-D-3DS4A	2
358	0x166	0.125	0	0	0.001	44.75	DDR4-2666T-3DS3A	
370	0x172	0.125	0	0	0.001	46.25	DDR4-2666V-3DS3A	
376	0x178	0.125	0	0	0.001	47.00	DDR4-2666W-3DS4A	
376	0x178	0.125	0	0	0.001	47.00	DDR4-2666W-D-3DS4A	2
		0.125			0.001	tbd	DDR4-2933tbd-3DS tbd	
		0.125			0.001	tbd	DDR4-2933tbd-3DS tbd	
		0.125			0.001	tbd	DDR4-2933tbd-3DS tbd	
356	0x164	0.125	0	0	0.001	44.50	DDR4-3200W-3DS4A	
363	0x16B	0.125	-105	0x97	0.001	45.27	DDR4-3200AA-3DS4A	
376	0x178	0.125	0	0	0.001	47.00	DDR4-3200AC-3DS4A	
376	0x178	0.125	0	0	0.001	47.00	DDR4-3200AC-D-3DS4A	2
NOTE 1 See SPD byte 120								
NOTE 2 Device supports downbinning in lower frequency applications; see supplier data sheet								

8.1.28 Byte 30 (0x01E): Minimum Refresh Recovery Delay Time ($t_{RFC1min}$), LSB

Byte 31 (0x01F): Minimum Refresh Recovery Delay Time ($t_{RFC1min}$), MSB

This word, shown in Table 58, defines the minimum SDRAM Refresh Recovery Time Delay in medium timebase (MTB) units. These values come from the DDR4 SDRAM data sheet.

Table 58 — Bytes 30 and 31, Minimum Refresh Recovery Delay Time ($t_{RFC1min}$)

Minimum SDRAM Refresh Recovery Delay Time ($t_{RFC1min}$) MTB Units	
Byte 31	Byte 30
Bits 15~8	Bits 7~0
Values defined from 1 to 65535	

8.1 General Configuration Section: Bytes 0~127 (0x000~0x07F) (Cont'd)
8.1.28 Bytes 30 and 31: Minimum Refresh Recovery Delay Time ($t_{RFC1min}$) (Cont'd)

Examples: See Table 59.

Table 59 — Examples of Minimum Refresh Recovery Delay Time ($t_{RFC1min}$)

$t_{RFC1min}$ (MTB units)	MTB (ns)	$t_{RFC1min}$ Result (ns)	Use
1280	0x0500	160	2 Gb DDR4 SDRAM
2080	0x0820	260	4 Gb DDR4 SDRAM
2800	0x0AF0	350	8 Gb DDR4 SDRAM
4400	0x1130	550	16 Gb DDR4 SDRAM
NOTE 1 For 3DS, t_{RFC1} refers to the t_{RFC_slr1} . See the DDR4 3DS Addendum for details.			

8.1.29 Byte 32 (0x020): Minimum Refresh Recovery Delay Time ($t_{RFC2min}$), LSB
Byte 33 (0x021): Minimum Refresh Recovery Delay Time ($t_{RFC2min}$), MSB

This word, shown in Table 60, defines the minimum SDRAM Refresh Recovery Time Delay in medium timebase (MTB) units. These values come from the DDR4 SDRAM data sheet.

Table 60 — Bytes 32 and 33, Minimum Refresh Recovery Delay Time ($t_{RFC2min}$)

Minimum SDRAM Refresh Recovery Delay Time ($t_{RFC2min}$) MTB Units	
Byte 33	Byte 32
Bits 15~8	Bits 7~0
Values defined from 1 to 65535	

Examples: See Table 61.

Table 61 — Examples of Minimum Refresh Recovery Delay Time ($t_{RFC2min}$)

$t_{RFC2min}$ (MTB units)	MTB (ns)	$t_{RFC2min}$ Result (ns)	Use
880	0x0370	110	2 Gb DDR4 SDRAM
1280	0x0500	160	4 Gb DDR4 SDRAM
2080	0x0820	260	8 Gb DDR4 SDRAM
2800	0x0AF0	350	16 Gb DDR4 SDRAM
NOTE 1 For 3DS, t_{RFC2} refers to the t_{RFC_slr2} . See the DDR4 3DS Addendum for details.			

8.1.30 Byte 34 (0x022): Minimum Refresh Recovery Delay Time ($t_{RFC4min}$), LSB
Byte 35 (0x023): Minimum Refresh Recovery Delay Time ($t_{RFC4min}$), MSB

This word, shown in Table 62, defines the minimum SDRAM Refresh Recovery Time Delay in medium timebase (MTB) units. These values come from the DDR4 SDRAM data sheet.

Table 62 — Bytes 34 and 35, Minimum Refresh Recovery Delay Time ($t_{RFC4min}$)

Minimum SDRAM Refresh Recovery Delay Time ($t_{RFC4min}$) MTB Units	
Byte 35	Byte 34
Bits 15~8	Bits 7~0
Values defined from 1 to 65535	

Examples: See Table 63.

Table 63 — Examples of Minimum Refresh Recovery Delay Time ($t_{RFC4min}$)

$t_{RFC4min}$ (MTB units)	MTB (ns)	$t_{RFC4min}$ Result (ns)	Use
720	0x02D0	90	2 Gb DDR4 SDRAM
880	0x0370	110	4 Gb DDR4 SDRAM
1280	0x0500	160	8 Gb DDR4 SDRAM
2080	0x0820	260	16 Gb DDR4 SDRAM
NOTE 1 For 3DS, t_{RFC4} refers to the t_{RFC_slr4} . See the DDR4 3DS Addendum for details.			

8.1.31 Byte 36 (0x024): Upper Nibble for t_{FAW}

This byte, shown in Table 64, defines the most significant nibble for the value of t_{FAW} (SPD byte 37). This value comes from the DDR4 SDRAM data sheet.

Table 64 — Byte 36 (0x024): Upper Nibble for t_{FAW}

Bits 7 ~ 4	Bits 3 ~ 0
Reserved	t_{FAW} Most Significant Nibble
Reserved; must be coded as 0000	See Byte 37 description

8.1.32 Byte 37 (0x025): Minimum Four Activate Window Delay Time (t_{FAWmin}), Least Significant Byte

The lower nibble of Byte 36 and the contents of Byte 37, shown in Table 65, combined create a 12-bit value which defines the minimum SDRAM Four Activate Window Delay Time in medium timebase (MTB) units. This value comes from the DDR4 SDRAM data sheet. The value of this number may be dependent on the SDRAM page size; please refer to the DDR4 SDRAM data sheet section on Addressing to determine the page size for these devices.

Table 65 — Byte 37 (0x025): Minimum Four Activate Window Delay Time (t_{FAWmin}), Least Significant Byte

Byte 36 Bits 3 ~ 0, Byte 37 Bits 7 ~ 0
Minimum Four Activate Window Delay Time (t_{FAW}) MTB Units
Values defined from 1 to 4095

8.1 General Configuration Section: Bytes 0~127 (0x000~0x07F) (Cont'd)**8.1.32 Byte 37 (0x025): Minimum Four Activate Window Delay Time (t_{FAWmin}), Least Significant Byte (Cont'd)**

Examples: See Table 66.

Table 66 — Examples of Minimum Four Activate Window Delay Time

tFAW (MTB units)		Timebase (ns)	tFAW Result (ns)	Use	Notes
280	0x118	0.125	35	DDR4-1600, 2 KB page size	
200	0x0C8	0.125	25	DDR4-1600, 1 KB page size	
160	0x0A0	0.125	20	DDR4-1600, 1/2 KB page size	
240	0x0F0	0.125	30	DDR4-1866, 2 KB page size	
184	0x0B8	0.125	23	DDR4-1866, 1 KB page size	
136	0x088	0.125	17	DDR4-1866, 1/2 KB page size	
240	0x0F0	0.125	30	DDR4-2133, 2 KB page size	
168	0x0A8	0.125	21	DDR4-2133, 1 KB page size	
120	0x078	0.125	15	DDR4-2133, 1/2 KB page size	
240	0x0F0	0.125	30	DDR4-2400, 2 KB page size	
168	0x0A8	0.125	21	DDR4-2400, 1 KB page size	
104	0x068	0.125	13	DDR4-2400, 1/2 KB page size	
240	0x0F0	0.125	30	DDR4-2666, 2 KB page size	
168	0x0A8	0.125	21	DDR4-2666, 1 KB page size	
96	0x060	0.125	12	DDR4-2666, 1/2 KB page size	
240	0x0F0	0.125	30	DDR4-2933, 2 KB page size	
168	0x0A8	0.125	21	DDR4-2933, 1 KB page size	
88	0x058	0.125	10.875	DDR4-2933, 1/2 KB page size	
240	0x0F0	0.125	30	DDR4-3200, 2 KB page size	
168	0x0A8	0.125	21	DDR4-3200, 1 KB page size	
80	0x050	0.125	10	DDR4-3200, 1/2 KB page size	
160	0x0A0	0.125	20	DDR4-1600-3DS, x4	1
200	0x0C8	0.125	25	DDR4-1600-3DS, x8	1
136	0x088	0.125	17	DDR4-1866-3DS, x4	1
184	0x0B8	0.125	23	DDR4-1866-3DS, x8	1
120	0x078	0.125	15	DDR4-2133-3DS, x4	1
168	0x0A8	0.125	21	DDR4-2133-3DS, x8	1
104	0x068	0.125	13	DDR4-2400-3DS, x4	1
168	0x0A8	0.125	21	DDR4-2400-3DS, x8	1
		0.125	tbd	DDR4-2666-3DS, x4	1
		0.125	tbd	DDR4-2666-3DS, x8	1
		0.125	tbd	DDR4-2933-3DS, x4	1
		0.125	tbd	DDR4-2933-3DS, x8	1
		0.125	tbd	DDR4-3200-3DS, x4	1
		0.125	tbd	DDR4-3200-3DS, x8	1

NOTE 1 For 3DS, tFAW refers to the tFAW_slr_x4 or tFAW_slr_x8 parameter depending on the I/O organization of the device. See the DDR4 3DS Addendum for details.

8.1.33 Byte 38 (0x026): Minimum Activate to Activate Delay Time (t_{RRD_smin}), Different Bank Group

This byte, shown in Table 67, defines the minimum SDRAM Activate to Activate Delay Time to different bank groups in medium timebase (MTB) units. This value comes from the DDR4 SDRAM data sheet. Controller designers must also note that at some frequencies, a minimum number of clocks may be required resulting in a larger t_{RRD_smin} value than indicated in the SPD. For example, t_{RRD_smin} for DDR4-1600 must be 4 clocks.

8.1 General Configuration Section: Bytes 0~127 (0x000~0x07F) (Cont'd)

8.1.33 Byte 38 (0x026): Minimum Activate to Activate Delay Time (t_{RRD_Smin}), Different Bank Group (Cont'd)

Table 67 — Byte 38 (0x026): Minimum Activate to Activate Delay Time (t_{RRD_Smin}), Different Bank Group

Bits 7~0
Minimum Active to Active/Refresh Time (t_{RRD_Smin})
MTB Units
Values defined from 1 to 255

If t_{RRD_Smin} cannot be divided evenly by the MTB, this byte must be rounded up to the next larger integer and the Fine Offset for t_{RRD_Smin} (SPD byte 119) used for correction to get the actual value.

Examples: See Table 68.

Table 68 — Examples of Minimum Activate to Activate Delay Time, Different Bank Group

t _{RRD_S} min (MTB units)		MTB (ns)	t _{RRD_S} min Offset (FTB units) ¹		FTB (ns)	t _{RRD_S} min Result (ns)	Use	Notes
48	0x30	0.125	0	0	0.001	6.00	DDR4-1600, 2 KB page size	
40	0x28	0.125	0	0	0.001	5.00	DDR4-1600, 1 KB page size	
40	0x28	0.125	0	0	0.001	5.00	DDR4-1600, 1/2 KB page size	
43	0x2B	0.125	-76	0xB4	0.001	5.30	DDR4-1866, 2 KB page size	
34	0x22	0.125	-50	0xCE	0.001	4.20	DDR4-1866, 1 KB page size	
34	0x22	0.125	-50	0xCE	0.001	4.20	DDR4-1866, 1/2 KB page size	
43	0x2B	0.125	-76	0xB4	0.001	5.30	DDR4-2133, 2 KB page size	
30	0x1E	0.125	-50	0xCE	0.001	3.70	DDR4-2133, 1 KB page size	
30	0x1E	0.125	-50	0xCE	0.001	3.70	DDR4-2133, 1/2 KB page size	
43	0x2B	0.125	-76	0xB4	0.001	5.30	DDR4-2400, 2 KB page size	
27	0x1B	0.125	-76	0xB4	0.001	3.30	DDR4-2400, 1 KB page size	
27	0x1B	0.125	-76	0xB4	0.001	3.30	DDR4-2400, 1/2 KB page size	
43	0x2B	0.125	-76	0xB4	0.001	5.30	DDR4-2666, 2 KB page size	
24	0x18	0.125	0	0	0.001	3.00	DDR4-2666, 1 KB page size	
24	0x18	0.125	0	0	0.001	3.00	DDR4-2666, 1/2 KB page size	
43	0x2B	0.125	-76	0xB4	0.001	5.30	DDR4-2933, 2 KB page size	
22	0x16	0.125	-50	0xCE	0.001	2.70	DDR4-2933, 1 KB page size	
22	0x16	0.125	-50	0xCE	0.001	2.70	DDR4-2933, 1/2 KB page size	
43	0x2B	0.125	-76	0xB4	0.001	5.30	DDR4-3200, 2 KB page size	
20	0x14	0.125	0	0	0.001	2.50	DDR4-3200, 1 KB page size	
20	0x14	0.125	0	0	0.001	2.50	DDR4-3200, 1/2 KB page size	
40	0x28	0.125	0	0	0.001	5.00	DDR4-1600-3DS	2
34	0x22	0.125	-50	0xCE	0.001	4.20	DDR4-1866-3DS	2
30	0x1E	0.125	-50	0xCE	0.001	3.70	DDR4-2133-3DS	2
27	0x1B	0.125	-75	0xB4	0.001	3.30	DDR4-2400-3DS	2
		0.125			0.001	tbd	DDR4-2666-3DS	2
		0.125			0.001	tbd	DDR4-2933-3DS	2
		0.125			0.001	tbd	DDR4-3200-3DS	2
NOTE 1 See SPD byte 119.								
NOTE 2 For 3DS, t _{RRD_S} refers to the t _{RRD_S_slr} parameter. See the DDR4 3DS Addendum for details.								

8.1.34 Byte 39 (0x027): Minimum Activate to Activate Delay Time (t_{RRD_Lmin}), Same Bank Group

This byte, shown in Table 69, defines the minimum SDRAM Activate to Activate Delay Time to the same bank group in medium timebase (MTB) units. This value comes from the DDR4 SDRAM data sheet. Controller designers must also note that at some

8.1 General Configuration Section: Bytes 0~127 (0x000~0x07F) (Cont'd)**8.1.34 Byte 39 (0x027): Minimum Activate to Activate Delay Time (t_{RRD_Lmin}), Same Bank Group (Cont'd)**

frequencies, a minimum number of clocks may be required resulting in a larger t_{RRD_Lmin} value than indicated in the SPD. For example, t_{RRD_Lmin} for DDR4-1600 must be 4 clocks.

Table 69 — Byte 39 (0x027): Minimum Activate to Activate Delay Time (t_{RRD_Lmin}), Same Bank Group

Bits 7~0
Minimum Active to Active/Refresh Time (t_{RRD_Lmin})
MTB Units
Values defined from 1 to 255

If t_{RRD_Lmin} cannot be divided evenly by the MTB, this byte must be rounded up to the next larger integer and the Fine Offset for t_{RRD_Lmin} (SPD byte 118) used for correction to get the actual value.

Examples: See Table 70.

Table 70 — Examples of Minimum Activate to Activate Delay Time, Same Bank Group

t _{RRD_L} min (MTB units)		MTB (ns)	t _{RRD_L} min Offset (FTB units) ¹		FTB (ns)	t _{RRD_L} min Result (ns)	Use
60	0x3C	0.125	0	0	0.001	7.50	DDR4-1600, 2 KB page size
48	0x30	0.125	0	0	0.001	6.00	DDR4-1600, 1 KB page size
48	0x30	0.125	0	0	0.001	6.00	DDR4-1600, 1/2 KB page size
52	0x34	0.125	-100	0x9C	0.001	6.40	DDR4-1866, 2 KB page size
43	0x2B	0.125	-76	0xB4	0.001	5.30	DDR4-1866, 1 KB page size
43	0x2B	0.125	-76	0xB4	0.001	5.30	DDR4-1866, 1/2 KB page size
52	0x34	0.125	-100	0x9C	0.001	6.40	DDR4-2133, 2 KB page size
43	0x2B	0.125	-76	0xB4	0.001	5.30	DDR4-2133, 1 KB page size
43	0x2B	0.125	-76	0xB4	0.001	5.30	DDR4-2133, 1/2 KB page size
52	0x34	0.125	-100	0x9C	0.001	6.40	DDR4-2400, 2 KB page size
40	0x28	0.125	-100	0x9C	0.001	4.90	DDR4-2400, 1 KB page size
40	0x28	0.125	-100	0x9C	0.001	4.90	DDR4-2400, 1/2 KB page size
52	0x34	0.125	-100	0x9C	0.001	6.40	DDR4-2666, 2 KB page size
40	0x28	0.125	-100	0x9C	0.001	4.90	DDR4-2666, 1 KB page size
40	0x28	0.125	-100	0x9C	0.001	4.90	DDR4-2666, 1/2 KB page size
52	0x34	0.125	-100	0x9C	0.001	6.40	DDR4-2933, 2 KB page size
40	0x28	0.125	-100	0x9C	0.001	4.90	DDR4-2933, 1 KB page size
40	0x28	0.125	-100	0x9C	0.001	4.90	DDR4-2933, 1/2 KB page size
52	0x34	0.125	-100	0x9C	0.001	6.40	DDR4-3200, 2 KB page size
40	0x28	0.125	-100	0x9C	0.001	4.90	DDR4-3200, 1 KB page size
40	0x28	0.125	-100	0x9C	0.001	4.90	DDR4-3200, 1/2 KB page size
48	0x30	0.125	0	0	0.001	6.00	DDR4-1600-3DS
43	0x2B	0.125	-76	0xB4	0.001	5.30	DDR4-1866-3DS
43	0x2B	0.125	-76	0xB4	0.001	5.30	DDR4-2133-3DS
40	0x28	0.125	-100	0x9C	0.001	4.90	DDR4-2400-3DS
		0.125			0.001	tbd	DDR4-2666-3DS
		0.125			0.001	tbd	DDR4-2933-3DS
		0.125			0.001	tbd	DDR4-3200-3DS

NOTE 1

See SPD byte 118

NOTE 2

For 3DS, t_{RRD_L} refers to the t_{RRD_L_slr} parameter. See the DDR4 3DS Addendum for details.

8.1.35 Byte 40 (0x028): Minimum CAS to CAS Delay Time (t_{CCD_Lmin}), Same Bank Group

This byte, shown in Table 71, defines the minimum SDRAM CAS to CAS Delay Time to the same bank group in medium timebase (MTB) units. This value comes from the DDR4 SDRAM data sheet. Controller designers must also note that at some frequencies, a

8.1 General Configuration Section: Bytes 0~127 (0x000~0x07F) (Cont'd)

8.1.35 Byte 40 (0x028): Minimum CAS to CAS Delay Time (t_{CCD_Lmin}), Same Bank Group (Cont'd)

minimum number of clocks may be required resulting in a larger t_{CCD_Lmin} value than indicated in the SPD. For example, t_{CCD_Lmin} for DDR4-2133 must be 6 clocks.

Table 71 — Byte 40 (0x028): Minimum CAS to CAS Delay Time (t_{CCD_Lmin}), Same Bank Group

Bits 7~0
Minimum CAS to CAS Time (t_{CCD_Lmin}) MTB Units
Values defined from 1 to 255

If t_{CCD_Lmin} cannot be divided evenly by the MTB, this byte must be rounded up to the next larger integer and the Fine Offset for t_{CCD_Lmin} (SPD byte 117) used for correction to get the actual value.

Examples: See Table 72.

Table 72 — Examples of Minimum CAS to CAS Delay Time, Same Bank Group

t _{CCD_L} min (MTB units)		MTB (ns)	t _{CCD_L} min Offset (FTB units) ¹		FTB (ns)	t _{CCD_L} min Result (ns)	Use	Notes
50	0x32	0.125	0	0	0.001	6.250	DDR4-1600	
43	0x2B	0.125	-20	0xEC	0.001	5.355	DDR4-1866	
43	0x2B	0.125	-20	0xEC	0.001	5.355	DDR4-2133	
40	0x28	0.125	0	0	0.001	5.000	DDR4-2400	
40	0x28	0.125	0	0	0.001	5.000	DDR4-2666	
40	0x28	0.125	0	0	0.001	5.000	DDR4-2933	
40	0x28	0.125	0	0	0.001	5.000	DDR4-3200	
50	0x32	0.125	0	0	0.001	6.250	DDR4-1600-3DS	2, 3
43	0x2B	0.125	-20	0xEC	0.001	5.355	DDR4-1866-3DS	2, 3
43	0x2B	0.125	-20	0xEC	0.001	5.355	DDR4-2133-3DS	2, 3
40	0x28	0.125	0	0	0.001	5.000	DDR4-2400-3DS	2, 3
		0.125			0.001	tbd	DDR4-2666-3DS	2, 3
		0.125			0.001	tbd	DDR4-2933-3DS	2, 3
		0.125			0.001	tbd	DDR4-3200-3DS	2, 3

NOTE 1

See SPD byte 117.

NOTE 2

For 3DS, t_{CCD_L} refers to the t_{CCD_L_slr} parameter. See the DDR4 3DS Addendum for details.

NOTE 3

DS timings are for 4 die per stack or less. Timings for greater than 4 die per stack at TBD.

8.1.36 Byte 41 (0x029): Upper Nibble for t_{WRmin}

This byte, shown in Table 73, defines the most significant nibble for the value of t_{WRmin} (byte 42). This value comes from the DDR4 SDRAM data sheet.

Table 73 — Byte 41 (0x029): Upper Nibble for t_{WRmin}

Bits 7~4	Bits 3~0
Reserved	t_{WRmin} Most Significant Nibble
Reserved; must be coded as 0000	See Byte 42 description

8.1.37 Byte 42 (0x02A): Minimum Write Recovery Time (t_{WRmin})

The lower nibble of Byte 41 and the contents of Byte 42, shown in Table 74, combined create a 12-bit value which defines the minimum SDRAM Write Recovery Time in medium timebase (MTB) units. The most significant bit is Bit 3 of Byte 41, and the least

8.1 General Configuration Section: Bytes 0~127 (0x000~0x07F) (Cont'd)**8.1.37 Byte 42 (0x02A): Minimum Write Recovery Time (t_{WRmin}) (Cont'd)**

significant bit is Bit 0 of Byte 42. This value comes from the DDR4 SDRAM data sheet.

Table 74 — Byte 42 (0x02A): Minimum Write Recovery Time (t_{WRmin})

Byte 41 Bits 3~0, Byte 42 Bits 7~0
Minimum Write Recovery Time (t_{WRmin}) MTB Units
Values defined from 1 to 4095

Examples: See Table 75.

Table 75 — Examples of Minimum Write Recovery Time

t _{WR} min (MTB units)		MTB (ns)	t _{WR} min Result (ns)	Use
120	0x78	0.125	15	All DDR4 bins
360	0x168	0.125	45	All DDR4E bins

8.1.38 Byte 43 (0x029): Upper Nibbles for t_{WTRmin}

This byte, shown in Table 76, defines the most significant nibbles for the values of t_{WTR_Smin} (byte 44) and t_{WTR_Lmin} (byte 45). This value comes from the DDR4 SDRAM data sheet.

Table 76 — Byte 43 (0x029): Upper Nibbles for t_{WTRmin}

Bits 7~4	Bits 3~0
t_{WTR_Lmin} Most Significant Nibble	t_{WTR_Smin} Most Significant Nibble
See Byte 45 description	See Byte 44 description

8.1.39 Byte 44 (0x02C): Minimum Write to Read Time (t_{WTR_Smin}), Different Bank Group

The lower nibble of Byte 43 and the contents of Byte 44, shown in Table 77, combined create a 12-bit value which defines the minimum SDRAM Write to Read Time to different bank groups in medium timebase (MTB) units. The most significant bit is Bit 3 of Byte 43, and the least significant bit is Bit 0 of Byte 44. This value comes from the DDR4 SDRAM data sheet. Controller designers must also note that at some frequencies, a minimum number of clocks may be required resulting in a larger t_{WTR_Smin} value than indicated in the SPD. For example, t_{WTR_Smin} must be at least 2 clocks.

Table 77 — Byte 44 (0x02C): Minimum Write to Read Time (t_{WTR_Smin}), Different Bank Group

Byte 43 Bits 3~0, Byte 44 Bits 7~0
Minimum Write Recovery Time (t_{WTR_Smin}), different bank group MTB Units
Values defined from 1 to 4095

Examples: See Table 78.

Table 78 — Examples of Minimum Write to Read Time, Different Bank Group

t _{WTR} min (MTB units)		MTB (ns)	t _{WTR_S} min Result (ns)	Use
20	0x14	0.125	2.5	All DDR4 bins
			TBD	All DDR4E bins

8 Details of Each Byte (Cont'd)

8.1 General Configuration Section: Bytes 0~127 (0x000~0x07F) (Cont'd)

8.1.40 Byte 45 (0x02D): Minimum Write to Read Time (t_{WTR_Lmin}), Same Bank Group

The upper nibble of Byte 43 and the contents of Byte 45, shown in Table 79, combined create a 12-bit value which defines the minimum SDRAM Write to Read Time for access to the same bank group in medium timebase (MTB) units. The most significant bit is Bit 7 of Byte 43, and the least significant bit is Bit 0 of Byte 45. This value comes from the DDR4 SDRAM data sheet. Controller designers must also note that at some frequencies, a minimum number of clocks may be required resulting in a larger t_{WTR_Lmin} value than indicated in the SPD. For example, t_{WTR_Lmin} must be at least 4 clocks.

Table 79 — Byte 45 (0x02D): Minimum Write to Read Time (t_{WTR_Lmin}), Same Bank Group

Byte 43 Bits 7~4, Byte 45 Bits 7~0
Minimum Write Recovery Time (t_{WTR_Lmin}), same bank group MTB Units
Values defined from 1 to 4095

Examples: See Table 80.

Table 80 — Examples of Minimum Write to Read Time, Same Bank Group

t_{WTRmin} (MTB units)	MTB (ns)	t_{WTR_Lmin} Result (ns)	Use
60	0x3C	0.125	7.5
			All DDR4 bins
			TBD
			All DDR4E bins

8.1.41 Byte 46~59 (0x02E~0x03B): Reserved, Base Configuration Section

Must be coded as 0x00

8.1.42 Bytes 60~77 (0x03C~0x04D): Connector to SDRAM Bit Mapping

These bytes, shown in Tables 81 and 82, document the connection between data signals at the edge connector of a module to the DDR4 SDRAM inputs pins for package rank 0 of the module. This information is used by the controller to route data onto the correct bit lines for CRC transmission as described in the DDR4 SDRAM data sheet JESD79-4. Each byte describes the mapping for one nibble (four bits) of data. In addition, each SPD byte describes the mapping between package rank 0 bits and equivalent bits in other ranks.

Table 81 — Bytes 60~77 (0x03C~0x04D): Connector to SDRAM Bit Mapping

SPD Byte		Connector Bits	SPD Byte		Connector Bits	SPD Byte		Connector Bits
60	0x03C	DQ0-3	66	0x042	DQ24-27	72	0x048	DQ40-43
61	0x03D	DQ4-7	67	0x043	DQ28-31	73	0x049	DQ44-47
62	0x03E	DQ8-11	68	0x044	CB0-3	74	0x04A	DQ48-51
63	0x03F	DQ12-15	69	0x045	CB4-7	75	0x04B	DQ52-55
64	0x040	DQ16-19	70	0x046	DQ32-35	76	0x04C	DQ56-59
65	0x041	DQ20-23	71	0x047	DQ36-39	77	0x04D	DQ60-63

The mapping rules are as follows:

- All bits within a nibble at the edge connector must be wired to the same SDRAM.
- All bits within a byte at the edge connector must be wired to the same SDRAM for x8 and wider SDRAMs.
- Bits within a nibble may be swapped in any order.
- Nibbles may be swapped within a byte.
- Bytes may be wired in any order within the SDRAM width for x16 and wider SDRAMs.

Each SPD byte in the Bit Mapping array is encoded as follows. If the nibble at the edge connector is wired to the lower nibble of a byte at the SDRAM (x8 and wider), then bit 5 is coded as 0. If the nibble at the edge connector is wired to the upper nibble of a byte at the SDRAM, then bit 5 is coded as 1. Bit 5 = 0 for all x4 based modules. Bits 6~7 define the connectivity between bits in different

8.1 General Configuration Section: Bytes 0~127 (0x000~0x07F) (Cont'd)
8.1.42 Bytes 60~77 (0x03C~0x04D): Connector to SDRAM Bit Mapping (Cont'd)

package ranks.

Table 82 — Bit Mapping Array

Bits 7 ~ 6	Bit 5	Bits 4 ~ 0
Package Rank Map	Wired to Upper/Lower Nibble	Bit Order at SDRAM
See Package Rank Map table	0 = lower nibble at SDRAM 1 = upper nibble at SDRAM	See Nibble Map table

Package Rank Map: Bits 7~6 in each SPD byte define the mapping between bits in Package Rank 0 and other package ranks on the module. The mapping rules are defined in Table 83:

Table 83 — Mapping Rules

Package Rank Map				
Bits 7 ~ 6	Bit Order at SDRAM			
00	Even package ranks (0, 2, etc.) have the same mapping Odd package ranks (1, 3, etc) map SDRAM data bits relative to Package Rank 0 as follows:			
	DQ0 → DQ1	DQ8 → DQ9	DQ16 → tbd	DQ24 → tbd
	DQ1 → DQ0	DQ9 → DQ8	DQ17 → tbd	DQ25 → tbd
	DQ2 → DQ3	DQ10 → DQ11	DQ18 → tbd	DQ26 → tbd
	DQ3 → DQ2	DQ11 → DQ10	DQ19 → tbd	DQ27 → tbd
	DQ4 → DQ5	DQ12 → DQ13	DQ20 → tbd	DQ28 → tbd
	DQ5 → DQ4	DQ13 → DQ12	DQ21 → tbd	DQ29 → tbd
	DQ6 → DQ7	DQ14 → DQ15	DQ22 → tbd	DQ30 → tbd
	DQ7 → DQ6	DQ15 → DQ14	DQ23 → tbd	DQ31 → tbd
01 10 11	Reserved			

The Nibble Map is coded in Table 84:

Table 84 — Nibble Map Encoding

Nibble Map					
Nibble Bit Order at Connector	Bit Map Index Bits 4 ~ 0	Bit 5 = 0: Wired to Lower Nibble Bit Order within SDRAM Byte		Bit 5 = 1: Wired to Upper Nibble Bit Order within SDRAM Byte	
	00000	0x00	Bit Map not specified	0x20	Bit Map not specified
0, 1, 2, 3 (4, 5, 6, 7) ...	00001	0x01	0, 1, 2, 3	0x21	4, 5, 6, 7
	00010	0x02	0, 1, 3, 2	0x22	4, 5, 7, 6
	00011	0x03	0, 2, 1, 3	0x23	4, 6, 5, 7
	00100	0x04	0, 2, 3, 1	0x24	4, 6, 7, 5
	00101	0x05	0, 3, 1, 2	0x25	4, 7, 5, 6
	00110	0x06	0, 3, 2, 1	0x26	4, 7, 6, 5
	00111	0x07	1, 0, 2, 3	0x27	5, 4, 6, 7

8.1 General Configuration Section: Bytes 0~127 (0x000~0x07F) (Cont'd)
8.1.42 Bytes 60~77 (0x03C~0x04D): Connector to SDRAM Bit Mapping (Cont'd)

Table 84 — Nibble Map Encoding (Cont'd)

Nibble Map					
Nibble Bit Order at Connector	Bit Map Index Bits 4 ~ 0	Bit 5 = 0: Wired to Lower Nibble Bit Order within SDRAM Byte		Bit 5 = 1: Wired to Upper Nibble Bit Order within SDRAM Byte	
0, 1, 2, 3 (4, 5, 6, 7) ...	01000	0x08	1, 0, 3, 2	0x28	5, 4, 7, 6
	01001	0x09	1, 2, 0, 3	0x29	5, 6, 4, 7
	01010	0x0A	1, 2, 3, 0	0x2A	5, 6, 7, 4
	01011	0x0B	1, 3, 0, 2	0x2B	5, 7, 4, 6
	01100	0x0C	1, 3, 2, 0	0x2C	5, 7, 6, 4
	01101	0x0D	2, 0, 1, 3	0x2D	6, 4, 5, 7
	01110	0x0E	2, 0, 3, 1	0x2E	6, 4, 7, 5
	01111	0x0F	2, 1, 0, 3	0x2F	6, 5, 4, 7
0, 1, 2, 3 (4, 5, 6, 7) ...	10000	0x10	2, 1, 3, 0	0x30	6, 5, 7, 4
	10001	0x11	2, 3, 0, 1	0x31	6, 7, 4, 5
	10010	0x12	2, 3, 1, 0	0x32	6, 7, 5, 4
	10011	0x13	3, 0, 1, 2	0x33	7, 4, 5, 6
	10100	0x14	3, 0, 2, 1	0x34	7, 4, 6, 5
	10101	0x15	3, 1, 0, 2	0x35	7, 5, 4, 6
	10110	0x16	3, 1, 2, 0	0x36	7, 5, 6, 4
	10111	0x17	3, 2, 0, 1	0x37	7, 6, 4, 5
	11000	0x18	3, 2, 1, 0	0x38	7, 6, 5, 4
	All other codes		Reserved		Reserved

NOTE 1 1. Hex codes shown in this table are for bits 5~0 only and must be added to bits 7~6 (Package Rank Map bits) for the SPD byte entry

NOTE 2 2. Devices with x16 are treated as two separate x8 devices for CRC bit mapping within each byte. Similarly, x32 devices are treated as four separate x8 devices for CRC bit mapping within each byte.

Example: See Figure 3 and Table 49.

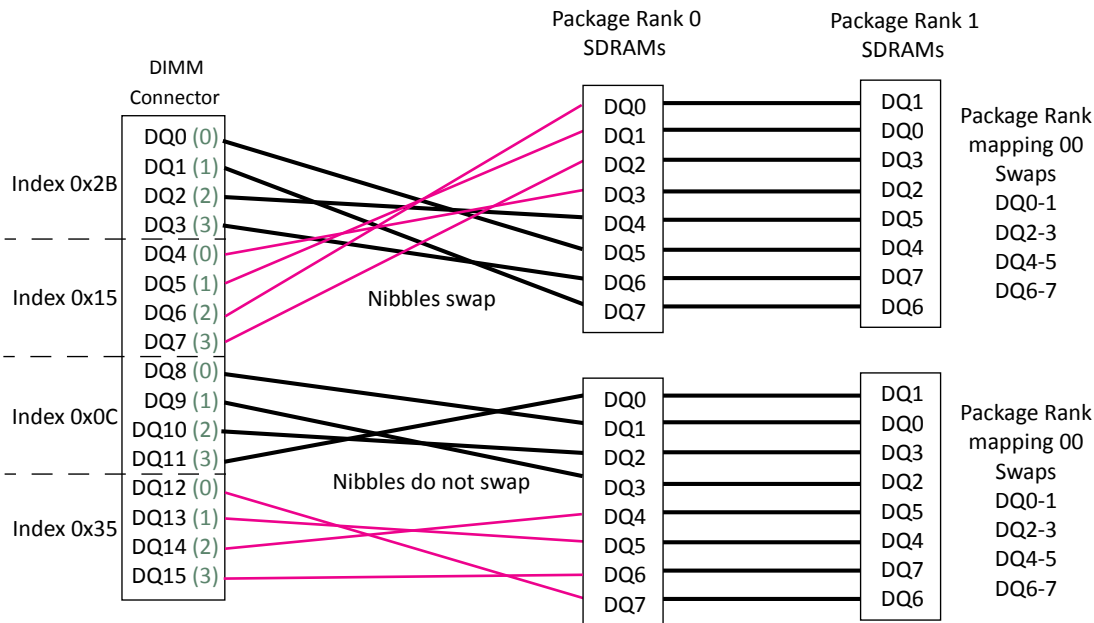


Figure 3 — Two Package Rank x8 Module (example only; may not represent a specific design)

8.1 General Configuration Section: Bytes 0~127 (0x000~0x07F) (Cont'd)

8.1.42 Bytes 60~77 (0x03C~0x04D): Connector to SDRAM Bit Mapping (Cont'd)

Table 85 — Two Package Rank x8 Module (example only; may not represent a specific design)

DQ bit at DIMM Connector																					
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19		
First SDRAM																					
5	7	4	6	Nibble wired to upper nibble of SDRAM byte (bit 5 = 1) using Nibble Map 01011 (bits 4~0)																	
4	6	5	7	Code 0x2B stored in the SPD table for the first nibble																	
				3	1	0	2	Nibble wired to lower nibble of SDRAM byte (bit 5 = 0) using Nibble													
				2	0	1	3	Map 1010 (bits 4~0). Code 0x15 stored for the second nibble													
								Second SDRAM													
								1	3	2	0	Code 0x0C for the 3rd nibble									
								0	2	3	1										
Package Rank Map 00																					
Even Package Ranks																					
Odd Package Ranks																					
												7	5	4	6	Code 0x35 for the					
												6	4	5	7	fourth nibble					

8.1.43 Bytes 78~116 (0x04E~0x074): Reserved, Base Configuration Section

Must be coded as 0x00

8.1.44 Byte 117 (0x075): Fine Offset for Minimum CAS to CAS Delay Time (t_{CCD_Lmin}), Same Bank Group

This byte modifies the calculation of SPD Byte 40 with a fine correction using FTB units. The value of t_{CCD_min} comes from the SDRAM data sheet. This value is a two's complement multiplier for FTB units, ranging from +127 to -128.

Examples: See SPD byte 40, in Section 8.1.35. For Two's Complement encoding, see Section 8.1.18.1, Relating the MTB and FTB.

8.1.45 Byte 118 (0x076): Fine Offset for Minimum Activate to Activate Delay Time ($t_{RRD_L_min}$), Same Bank Group

This byte modifies the calculation of SPD Byte 39 with a fine correction using FTB units. The value of t_{RRD_min} comes from the SDRAM data sheet. This value is a two's complement multiplier for FTB units, ranging from +127 to -128.

Examples: See SPD byte 39, in Section 8.1.34. For Two's Complement encoding, see Section 8.1.18.1, Relating the MTB and FTB.

8.1.46 Byte 119 (0x077): Fine Offset for Minimum Activate to Activate Delay Time (t_{RRD_Smin}), Different Bank Group

This byte modifies the calculation of SPD Byte 38 (MTB units) with a fine correction using FTB units. The value of $t_{\text{RRD_Smin}}$ comes from the SDRAM data sheet. This value is a two's complement multiplier for FTB units, ranging from +127 to -128.

Examples: See SPD byte 38, in Section 8.1.33. For Two's Complement encoding, see Section 8.1.18.1, Relating the MTB and FTB.

8.1.47 Byte 120 (0x078): Fine Offset for Minimum Active to Active/Refresh Delay Time (t_{RCmin})

This byte modifies the calculation of SPD Bytes 27 and 29 (MTB units) with a fine correction using FTB units. The value of t_{RCmin} comes from the SDRAM data sheet. This value is a two's complement multiplier for FTB units, ranging from +127 to -128.

Examples: See SPD bytes 27 and 29, in Sections 8.1.25 and 8.27. For Two's Complement encoding, see Section 8.1.18.1, Relating the MTB and FTB.

8 Details of Each Byte (Cont'd)

8.1 General Configuration Section: Bytes 0~127 (0x000~0x07F) (Cont'd)

8.1.48 Byte 121 (0x079): Fine Offset for Minimum Row Precharge Delay Time (t_{RPMIN})

This byte modifies the calculation of SPD Byte 26 (MTB units) with a fine correction using FTB units. The value of t_{RPMIN} comes from the SDRAM data sheet. This value is a two's complement multiplier for FTB units, ranging from +127 to -128.

Examples: See SPD byte 26, in Section 8.1.24. For Two's Complement encoding, see Section 8.1.18.1, Relating the MTB and FTB.

8.1.49 Byte 122 (0x07A): Fine Offset for Minimum RAS to CAS Delay Time (t_{RCDMIN})

This byte modifies the calculation of SPD Byte 25 (MTB units) with a fine correction using FTB units. The value of t_{RCDMIN} comes from the SDRAM data sheet. This value is a two's complement multiplier for FTB units, ranging from +127 to -128.

Examples: See SPD byte 25, in Section 8.1.23. For Two's Complement encoding, see Section 8.1.18.1, Relating the MTB and FTB.

8.1.50 Byte 123 (0x07B): Fine Offset for Minimum CAS Latency Time (t_{AAMIN})

This byte modifies the calculation of SPD Byte 24 (MTB units) with a fine correction using FTB units. The value of t_{AAMIN} comes from the SDRAM data sheet. This value is a two's complement multiplier for FTB units, ranging from +127 to -128.

Examples: See SPD byte 24, in Section 8.1.22. For Two's Complement encoding, see Section 8.1.18.1, Relating the MTB and FTB.

8.1.51 Byte 124 (0x07C): Fine Offset for SDRAM Maximum Cycle Time ($t_{CKAVGMAX}$)

This byte modifies the calculation of SPD Byte 19 (MTB units) with a fine correction using FTB units. The value of $t_{CKAVGMAX}$ comes from the SDRAM data sheet. This value is a two's complement multiplier for FTB units, ranging from +127 to -128.

Examples: See SPD byte 19, in Section 8.1.20. For Two's Complement encoding, see Section 8.1.18.1, Relating the MTB and FTB.

8.1.52 Byte 125 (0x07D): Fine Offset for SDRAM Minimum Cycle Time ($t_{CKAVGMIN}$)

This byte modifies the calculation of SPD Byte 18 (MTB units) with a fine correction using FTB units. The value of $t_{CKAVGMIN}$ comes from the SDRAM data sheet. This value is a two's complement multiplier for FTB units, ranging from +127 to -128.

Examples: See SPD byte 19, in Section 8.1.20. For Two's Complement encoding, see Section 8.1.18.1, Relating the MTB and FTB.

**8.1.53 Byte 126 (0x07E): Cyclical Redundancy Code (CRC) for Base Configuration Section, LSB
Byte 127 (0x07F): Cyclical Redundancy Code (CRC) for Base Configuration Section, MSB**

This two-byte field contains the calculated CRC for bytes 0~125 (0x000~0x07D) in the SPD. The following algorithm and data structures (shown in C) are to be followed in calculating and checking the code.

```
int Crc16 (char *ptr, int count)
{
    int crc, i;

    crc = 0;
    while (--count >= 0) {
        crc = crc ^ (int)*ptr++ << 8;
        for (i = 0; i < 8; ++i)
            if (crc & 0x8000)
                crc = crc << 1 ^ 0x1021;
            else
                crc = crc << 1;
    }
    return (crc & 0xFFFF);
}
```

```
char spdBytes[] = { SPD_byte_0, SPD_byte_1, ..., SPD_byte_N-1 };
```

8.1 General Configuration Section: Bytes 0~127 (0x000~0x07F) (Cont'd)**8.1.53 ByteS 126 and 127: Cyclical Redundancy Code (CRC) for Base Configuration Section (Cont'd)**

```
int data16;

data16 = Crc16 (spdBytes, sizeof(spdBytes));
SPD_byte_126 = (char) (data16 & 0xFF);
SPD_byte_127 = (char) (data16 >> 8);
```

8.2 Module-Specific Section: Bytes 128~191 (0x080~0x0BF)

This section of the SPD contains bytes that are specific to DDR4 module families. Module Type Key Byte 3 is used as an index for the encoding of bytes 128~191. The content of bytes 128~191 are described in the separate sections, one for each memory module family.

- 9.1 – UDIMMs
- 9.2 – RDIMMs
- 9.3 – LRDIMMs

8.3 Hybrid Memory Architecture Specific Parameters: Bytes 192~255 (0x0C0~0x0FF)

This section of the SPD contains bytes that are specific to hybrid memory module families. Module Type Key Byte 3 bits 7~4 are used as an index for the encoding of bytes 192~255. The content of bytes 192~255 are described in separate sections, one for each hybrid module family.

- 10.1 – NVIMMs

8.4 Extended Function Parameter Block: Bytes 256~319 (0x100~0x13F)

This section of the SPD contains zero or more blocks of extended function and interface protocol parameter sets for Hybrid memory modules. The content of bytes 256~319 are described in separate sections, one for each hybrid module function type.

- 11.1 – Extended Function Parameter Block for Byte Addressable Energy Backed NVDIMM (BYTE-E)
- 11.2 – Extended Function Parameter Block for Block Addressed NVDIMM (BLOCK)
- 11.3 – Extended Function Parameter Block for Byte Addressable No Energy Backed NVDIMM (BYTE-NOE)

8.5 Module Supplier's Data: Bytes 320~383 (0x140~0x17F)**8.5.1 Byte 320 (0x140): Module Manufacturer ID Code, LSB****Byte 321 (0x141): Module Manufacturer ID Code, MSB**

This two-byte field, shown in Table 86, indicates the manufacturer of the module, encoded as follows: the first byte is the number of continuation bytes indicated in JEP-106; the second byte is the last non-zero byte of the manufacturer's ID code, again as indicated in JEP-106.

Table 86 — Bytes 320 and 321: Module Manufacturer ID Code

Byte 321, Bits 7~0	Byte 320, Bit 7	Byte 320, Bits 6~0
Last non-zero byte, Module Manufacturer	Odd Parity for Byte 320, bits 6~0	Number of continuation codes, Module Manufacturer
See JEP-106		See JEP-106

8.5 Module Supplier's Data: Bytes 320~383 (0x140~0x17F) (Cont'd)

8.5.1 Bytes 320 and 321: Module Manufacturer ID Code (Cont'd)

Examples: See Table 87.

Table 87 — Examples of Module Manufacturer ID Code

Company	JEP-106		# continuation codes	SPD	
	Bank	Code		Byte 320	Byte 321
Fujitsu	1	04	0	0x80	0x04
US Modular	5	A8	4	0x04	0xA8

8.5.2 Byte 322 (0x142): Module Manufacturing Location

The module manufacturer includes an identifier that uniquely defines the manufacturing location of the memory module. While the SPD specification will not attempt to present a decode table for manufacturing sites, the individual manufacturer may keep track of manufacturing location and its appropriate decode represented in this byte.

8.5.3 Bytes 323~324 (0x143~0x144): Module Manufacturing Date

The module manufacturer includes a date code for the module. The JEDEC definitions for bytes 323 and 324 are year and week respectively. These bytes must be represented in Binary Coded Decimal (BCD). For example, week 47 in year 2014 would be coded as 0x14 (0001 0100) in byte 323 and 0x47 (0100 0111) in byte 324.

8.5.4 Bytes 325~328 (0x145~0x148): Module Serial Number

The supplier must include a unique serial number for the module. The supplier may use whatever decode method desired to maintain a unique serial number for each module.

One method of achieving this is by assigning a byte in the field from 325~328 as a tester ID byte and using the remaining bytes as a sequential serial number. Bytes 320~328 will then result in a nine-byte unique module identifier. Note that part number is not included in this identifier: the supplier may not give the same value for Bytes 320~328 to more than one DIMM even if the DIMMs have different part numbers.

8.5.5 Bytes 329~348 (0x149~15C): Module Part Number

The manufacturer's part number is written in ASCII format within these bytes. Unused digits are coded as ASCII blanks (0x20).

8.5.6 Bytes 349 (0x15D): Module Revision Code

This refers to the module revision code. While the SPD specification will not attempt to define the format for this information, the individual manufacturer may keep track of the revision code and its appropriate decode represented in this byte. This revision code refers to the manufacturer's assembly revision level and may be different than the raw card revision in SPD bytes 128 and 130.

8.5.7 Byte 350 (0x15E): DRAM Manufacturer ID Code, LSB

Byte 351 (0x15F): DRAM Manufacturer ID Code, MSB

This two-byte field, shown in Table 88, indicates the manufacturer of the DRAM on the module, encoded as follows: the first byte is the number of continuation bytes indicated in JEP-106; the second byte is the last non-zero byte of the manufacturer's ID code, again as indicated in JEP-106.

Table 88 — Bytes 350 and 351: DRAM Manufacturer ID Code

Byte 351, Bits 7~0	Byte 350, Bit 7	Byte 350, Bits 6~0
Last non-zero byte, DRAM Manufacturer	Odd Parity for Byte 350, bits 6~0	Number of continuation codes, DRAM Manufacturer
See JEP-106		See JEP-106

Example: See bytes 320~321, in Section 3.5.1, for example manufacturer codes.

8 Details of Each Byte (Cont'd)**8.5 Module Supplier's Data: Bytes 320~383 (0x140~0x17F) (Cont'd)****8.5.8 Byte 352 (0x160): DRAM Stepping**

This byte, shown in Table 89, defines the vendor die revision level (often called the “stepping”) of the DRAMs on the module. This byte is optional. For modules without DRAM stepping information, this byte should be programmed to 0xFF.

Table 89 — Byte 352 (0x160): DRAM Stepping

Bits 7~0
DRAM Stepping
Programmed in straight Hex format - no conversion needed.
00 - Valid
01 - Valid
..
FE - Valid
FF - Undefined (No Stepping Number Provided)

Examples: See Table 90.

Table 90 — Examples of DRAM Stepping

Code	Meaning
0x00	Stepping 0
0x01	Stepping 1
0x31	Stepping 3.1
0xA3	Stepping A3
0xB1	Stepping B1
0xFF	Stepping information not provided

8.5.9 Bytes 353~381 (0x161~0x17D): Manufacturer's Specific Data

The module manufacturer may include any additional information desired into the module within these locations.

8.5.10 Byte 382~383 (0x17E~0x17F): Reserved

Must be coded as 0x00.

8.6 ASCII Decode Matrix for SPDs

Table 91 is a subset of the full ASCII standard which is used for coding bytes in the Serial Presence Detect EEPROM that require ASCII characters:

Table 91 — ASCII Decode Matrix for SPDs

First Hex Digit in Pair	Second Hex Digit in Pair															
	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
2	Blank Space								(	)				- Dash	. Period	
3	0	1	2	3	4	5	6	7	8	9						
4		A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
5	P	Q	R	S	T	U	V	W	X	Y	Z					
6		a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
7	p	q	r	s	t	u	v	w	x	y	z					

Examples: See Table 92.

0x20 = Blank Space
0x34 = '4'
0x41 = 'A'

Table 92 — SPD Bytes 329~348

SPD Bytes 329~348	
Manufacturer's PN	Coded in ASCII
13M32734BCD-260Y	31334D33323733344243442D323630592020

9 SPD Bytes Specific to DDR4 Module Families

9.1 Module Specific Bytes for Unbuffered Memory Module Types (Bytes 128~191, 0x080~0x0BF)

This section defines the encoding of SPD bytes 128~191 when Memory Technology Key Byte 2 contains the value 0x0C and Module Type Key Byte 3 contains any of the following:

- 0xH2, UDIMM
- 0xH3, SO-DIMM
- 0xH6, Mini-UDIMM
- 0xH9, 72b-SO-UDIMM
- 0xHC, 16b-SO-DIMM
- 0xHD, 32b-SO-DIMM
- where H refers to the hybrid memory architecture, if any present on the module

Table 93 is the SPD address map for the module specific section, bytes 128~255, of the SPD for Unbuffered Module Types.

Table 93 — Module Specific SPD Bytes for Unbuffered Module Types

Byte Number		Function Described	Notes
128	0x080	Raw Card Extension, Module Nominal Height	
129	0x081	Module Maximum Thickness	
130	0x082	Reference Raw Card Used	
131	0x083	Address Mapping from Edge Connector to DRAM	
132~191	0x084~0x0BF	Reserved -- Must be coded as 0x00	
254	0x0FE	CRC for SPD Block 1, Least Significant Byte	1
255	0x0FF	CRC for SPD Block 1, Most Significant Byte	1
NOTE 1 CRC in bytes 254~255 covers all of SPD Block 1, bytes 128~255 (0x080~0x0FF)			

9.1.1 Byte 128 (0x080) (Unbuffered): Raw Card Extension, Module Nominal Height

The upper 3 bits of this byte, shown in Table 94, define extensions to the Raw Card Revision in Byte 130. The lower 5 bits of this byte define the nominal height (A dimension) in millimeters of the fully assembled module including heat spreaders or other added components. Refer to the relevant JEDEC JC-11 module outline (MO) documents for dimension definitions.

9.1 Module Specific Bytes for Unbuffered Memory Module Types (Bytes 128~191, 0x080~0x0BF) (Cont'd)

9.1.1 Byte 128 (0x080) (Unbuffered): Raw Card Extension, Module Nominal Height (Cont'd)

Table 94 — Byte 128 (0x080) (Unbuffered): Raw Card Extension, Module Nominal Height

Bits 7~5	Bits 4~0
Raw Card Extension	Module Nominal Height max, in mm (baseline height = 15 mm)
000 = raw card revisions 0 to 3; see byte 130 001 = raw card revision 4 010 = raw card revision 5 011 = raw card revision 6 100 = raw card revision 7 101 = raw card revision 8 110 = raw card revision 9 111 = raw card revision 10	00000 = height ≤ 15 mm 00001 = 15 < height ≤ 16 mm 00010 = 16 < height ≤ 17 mm 00011 = 17 < height ≤ 18 mm 00100 = 18 < height ≤ 19 mm ... 01010 = 24 < height ≤ 25 mm 01011 = 25 < height ≤ 26 mm ... 01111 = 29 < height ≤ 30 mm 10000 = 30 < height ≤ 31 mm 10001 = 31 < height ≤ 32 mm ... 11111 = 45 mm < height

Examples: See Table 95.

Table 95 — Examples of Byte 128 (0x080)

Nominal Module Height	Coding, bits 4~0	Meaning
mm	Binary	mm
18.75	00100	18 < height ≤ 19 mm
25.40	01011	25 < height ≤ 26 mm
30.00	01111	29 < height ≤ 30 mm
30.25	10000	30 < height ≤ 31 mm
31.25	10001	30 < height ≤ 31 mm

9.1.2 Byte 129 (0x081) (Unbuffered): Module Maximum Thickness

This byte, shown in Table 96, defines the maximum thickness (E dimension) in millimeters of the fully assembled module including heat spreaders or other added components above the module circuit board surface. Thickness of the front of the module is calculated as the E1 dimension minus the PCB thickness. Thickness of the back of the module is calculated as the E dimension minus the E1 dimension, rounding up to the next integer. Refer to the relevant JEDEC JC-11 module outline (MO) documents for dimension definitions.

Table 96 — Byte 129 (0x081) (Unbuffered): Module Maximum Thickness

Bits 7~4	Bits 3~0
Module Maximum Thickness max, Back, in mm (baseline thickness = 1 mm)	Module Maximum Thickness max, Front, in mm (baseline thickness = 1 mm)
0000 = thickness ≤ 1 mm 0001 = 1 < thickness ≤ 2 mm 0010 = 2 < thickness ≤ 3 mm 0011 = 3 < thickness ≤ 4 mm ... 1110 = 14 < thickness ≤ 15 mm 1111 = 15 < thickness	0000 = thickness ≤ 1 mm 0001 = 1 < thickness ≤ 2 mm 0010 = 2 < thickness ≤ 3 mm 0011 = 3 < thickness ≤ 4 mm ... 1110 = 14 < thickness ≤ 15 mm 1111 = 15 < thickness
Note: Thickness = E - E1	Note: Thickness = E1 - PCB

9 SPD Bytes Specific to DDR4 Module Families (Cont'd)

9.1 Module Specific Bytes for Unbuffered Memory Module Types (Bytes 128~191, 0x080~0x0BF) (Cont'd)

9.1.3 Byte 130 (0x082) (Unbuffered): Reference Raw Card Used

This byte, shown in Tables 97 and 98, indicates which JEDEC reference design raw card was used as the basis for the module assembly, if any. Bits 4~0 describe the raw card and bits 6~5 describe the revision level of that raw card. Special reference raw card indicator, ZZ, is used when no JEDEC standard raw card reference design was used as the basis for the module design. Pre-production modules should be encoded as revision 0 in bits 6~5.

Table 97 — Byte 130 (0x082) (Unbuffered): Reference Raw Cards A through AL

Bit 7	Bits 6~5	Bits 4~0
Reference Raw Card Extension	Reference Raw Card Revision	Reference Raw Card
0 = Reference raw cards A through AL	00 = revision 0 01 = revision 1 10 = revision 2 11 = revision 3 See byte 128 for extensions beyond revision 3; these bits must be coded as 11 for all revisions greater than 3	When bit 7 = 0, 00000 = Reference raw card A 00001 = Reference raw card B 00010 = Reference raw card C 00011 = Reference raw card D 00100 = Reference raw card E 00101 = Reference raw card F 00110 = Reference raw card G 00111 = Reference raw card H 01000 = Reference raw card J 01001 = Reference raw card K 01010 = Reference raw card L 01011 = Reference raw card M 01100 = Reference raw card N 01101 = Reference raw card P 01110 = Reference raw card R 01111 = Reference raw card T 10000 = Reference raw card U 10001 = Reference raw card V 10010 = Reference raw card W 10011 = Reference raw card Y 10100 = Reference raw card AA 10101 = Reference raw card AB 10110 = Reference raw card AC 10111 = Reference raw card AD 11000 = Reference raw card AE 11001 = Reference raw card AF 11010 = Reference raw card AG 11011 = Reference raw card AH 11100 = Reference raw card AJ 11101 = Reference raw card AK 11110 = Reference raw card AL 11111 = ZZ (no JEDEC reference raw card design used)

9.1 Module Specific Bytes for Unbuffered Memory Module Types (Bytes 128~191, 0x080~0x0BF) (Cont'd)

9.1.3 Byte 130 (0x082) (Unbuffered): Reference Raw Card Used (Cont'd)

Table 98 — Byte 130 (0x082) (Unbuffered): Reference Raw Cards AM through CB

Bit 7	Bits 6~5	Bits 4~0
Reference Raw Card Extension	Reference Raw Card Revision	Reference Raw Card
1 = Reference raw cards AM through CB	00 = revision 0 01 = revision 1 10 = revision 2 11 = revision 3 See byte 128 for extensions beyond revision 3; these bits must be coded as 11 for all revisions greater than 3	When bit 7 = 1, 00000 = Reference raw card AM 00001 = Reference raw card AN 00010 = Reference raw card AP 00011 = Reference raw card AR 00100 = Reference raw card AT 00101 = Reference raw card AU 00110 = Reference raw card AV 00111 = Reference raw card AW 01000 = Reference raw card AY 01001 = Reference raw card BA 01010 = Reference raw card BB 01011 = Reference raw card BC 01100 = Reference raw card BD 01101 = Reference raw card BE 01110 = Reference raw card BF 01111 = Reference raw card BG 10000 = Reference raw card BH 10001 = Reference raw card BJ 10010 = Reference raw card BK 10011 = Reference raw card BL 10100 = Reference raw card BM 10101 = Reference raw card BN 10110 = Reference raw card BP 10111 = Reference raw card BR 11000 = Reference raw card BT 11001 = Reference raw card BU 11010 = Reference raw card BV 11011 = Reference raw card BW 11100 = Reference raw card BY 11101 = Reference raw card CA 11110 = Reference raw card CB 11111 = ZZ (no JEDEC reference raw card design used)

9.1.4 Byte 131 (0x083) (Unbuffered): Address Mapping from Edge Connector to DRAM

This byte, shown in Table 99, describes the connection of edge connector pins for address bits to the corresponding input pins of the DDR4 SDRAMs for rank 1 only; rank 0 is always assumed to use standard mapping. Only two connection types are supported, standard or mirrored, as described in the mapping table below. System software must compensate for this mapping when issuing mode register set commands to the ranks of DDR4 SDRAMs on this module.

Table 99 — Byte 131 (0x083) (Unbuffered): Address Mapping from Edge Connector to DRAM

Bits 7~1	Bit 0
Reserved	Rank 1 Mapping
Reserved; must be coded as 0000000	0 = standard 1 = mirrored

9.1 Module Specific Bytes for Unbuffered Memory Module Types (Bytes 128~191, 0x080~0x0BF) (Cont'd)

9.1.4 Byte 131 (0x083) (Unbuffered): Address Mapping from Edge Connector to DRAM (Cont'd)

The definition of standard and mirrored address connection mapping is detailed in Table 100; highlighted rows in the table indicate which signals change between mappings.

Table 100 — Mirrored Address Connection Mapping

Edge Connector Pin	DRAM Pin, Non-mirrored	DRAM Pin, Mirrored
A0	A0	A0
A1	A1	A1
A2	A2	A2
A3	A3	A4
A4	A4	A3
A5	A5	A6
A6	A6	A5
A7	A7	A8
A8	A8	A7
A9	A9	A9
A10	A10	A10
A11	A11	A13
A13	A13	A11
A12	A12	A12
A14	A14	A14
A15	A15	A15
A16	A16	A16
A17	A17	A17
BA0	BA0	BA1
BA1	BA1	BA0
BG0	BG0	BG1
BG1	BG1	BG0

9.1.5 Bytes 132~191 (0x084~0x0BF) (Unbuffered):

Reserved – must be coded as 0x00

9.1.6 Byte 254 (0x0FE) (Unbuffered): Cyclical Redundancy Code (CRC) for SPD Block 1, LSB Byte 255 (0x0FF) (Unbuffered): Cyclical Redundancy Code (CRC) for SPD Block 1, MSB

This two-byte field contains the calculated CRC for bytes 128~253 (0x080~0x0FD) in the SPD. See bytes 126~127, Section 8.1.53, for a coding example.

9.2 Module Specific Bytes for Registered Memory Module Types (Bytes 128~191, 0x080~0x0BF)

This section defines the encoding of SPD bytes 128~191 when Memory Technology Key Byte 2 contains the value 0x0C and Module Type Key Byte 3 contains any of the following:

- 0xH1, RDIMM
- 0xH5, Mini-RDIMM
- 0xH8, 72b-SO-RDIMM
- where H refers to the hybrid memory architecture, if any present on the module

Table 101 provides the SPD address map for the module specific section, bytes 128~191, of the SPD for Registered Module Types.

9 SPD Bytes Specific to DDR4 Module Families (Cont'd)

9.2 Module Specific Bytes for Registered Memory Module Types (Bytes 128~191, 0x080~0x0BF) (Cont'd)

Table 101 — Module Specific SPD Bytes for Registered Module Types

Byte Number		Function Described	Notes
128	0x080	Raw Card Extension, Module Nominal Height	
129	0x081	Module Maximum Thickness	
130	0x082	Reference Raw Card Used	
131	0x083	DIMM Attributes	
132	0x084	RDIMM Thermal Heat Spreader Solution	
133	0x085	Register Manufacturer ID Code, Least Significant Byte	
134	0x086	Register Manufacturer ID Code, Most Significant Byte	
135	0x087	Register Revision Number	
136	0x088	Address Mapping from Register to DRAM	
137	0x89	Register Output Drive Strength for Control and Command/Address	
138	0x8A	Register Output Drive Strength for Clock	
139~191	0x089~0x0BF	Reserved -- must be coded as 0x00	
254	0x0FE	CRC for SPD Block 1, Least Significant Byte	1
255	0x0FF	CRC for SPD Block 1, Most Significant Byte	1

1. CRC in bytes 254~255 covers all of SPD Block 1, bytes 128~255 (0x080~0x0FF)

9.2.1 Byte 128 (0x080) (Registered): Raw Card Extension, Module Nominal Height

The upper 3 bits of this byte, shown in Table 102, define extensions to the Raw Card Revision in Byte 130. The lower 5 bits of this byte define the nominal height (A dimension) in millimeters of the fully assembled module including heat spreaders or other added components. Refer to the relevant JEDEC JC-11 module outline (MO) documents for dimension definitions.

Table 102 — Byte 128 (0x080) (Registered): Raw Card Extension, Module Nominal Height

Bits 7~5	Bits 4~0
Raw Card Extension	Module Nominal Height max, in mm (baseline height = 15 mm)
000 = raw card revisions 0 to 3; see byte 130 001 = raw card revision 4 010 = raw card revision 5 011 = raw card revision 6 100 = raw card revision 7 101 = raw card revision 8 110 = raw card revision 9 111 = raw card revision 10	00000 = height ≤ 15 mm 00001 = 15 < height ≤ 16 mm 00010 = 16 < height ≤ 17 mm 00011 = 17 < height ≤ 18 mm 00100 = 18 < height ≤ 19 mm ... 01010 = 24 < height ≤ 25 mm 01011 = 25 < height ≤ 26 mm ... 01111 = 29 < height ≤ 30 mm 10000 = 30 < height ≤ 31 mm 10001 = 31 < height ≤ 32 mm ... 11111 = 45 mm < height

Examples: See Table 103.

9.2 Module Specific Bytes for Registered Memory Module Types (Bytes 128~191, 0x080~0x0BF) (Cont'd)

9.2.1 Byte 128 (0x080) (Registered): Raw Card Extension, Module Nominal Height (Cont'd)

Table 103 — Examples of Raw Card Extension, Module Nominal Height

Nominal Module Height	Coding, bits 4~0	Meaning
mm	Binary	mm
18.75	00100	18 < height ≤ 19 mm
25.40	01011	25 < height ≤ 26 mm
30.00	01111	29 < height ≤ 30 mm
30.25	10000	30 < height ≤ 31 mm
31.25	10001	31 < height ≤ 32 mm

9.2.2 Byte 129 (0x081) (Registered): Module Maximum Thickness

This byte, shown in Table 104, defines the maximum thickness (E dimension) in millimeters of the fully assembled module including heat spreaders or other added components above the module circuit board surface, rounding up to the next integer. Thickness of the front of the module is calculated as the E1 dimension minus the PCB thickness. Thickness of the back of the module is calculated as the E dimension minus the E1 dimension. Refer to the relevant JEDEC JC-11 module outline (MO) documents for dimension definitions.

Table 104 — Byte 129 (0x081) (Registered): Module Maximum Thickness

Bits 7~4	Bits 3~0
Module Maximum Thickness max, Back, in mm (baseline thickness = 1 mm)	Module Maximum Thickness max, Front, in mm (baseline thickness = 1 mm)
0000 = thickness ≤ 1 mm 0001 = 1 < thickness ≤ 2 mm 0010 = 2 < thickness ≤ 3 mm 0011 = 3 < thickness ≤ 4 mm ... 1110 = 14 < thickness ≤ 15 mm 1111 = 15 < thickness	0000 = thickness ≤ 1 mm 0001 = 1 < thickness ≤ 2 mm 0010 = 2 < thickness ≤ 3 mm 0011 = 3 < thickness ≤ 4 mm ... 1110 = 14 < thickness ≤ 15 mm 1111 = 15 < thickness
NOTE Thickness = E - E1	NOTE Thickness = E1 - PCB

9.2.3 Byte 130 (0x082) (Registered): Reference Raw Card Used

This byte, shown in Tables 105 and 106, indicates which JEDEC reference design raw card was used as the basis for the module assembly, if any. Bits 4~0 describe the raw card and bits 6~5 describe the revision level of that raw card. Special raw card indicator, ZZ, is used when no JEDEC standard raw card was used as the basis for the design. Pre-production modules should be encoded as revision 0 in bits 6~5.

9.2 Module Specific Bytes for Registered Memory Module Types (Bytes 128~191, 0x080~0x0BF) (Cont'd)

9.2.3 Byte 130 (0x082) (Registered): Reference Raw Card Used (Cont'd)

Table 105 — Byte 130 (0x082) (Registered): Reference Raw Cards A through AL

Bit 7	Bits 6~5	Bits 4~0
Reference Raw Card Extension	Reference Raw Card Revision	Reference Raw Card
0 = Reference raw cards A through AL	00 = revision 0 01 = revision 1 10 = revision 2 11 = revision 3 See byte 128 for extensions beyond revision 3; these bits must be coded as 11 for all revisions greater than 3	When bit 7 = 0, 00000 = Reference raw card A 00001 = Reference raw card B 00010 = Reference raw card C 00011 = Reference raw card D 00100 = Reference raw card E 00101 = Reference raw card F 00110 = Reference raw card G 00111 = Reference raw card H 01000 = Reference raw card J 01001 = Reference raw card K 01010 = Reference raw card L 01011 = Reference raw card M 01100 = Reference raw card N 01101 = Reference raw card P 01110 = Reference raw card R 01111 = Reference raw card T 10000 = Reference raw card U 10001 = Reference raw card V 10010 = Reference raw card W 10011 = Reference raw card Y 10100 = Reference raw card AA 10101 = Reference raw card AB 10110 = Reference raw card AC 10111 = Reference raw card AD 11000 = Reference raw card AE 11001 = Reference raw card AF 11010 = Reference raw card AG 11011 = Reference raw card AH 11100 = Reference raw card AJ 11101 = Reference raw card AK 11110 = Reference raw card AL 11111 = ZZ (no JEDEC reference raw card design used)

9.2 Module Specific Bytes for Registered Memory Module Types (Bytes 128~191, 0x080~0x0BF) (Cont'd)

9.2.3 Byte 130 (0x082) (Registered): Reference Raw Card Used (Cont'd)

Table 106 — Byte 130 (0x082) (Registered): Reference Raw Cards AM through CB

Bit 7	Bits 6~5	Bits 4~0
Reference Raw Card Extension	Reference Raw Card Revision	Reference Raw Card
1 = Reference raw cards AM through CB	00 = revision 0 01 = revision 1 10 = revision 2 11 = revision 3 See byte 128 for extensions beyond revision 3; these bits must be coded as 11 for all revisions greater than 3	When bit 7 = 1, 00000 = Reference raw card AM 00001 = Reference raw card AN 00010 = Reference raw card AP 00011 = Reference raw card AR 00100 = Reference raw card AT 00101 = Reference raw card AU 00110 = Reference raw card AV 00111 = Reference raw card AW 01000 = Reference raw card AY 01001 = Reference raw card BA 01010 = Reference raw card BB 01011 = Reference raw card BC 01100 = Reference raw card BD 01101 = Reference raw card BE 01110 = Reference raw card BF 01111 = Reference raw card BG 10000 = Reference raw card BH 10001 = Reference raw card BJ 10010 = Reference raw card BK 10011 = Reference raw card BL 10100 = Reference raw card BM 10101 = Reference raw card BN 10110 = Reference raw card BP 10111 = Reference raw card BR 11000 = Reference raw card BT 11001 = Reference raw card BU 11010 = Reference raw card BV 11011 = Reference raw card BW 11100 = Reference raw card BY 11101 = Reference raw card CA 11110 = Reference raw card CB 11111 = ZZ (no JEDEC reference raw card design used)

9.2.4 Byte 131 (0x083) (Registered): DIMM Attributes

This byte, shown in Table 107, indicates number of registers used on a module, and the register type. Further it indicates number of rows of DRAM packages (monolithic, DDP or other stacked) parallel to edge connector (independent of DRAM orientation) on each side of the printed circuit board.

Table 107 — Byte 131 (0x083) (Registered): DIMM Attributes

Bits 7~4	Bits 3~2	Bits 1~0
Register Type	# of rows of DRAMs on RDIMM	# of Registers used on RDIMM
0000 = DDR4RCD01 0001 = DDR4RCD02 All other values reserved	00 = Undefined 01 = 1 row 10 = 2 rows 11 = 4 rows	00 = Undefined 01 = 1 register 10 = 2 registers 11 = 4 registers

9.2.5 Byte 132 (0x084) (Registered): RDIMM Thermal Heat Spreader Solution

This byte, shown in Table 108, describes the module's supported thermal heat spreader solution.

9.2 Module Specific Bytes for Registered Memory Module Types (Bytes 128~191, 0x080~0x0BF) (Cont'd)

9.2.5 Byte 132 (0x084) (Registered): RDIMM Thermal Heat Spreader Solution (Cont'd)

Table 108 — Byte 132 (0x084) (Registered): RDIMM Thermal Heat Spreader Solution

Bit 7	Bits 6~0
Heat Spreader Solution	Heat Spreader Thermal Characteristics
0 = Heat spreader solution is not incorporated onto this assembly 1 = Heat spreader solution is incorporated onto this assembly	0 = Undefined All other settings to be defined

9.2.6 Byte 133 (0x085) (Registered): Register Manufacturer ID Code, LSB

Byte 134 (0x086) (Registered): Register Manufacturer ID Code, MSB

This two-byte field, shown in Table 109, indicates the manufacturer of the register used on the module, encoded as follows: the first byte is the number of continuation bytes indicated in JEP-106; the second byte is the last non-zero byte of the manufacturer's ID code, again as indicated in JEP-106. These bytes are optional. For modules without the Register Manufacturer ID Code information both bytes should be programmed to 0x00.

Table 109 — Bytes 133 and 134, Register Manufacturer ID Code

Byte 134, Bits 7~0	Byte 133, Bit 7	Byte 133 Bits 6~0
Last non-zero byte, Register Manufacturer	Odd parity for Byte 133, bits 6~0	Number of continuation codes, Register Manufacturer
See JEP-106		See JEP-106

Example: See SPD bytes 320~321, Section 8.5.1, for example manufacturer codes.

9.2.7 Byte 135 (0x087) (Registered): Register Revision Number

This byte, shown in Table 110, defines the vendor die revision level of the registering clock driver component. This byte is optional. For modules without the Register Revision Number information, this byte should be programmed to 0xFF. See byte 131, Section 9.2.4, for the register type.

Table 110 — Byte 135 (0x087) (Registered): Register Revision Number

Bits 7~0
Register Revision Number
Programmed in straight Hex format - no conversion needed. 00 - Valid 01 - Valid .. FE - Valid FF - Undefined (No Revision Number Provided)

Examples: See Table 111.

Table 111 — Examples of Register Revision Number

Code	Meaning
0x00	Revision 0
0x01	Revision 1
0x31	Revision 3.1
0xA3	Revision A3
0xB1	Revision B1
0xFF	Revision information not supplied

9.2.8 Byte 136 (0x088) (Registered): Address Mapping from Register to DRAM

This byte, shown in Table 112, describes the connection of register output pins for address bits to the corresponding input pins of the DDR4 SDRAMs for rank 1 and rank 3 only; rank 0 and rank 2 are always assumed to use standard mapping. Only two connection types are supported, standard or mirrored, as described in the mapping table below. System software must compensate for this mapping when issuing mode register set commands to the ranks of DDR4 SDRAMs on this module.

Table 112 — Byte 136 (0x088) (Registered): Address Mapping from Register to DRAM

Bits 7~1	Bit 0
Reserved	Rank 1 Mapping
Reserved; must be coded as 0000000	0 = standard 1 = mirrored

The definition of standard and mirrored address connection mapping is detailed in Table 113; highlighted rows in the table indicate which signals change between mappings.

Table 113 — Standard and Mirrored Address Connection Mapping

Edge Connector Pin	DRAM Pin, Non-mirrored	DRAM Pin, Mirrored
A0	A0	A0
A1	A1	A1
A2	A2	A2
A3	A3	A4
A4	A4	A3
A5	A5	A6
A6	A6	A5
A7	A7	A8
A8	A8	A7
A9	A9	A9
A10	A10	A10
A11	A11	A13
A13	A13	A11
A12	A12	A12
A14	A14	A14
A15	A15	A15
A16	A16	A16
A17	A17	A17
BA0	BA0	BA1
BA1	BA1	BA0
BG0	BG0	BG1
BG1	BG1	BG0

9.2.9 Byte 137 (0x089) (Registered): Register Output Drive Strength for Control and Command/Address

This byte, shown in Table 114, defines the drive strength for the registering clock driver outputs for control and address/command signals.

9.2 Module Specific Bytes for Registered Memory Module Types (Bytes 128~191, 0x080~0x0BF) (Cont'd)**9.2.9 Byte 137 (0x089) (Registered): Register Output Drive Strength for Control and Command/Address (Cont'd)****Table 114 — Byte 137 (0x089) (Registered): Register Output Drive Strength for Control and Command/Address**

Chip Select		Command/Address		ODT		CKE	
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
00 = Light Drive 01 = Moderate Drive 10 = Strong Drive 11 = Very Strong Drive		00 = Light Drive 01 = Moderate Drive 10 = Strong Drive 11 = Very Strong Drive		00 = Light Drive 01 = Moderate Drive 10 = Strong Drive 11 = Very Strong Drive		00 = Light Drive 01 = Moderate Drive 10 = Strong Drive 11 = Very Strong Drive	
NOTE Standard values for drive strength are defined in the DDR4 Registered DIMM Reference Design Specification for JEDEC standard module reference designs.							

9.2.10 Byte 138 (0x08A) (Registered): Register Output Drive Strength for Clock

This byte, shown in Table 115, defines the drive strength for the registering clock driver outputs.

Table 115 — Byte 138 (0x08A) (Registered): Register Output Drive Strength for Clock

Reserved	RCD Output Slew Rate Control	Reserved		Y1, Y3		Y0, Y2	
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Reserved; must be coded as 0	0 = Not supported 1 = Supported	Reserved; must be coded as 00		00 = Light Drive 01 = Moderate Drive 10 = Strong Drive 11 = Very Strong Drive		00 = Light Drive 01 = Moderate Drive 10 = Strong Drive 11 = Very Strong Drive	
NOTE Standard values for drive strength are defined in the DDR4 Registered DIMM Reference Design Specification for JEDEC standard module reference designs.							

9.2.11 Bytes 139~191 (0x08B~0x0BF) (Registered):

Reserved — must be coded as 0x00

**9.2.12 Byte 254 (0x0FE) (Registered): Cyclical Redundancy Code (CRC) for SPD Block 1, LSB
Byte 255 (0x0FF) (Registered): Cyclical Redundancy Code (CRC) for SPD Block 1, MSB**

This two-byte field contains the calculated CRC for bytes 128~253 (0x080~0x0FD) in the SPD. See bytes 126~127, Section 8.1.53, for a coding example.

9.3 Module Specific Bytes for Load Reduced Memory Module Types (Bytes 128~191, 0x080~0x0BF)

This section defines the encoding of SPD bytes 128~191 when Memory Technology Key Byte 2 contains the value 0x0C and Module Type Key Byte 3 contains any of the following:

- 0xH4, LRDIMM
- where H refers to the hybrid memory architecture, if any present on the module

Table 116 provides the SPD address map for the module specific section, bytes 128~255, of the SPD for Load Reduced Module Types.

Table 116 — Module Specific SPD Bytes for Load Reduced Module Types

Byte Number		Function Described	Notes
128	0x080	Raw Card Extension, Module Nominal Height	
129	0x081	Module Maximum Thickness	
130	0x082	Reference Raw Card Used	
131	0x083	DIMM Module Attributes	
132	0x084	LRDIMM Thermal Heat Spreader Solution	
133	0x085	Register and Data Buffer Manufacturer ID Code, Least Significant Byte	

9 SPD Bytes Specific to DDR4 Module Families (Cont'd)

9.3 Module Specific Bytes for Load Reduced Memory Module Types (Bytes 128~191, 0x080~0x0BF) (Cont'd)

Table 116 — Module Specific SPD Bytes for Load Reduced Module Types (Cont'd)

Byte Number		Function Described	Notes
134	0x086	Register and Data Buffer Manufacturer ID Code, Most Significant Byte	
135	0x087	Register Revision Number	
136	0x088	Address Mapping from Register to DRAM	
137	0x089	Register Output Drive Strength for Control and Command/Address	
138	0x08A	Register Output Drive Strength for Clock and Data Buffer Control	
139	0x08B	Data Buffer Revision Number	
140	0x08C	DRAM VrefDQ for Package Rank 0	
141	0x08D	DRAM VrefDQ for Package Rank 1	
142	0x08E	DRAM VrefDQ for Package Rank 2	
143	0x08F	DRAM VrefDQ for Package Rank 3	
144	0x090	Data Buffer VrefDQ for DRAM Interface	
145	0x091	Data Buffer MDQ Drive Strength and RTT for data rate ≤ 1866	
146	0x092	Data Buffer MDQ Drive Strength and RTT for $1866 < \text{data rate} \leq 2400$	
147	0x093	Data Buffer MDQ Drive Strength and RTT for $2400 < \text{data rate} \leq 3200$	
148	0x094	DRAM Drive Strength (for data rates ≤ 1866 , $1866 < \text{data rate} < 2400$, and $2400 < \text{data rate} \leq 3200$)	
149	0x095	DRAM ODT (RTT_WR, RTT_NOM) for data rate ≤ 1866	
150	0x096	DRAM ODT (RTT_WR, RTT_NOM) for $1866 < \text{data rate} \leq 2400$	
151	0x097	DRAM ODT (RTT_WR, RTT_NOM) for $2400 < \text{data rate} \leq 3200$	
152	0x098	DRAM ODT (RTT_PARK) for data rate ≤ 1866	
153	0x099	DRAM ODT (RTT_PARK) for $1866 < \text{data rate} \leq 2400$	
154	0x09A	DRAM ODT (RTT_PARK) for $2400 < \text{data rate} \leq 3200$	
155	0x09B	Data buffer VrefDQ for DRAM interface range	
156	0x09C	Data buffer DQ Decision Feedback Equalization (DFE)	
157~191	0x09D~0x0BF	Reserved -- must be coded as 0x00	
254	0x0FE	CRC for SPD Block 1, Least Significant Byte	1
255	0x0FF	CRC for SPD Block 1, Most Significant Byte	1
1. CRC in bytes 254~255 covers all of SPD Block 1, bytes 128~255 (0x080~0x0FF)			

9.3.1 Byte 128 (0x080) (Load Reduced): Raw Card Extension, Module Nominal Height

The upper 3 bits of this byte, shown in Table 117, define extensions to the Raw Card Revision in Byte 130. The lower 5 bits of this byte define the nominal height (A dimension) in millimeters of the fully assembled module including heat spreaders or other added components. Refer to the relevant JEDEC JC-11 module outline (MO309A) documents for dimension definitions.

9.3 Module Specific Bytes for Load Reduced Memory Module Types (Bytes 128~191, 0x080~0x0BF) (Cont'd)

9.3.1 Byte 128 (0x080) (Load Reduced): Raw Card Extension, Module Nominal Height (Cont'd)

Table 117 — Byte 128 (0x080) (Load Reduced): Raw Card Extension, Module Nominal Height

Bits 7 ~ 5	Bits 4 ~ 0
Raw Card Extension	Module Nominal Height max, in mm (baseline height = 15 mm)
000 = raw card revisions 0 to 3; see byte 130 001 = raw card revision 4 010 = raw card revision 5 011 = raw card revision 6 100 = raw card revision 7 101 = raw card revision 8 110 = raw card revision 9 111 = raw card revision 10	00000 = height ≤ 15 mm 00001 = 15 < height ≤ 16 mm 00010 = 16 < height ≤ 17 mm 00011 = 17 < height ≤ 18 mm 00100 = 18 < height ≤ 19 mm ... 01010 = 24 < height ≤ 25 mm 01011 = 25 < height ≤ 26 mm ... 01111 = 29 < height ≤ 30 mm 10000 = 30 < height ≤ 31 mm ... 11111 = 45 mm < height

Examples: See Table 118.

Table 118 — Examples of Raw Card Extension, Module Nominal Height

Nominal Module Height	Coding, bits 4 ~ 0	Meaning
mm	Binary	mm
18.75	00100	18 < height ≤ 19 mm
25.40	01011	25 < height ≤ 26 mm
30.00	01111	29 < height ≤ 30 mm
30.25	10000	30 < height ≤ 31 mm
31.25	10001	31 < height ≤ 32 mm

9.3.2 Byte 129 (0x081) (Load Reduced): Module Maximum Thickness

This byte, shown in Table 119, defines the maximum thickness (E dimension) in millimeters of the fully assembled module including heat spreaders or other added components above the module circuit board surface, rounding up to the next integer. Thickness of the front of the module is calculated as the E1 dimension minus the PCB thickness. Thickness of the back of the module is calculated as the E dimension minus the E1 dimension. Refer to the relevant JEDEC JC-11 module outline (MO) documents for dimension definitions.

Table 119 — Byte 129 (0x081) (Load Reduced): Module Maximum Thickness

Bits 7 ~ 4	Bits 3 ~ 0
Module Maximum Thickness max, Back, in mm (baseline thickness = 1 mm)	Module Maximum Thickness max, Front, in mm (baseline thickness = 1 mm)
0000 = thickness ≤ 1 mm 0001 = 1 < thickness ≤ 2 mm 0010 = 2 < thickness ≤ 3 mm 0011 = 3 < thickness ≤ 4 mm ... 1110 = 14 < thickness ≤ 15 mm 1111 = 15 < thickness	0000 = thickness ≤ 1 mm 0001 = 1 < thickness ≤ 2 mm 0010 = 2 < thickness ≤ 3 mm 0011 = 3 < thickness ≤ 4 mm ... 1110 = 14 < thickness ≤ 15 mm 1111 = 15 < thickness
NOTE Thickness = E - E1	NOTE 1 Thickness = E1 - PCB

9 SPD Bytes Specific to DDR4 Module Families (Cont'd)

9.3 Module Specific Bytes for Load Reduced Memory Module Types (Bytes 128~191, 0x080~0x0BF) (Cont'd)

9.3.3 Byte 130 (0x082) (Load Reduced): Reference Raw Card Used

This byte, shown in Tables 120 and 121, indicates which JEDEC reference design raw card was used as the basis for the module assembly, if any. Bits 4~0 describe the raw card and bits 6~5 describe the revision level of that raw card. Special raw card indicator, ZZ, is used when no JEDEC standard raw card was used as the basis for the design. Pre-production modules should be encoded as revision 0 in bits 6~5.

Table 120 — Byte 130 (0x082) (Load Reduced): Reference Raw Cards A through AL

Bit 7	Bits 6 ~ 5	Bits 4 ~ 0
Reference Raw Card Extension	Reference Raw Card Revision	Reference Raw Card
0 = Reference raw cards A through AL	00 = revision 0 01 = revision 1 10 = revision 2 11 = revision 3 See byte 128 for extensions beyond revision 3; these bits must be coded as 11 for all revisions greater than 3	When bit 7 = 0, 00000 = Reference raw card A 00001 = Reference raw card B 00010 = Reference raw card C 00011 = Reference raw card D 00100 = Reference raw card E 00101 = Reference raw card F 00110 = Reference raw card G 00111 = Reference raw card H 01000 = Reference raw card J 01001 = Reference raw card K 01010 = Reference raw card L 01011 = Reference raw card M 01100 = Reference raw card N 01101 = Reference raw card P 01110 = Reference raw card R 01111 = Reference raw card T 10000 = Reference raw card U 10001 = Reference raw card V 10010 = Reference raw card W 10011 = Reference raw card Y 10100 = Reference raw card AA 10101 = Reference raw card AB 10110 = Reference raw card AC 10111 = Reference raw card AD 11000 = Reference raw card AE 11001 = Reference raw card AF 11010 = Reference raw card AG 11011 = Reference raw card AH 11100 = Reference raw card AJ 11101 = Reference raw card AK 11110 = Reference raw card AL 11111 = ZZ (no JEDEC reference raw card design used)

9.3 Module Specific Bytes for Load Reduced Memory Module Types (Bytes 128~191, 0x080~0x0BF) (Cont'd)**9.3.3 Byte 130 (0x082) (Load Reduced): Reference Raw Card Used (Cont'd)****Table 121 — Byte 130 (0x082) (Load Reduced): Reference Raw Cards AM through CB**

Bit 7	Bits 6 ~ 5	Bits 4 ~ 0
Reference Raw Card Extension	Reference Raw Card Revision	Reference Raw Card
1 = Reference raw cards AM through CB	00 = revision 0 01 = revision 1 10 = revision 2 11 = revision 3 See byte 128 for extensions beyond revision 3; these bits must be coded as 11 for all revisions greater than 3	When bit 7 = 1, 00000 = Reference raw card AM 00001 = Reference raw card AN 00010 = Reference raw card AP 00011 = Reference raw card AR 00100 = Reference raw card AT 00101 = Reference raw card AU 00110 = Reference raw card AV 00111 = Reference raw card AW 01000 = Reference raw card AY 01001 = Reference raw card BA 01010 = Reference raw card BB 01011 = Reference raw card BC 01100 = Reference raw card BD 01101 = Reference raw card BE 01110 = Reference raw card BF 01111 = Reference raw card BG 10000 = Reference raw card BH 10001 = Reference raw card BJ 10010 = Reference raw card BK 10011 = Reference raw card BL 10100 = Reference raw card BM 10101 = Reference raw card BN 10110 = Reference raw card BP 10111 = Reference raw card BR 11000 = Reference raw card BT 11001 = Reference raw card BU 11010 = Reference raw card BV 11011 = Reference raw card BW 11100 = Reference raw card BY 11101 = Reference raw card CA 11110 = Reference raw card CB 11111 = ZZ (no JEDEC reference raw card design used)

9.3.4 Byte 131 (0x083) (Load Reduced): DIMM Attributes

This byte, shown in Table 122, indicates number of registers used on a module, and the type of the registers and data buffers. Further it indicates number of rows of DRAM packages (monolithic, DDP or 3D stacked) parallel to edge connector (independent of DRAM orientation) on each side of the printed circuit board.

Table 122 — Byte 131 (0x083) (Load Reduced): DIMM Attributes

Bits 7~4	Bits 3~2	Bits 1~0
Register and Data Buffer Types	# of rows of DRAMs on LRDIMM	# of Registers used on LRDIMM
0000 = DDR4RCD01 and DDR4DB01 0001 = DDR4RCD02 and DDR4DB02 All other codes reserved	00 = Undefined 01 = 1 row 10 = 2 rows 11 = Reserved	00 = Undefined 01 = 1 register 10 = Reserved 11 = Reserved

9.3.5 Byte 132 (0x084) (Load Reduced): LRDIMM Thermal Heat Spreader Solution

This byte, shown in Table 123, describes if the module assembly incorporates a heat spreader.

9.3 Module Specific Bytes for Load Reduced Memory Module Types (Bytes 128~191, 0x080~0x0BF) (Cont'd)

9.3.5 Byte 132 (0x084) (Load Reduced): LRDIMM Thermal Heat Spreader Solution (Cont'd)

Table 123 — Byte 132 (0x084) (Load Reduced): LRDIMM Thermal Heat Spreader Solution

Bit 7	Bit 6~0
Heat Spreader	Heat Spreader Thermal Characteristics
0 = Heat spreader solution is not incorporated onto this assembly 1 = Heat spreader solution is incorporated onto this assembly	Undefined; must be coded as 0000000

9.3.6 Byte 133 (0x085) (Load Reduced): Register and Data Buffer Manufacturer ID Code, LSB
Byte 134 (0x086) (Load Reduced): Register and Data Buffer Manufacturer ID Code, MSB

This two-byte field, shown in Table 124, indicates the manufacturer of the memory buffer used on the module, encoded as follows: the first byte is the number of continuation bytes indicated in JEP-106; the second byte is the last non-zero byte of the manufacturer's ID code, again as indicated in JEP-106.

Table 124 — Bytes 133 and 133, (Load Reduced): Register and Data Buffer Manufacturer ID Code

Byte 134, Bits 7 ~ 0	Byte 133, Bit 7	Byte 133 Bits 6 ~ 0
Last non-zero byte, Memory Buffer Manufacturer	Odd parity for Byte 133, bits 6 ~ 0	Number of continuation codes, Memory Buffer Manufacturer
See JEP-106		See JEP-106

Example: See SPD byte 320~321, Section 8.5.1, for example manufacturer codes.

9.3.7 Byte 135 (0x087) (Load Reduced): Register Revision Number

This byte, shown in Table 125, defines the vendor die revision level of the registering clock driver component. See byte 131, Section 9.3.4, for the register type.

Table 125 — Byte 135 (0x087) (Load Reduced): Register Revision Number

Bits 7 ~ 0
Register Revision Number
Programmed in straight Hex format - no conversion needed. 00 - Valid 01 - Valid .. FE - Valid FF - Undefined

Examples: See Table 126

Table 126 — Examples of (Load Reduced) Register Revision Number

Code	Meaning
0x00	Revision 0
0x01	Revision 1
0x31	Revision 3.1
0xA3	Revision A3
0xB1	Revision B1

9.3.8 Byte 136 (0x088) (Load Reduced): Address Mapping from Register to DRAM

This byte, shown in Table 127, describes the connection of register output pins for address bits to the corresponding input pins of the DDR4 SDRAMs for odd ranks only; even ranks are always assumed to use standard mapping. Only two connection types are

9.3 Module Specific Bytes for Load Reduced Memory Module Types (Bytes 128~191, 0x080~0x0BF) (Cont'd)**9.3.8 Byte 136 (0x088) (Load Reduced): Address Mapping from Register to DRAM (Cont'd)**

currently supported, all rank non-mirrored or odd ranks mirrored, as described in the mapping table below. System software must compensate for this mapping when issuing mode register set commands to the ranks of DDR4 SDRAMs on this module.

Table 127 — Byte 136 (0x088) (Load Reduced): Address Mapping from Register to DRAM

Bits 7 ~ 1	Bits 0
Reserved	Odd Rank Mapping
Reserved; must be coded as 0000000	0 = standard 1 = mirrored

The definition of standard and mirrored address connection mapping is detailed in Table 128; highlighted rows in the table indicate which signals change between mappings.

Table 128 — Standard and Mirrored Address Connection Mapping

Edge Connector Pin	DRAM Pin, Non-mirrored	DRAM Pin, Mirrored
A0	A0	A0
A1	A1	A1
A2	A2	A2
A3	A3	A4
A4	A4	A3
A5	A5	A6
A6	A6	A5
A7	A7	A8
A8	A8	A7
A9	A9	A9
A10	A10	A10
A11	A11	A13
A13	A13	A11
A12	A12	A12
A14	A14	A14
A15	A15	A15
A16	A16	A16
A17	A17	A17
BA0	BA0	BA1
BA1	BA1	BA0
BG0	BG0	BG1
BG1	BG1	BG0

9.3.9 Byte 137 (0x089) (Load Reduced): Register Output Drive Strength for Control and Command/Address

This byte, shown in Table 129, defines the drive strength for control and command/address outputs of the registering clock driver component.

Table 129 — Byte 137 (0x089) (Load Reduced): Register Output Drive Strength for Control and Command/Address

Chip Select		Command/Address		ODT		CKE	
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
00 = Light Drive 01 = Moderate Drive 10 = Strong Drive 11 = Very Strong Drive		00 = Light Drive 01 = Moderate Drive 10 = Strong Drive 11 = Very Strong Drive		00 = Light Drive 01 = Moderate Drive 10 = Strong Drive 11 = Very Strong Drive		00 = Light Drive 01 = Moderate Drive 10 = Strong Drive 11 = Very Strong Drive	
NOTE Standard values are defined in the DDR4 LRDIMM Reference Design Specification for JEDEC standard module reference designs.							

9.3.10 Byte 138 (0x08A) (Load Reduced): Register Output Drive Strength for Clock and Data Buffer Control

This byte, shown in Table 130, defines the drive strength for clock outputs and data buffer control outputs of the registering clock driver component.

Table 130 — Byte 138 (0x08A) (Load Reduced): Register Output Drive Strength for Clock and Data Buffer Control

Reserved	RCD Output Slew Rate Control	BCK	BCOM, BODT, BCKE	Y1, Y3		Y0, Y2	
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Reserved; must be coded as 0	0 = Not supported 1 = Supported	0 = Moderate Drive 1 = Strong Drive	0 = Moderate Drive 1 = Strong Drive	00 = Light Drive 01 = Moderate Drive 10 = Strong Drive 11 = Very Strong Drive		00 = Light Drive 01 = Moderate Drive 10 = Strong Drive 11 = Very Strong Drive	
NOTE Standard values are defined in the DDR4 LRDIMM Reference Design Annexes for JEDEC standard module reference designs.							

9.3.11 Byte 139 (0x08B) (Load Reduced): Data Buffer Revision Number

This byte, shown in Table 131, defines the vendor die revision level of the data buffer component. See byte 131 for the data buffer type.

Table 131 — Byte 139 (0x08B) (Load Reduced): Data Buffer Revision Number

Bits 7 ~ 0
Data Buffer Revision Number
Programmed in straight Hex format - no conversion needed. 00 - Valid 01 - Valid .. FE - Valid FF - Undefined

Examples: See Table 132.

Table 132 — Examples of Data Buffer Revision Number

Code	Meaning
0x00	Revision 0
0x01	Revision 1
0x31	Revision 3.1
0xA3	Revision A3
0xB1	Revision B1

9.3.12 Byte 140 (0x08C) (Load Reduced): DRAM VrefDQ for Package Rank 0

This byte, shown in Table 133, defines the VrefDQ value for the package rank 0 DRAMs.

9.3 Module Specific Bytes for Load Reduced Memory Module Types (Bytes 128~191, 0x080~0x0BF) (Cont'd)**9.3.12 Byte 140 (0x08C) (Load Reduced): DRAM VrefDQ for Package Rank 0 (Cont'd)****Table 133 — Byte 140 (0x08C) (Load Reduced): DRAM VrefDQ for Package Rank 0**

Reserved		DRAM VrefDQ for package rank 0 DRAMs					
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Reserved; must be coded as 00		Encoding matches MR6 A5:A0 encoding in the JESD79-4 specification.					
NOTE 1 Standard values are defined in the DDR4 LRDIMM Reference Design Annexes for JEDEC standard module reference designs							
NOTE 2 See Byte 155, Section 9.3.27, for Range encoding							

9.3.13 Byte 141 (0x08D) (Load Reduced): DRAM VrefDQ for Package Rank 1

This byte, shown in Table 134, defines the VrefDQ value for the package rank 1 DRAMs.

Table 134 — Byte 141 (0x08D) (Load Reduced): DRAM VrefDQ for Package Rank 1

Reserved		DRAM VrefDQ for package rank 1 DRAMs					
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Reserved; must be coded as 00		Encoding matches MR6 A5:A0 encoding in the JESD79-4 specification.					
NOTE 1 Standard values are defined in the DDR4 LRDIMM Reference Design Annexes for JEDEC standard module reference designs							
NOTE 2 See Byte 155, Section 9.3.27, for Range encoding							

9.3.14 Byte 142 (0x08E) (Load Reduced): DRAM VrefDQ for Package Rank 2

This byte, shown in Table 135, defines the VrefDQ value for the package rank 2 DRAMs.

Table 135 — Byte 142 (0x08E) (Load Reduced): DRAM VrefDQ for Package Rank 2

Reserved		DRAM VrefDQ for package rank 2 DRAMs					
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Reserved; must be coded as 00		Encoding matches MR6 A5:A0 encoding in the JESD79-4 specification.					
NOTE 1 Standard values are defined in the DDR4 LRDIMM Reference Design Annexes for JEDEC standard module reference designs							
NOTE 2 See Byte 155, Section 9.3.27, for Range encoding							

9 SPD Bytes Specific to DDR4 Module Families (Cont'd)

9.3 Module Specific Bytes for Load Reduced Memory Module Types (Bytes 128~191, 0x080~0x0BF) (Cont'd)

9.3.15 Byte 143 (0x08F) (Load Reduced): DRAM VrefDQ for Package Rank 3

This byte, shown in Table 136, defines the VrefDQ value for the package rank 3 DRAMs.

Table 136 — Byte 143 (0x08F) (Load Reduced): DRAM VrefDQ for Package Rank 3

Reserved		DRAM VrefDQ for package rank 3 DRAMs					
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Reserved; must be coded as 00		Encoding matches MR6 A5:A0 encoding in the JESD79-4 specification.					
NOTE 1 Standard values are defined in the DDR4 LRDIMM Reference Design Annexes for JEDEC standard module reference designs							
NOTE 2 See Byte 155, Section 9.3.27, for Range encoding							

9.3.16 Byte 144 (0x090) (Load Reduced): Data Buffer VrefDQ for DRAM Interface

This byte, shown in Table 137, defines the DRAM interface VrefDQ value for the data buffer component.

Table 137 — Byte 144 (0x090) (Load Reduced): Data Buffer VrefDQ for DRAM Interface

Data Buffer VrefDQ for DRAM Interface							
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
For encoding see definition of F5BC6x (DRAM Interface VREF Control word) in the DDR4DB01 specification.							
NOTE 1 Standard values are defined in the DDR4 LRDIMM Reference Design Annexes for JEDEC standard module reference designs							
NOTE 2 See Byte 155, Section 9.3.27, for Range encoding							

9.3.17 Byte 145 (0x091) (Load Reduced): Data Buffer MDQ Drive Strength and RTT for data rate ≤ 1866

This byte, shown in Table 138, defines the drive strength for MDQ/MDQS outputs and the Read RTT termination strength of the data buffer component.

Table 138 — Byte 145 (0x091) (Load Reduced): Data Buffer MDQ Drive Strength and RTT for data rate ≤ 1866

Data Buffer MDQ Drive Strength and RTT							
DRAM Interface MDQ Drive Strength				DRAM Interface MDQ Read Termination Strength			
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Reserved; must be coded as 0	000 = RZQ/6 (40 Ω) 001 = RZQ/7 (34 Ω) 010 = RZQ/5 (48 Ω) 011 = Reserved 100 = Reserved 101 = RZQ/4 (60 Ω) 110 = Reserved 111 = Reserved			Reserved; must be coded as 0	000 = Disabled 001 = RZQ/4 (60 Ω) 010 = RZQ/2 (120 Ω) 011 = RZQ/6 (40 Ω) 100 = RZQ (240 Ω) 101 = RZQ/5 (48 Ω) 110 = RZQ/3 (80 Ω) 111 = RZQ/7 (34 Ω)		
NOTE Standard values are defined in the DDR4 LRDIMM Reference Design Annexes for JEDEC standard module reference designs.							

9 SPD Bytes Specific to DDR4 Module Families (Cont'd)

9.3 Module Specific Bytes for Load Reduced Memory Module Types (Bytes 128~191, 0x080~0x0BF) (Cont'd)

**9.3.18 Byte 146 (0x092) (Load Reduced): Data Buffer MDQ Drive Strength and RTT for
1866 < data rate ≤ 2400**

This byte, shown in Table 139, defines the drive strength for MDQ/MDQS outputs and the Read RTT termination strength of the data buffer component.

**Table 139 — Byte 146 (0x092) (Load Reduced): Data Buffer MDQ Drive Strength and RTT for
1866 < data rate ≤ 2400**

Data Buffer MDQ Drive Strength and RTT							
DRAM Interface MDQ Drive Strength				DRAM Interface MDQ Read Termination Strength			
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Reserved; must be coded as 0	000 = RZQ/6 (40 Ω)			Reserved; must be coded as 0	000 = Disabled		
	001 = RZQ/7 (34 Ω)				001 = RZQ/4 (60 Ω)		
	010 = RZQ/5 (48 Ω)				010 = RZQ/2 (120 Ω)		
	011 = Reserved				011 = RZQ/6 (40 Ω)		
	100 = Reserved				100 = RZQ (240 Ω)		
	101 = RZQ/4 (60 Ω)				101 = RZQ/5 (48 Ω)		
	110 = Reserved				110 = RZQ/3 (80 Ω)		
	111 = Reserved				111 = RZQ/7 (34 Ω)		
NOTE Standard values are defined in the DDR4 LRDIMM Reference Design Annexes for JEDEC standard module reference designs.							

**9.3.19 Byte 147 (0x093) (Load Reduced): Data Buffer MDQ Drive Strength and RTT for
2400 < data rate ≤ 3200**

This byte, shown in Table 140, defines the drive strength for MDQ/MDQS outputs and the Read RTT termination strength of the data buffer component.

**Table 140 — Byte 147 (0x093) (Load Reduced): Data Buffer MDQ Drive Strength and RTT for
2400 < data rate ≤ 3200**

Data Buffer MDQ Drive Strength and RTT							
DRAM Interface MDQ Drive Strength				DRAM Interface MDQ Read Termination Strength			
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Reserved; must be coded as 0	000 = RZQ/6 (40 Ω)			Reserved; must be coded as 0	000 = Disabled		
	001 = RZQ/7 (34 Ω)				001 = RZQ/6 (60 Ω)		
	010 = RZQ/5 (48 Ω)				010 = RZQ/2 (120 Ω)		
	011 = Reserved				011 = RZQ/4 (40 Ω)		
	100 = Reserved				100 = RZQ (240 Ω)		
	101 = RZQ/4 (60 Ω)				101 = RZQ/5 (48 Ω)		
	110 = Reserved				110 = RZQ/3 (80 Ω)		
	111 = Reserved				111 = RZQ/7 (34 Ω)		
NOTE Standard values are defined in the DDR4 LRDIMM Reference Design Annexes for JEDEC standard module reference designs.							

9 SPD Bytes Specific to DDR4 Module Families (Cont'd)

9.3 Module Specific Bytes for Load Reduced Memory Module Types (Bytes 128~191, 0x080~0x0BF) (Cont'd)

9.3.20 Byte 148 (0x094) (Load Reduced): DRAM Drive Strength (for data rates ≤ 1866 , $1866 < \text{data rate} \leq 2400$, and $2400 < \text{data rate} \leq 3200$)

This byte, shown in Table 141, defines the output buffer drive strength for the DRAMs.

Table 141 — Byte 148 (0x094) (Load Reduced): DRAM Drive Strength (for data rates ≤ 1866 , $1866 < \text{data rate} \leq 2400$, and $2400 < \text{data rate} \leq 3200$)

DRAM Drive Strength							
Reserved		2400 < Data rate ≤ 3200		1866 < Data rate ≤ 2400		Data rate ≤ 1866	
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Reserved; must be coded as 00		00 = RZQ/7 (34 Ω)		00 = RZQ/7 (34 Ω)		00 = RZQ/7 (34 Ω)	
		01 = RZQ/5 (48 Ω)		01 = RZQ/5 (48 Ω)		01 = RZQ/5 (48 Ω)	
		10 = Reserved		10 = Reserved		10 = Reserved	
		11 = Reserved		11 = Reserved		11 = Reserved	
NOTE Standard values are defined in the DDR4 LRDIMM Reference Design Annexes for JEDEC standard module reference designs.							

9.3.21 Byte 149 (0x095) (Load Reduced): DRAM ODT (RTT_WR and RTT_NOM) for data rate ≤ 1866

This byte, shown in Table 142, defines the ODT termination strength for the DRAMs.

Table 142 — Byte 149 (0x095) (Load Reduced): DRAM ODT (RTT_WR and RTT_NOM) for data rate ≤ 1866

DRAM ODT Strength							
Reserved		RTT_WR			RTT_NOM		
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Reserved; must be coded as 00		000 = Dynamic ODT Off			000 = Disabled		
		001 = RZQ/2 (120 Ω)			001 = RZQ/4 (60 Ω)		
		010 = RZQ (240 Ω)			010 = RZQ/2 (120 Ω)		
		011 = Hi-Impedance			011 = RZQ/6 (40 Ω)		
		100 = RZQ/3 (80 Ω)			100 = RZQ (240 Ω)		
		101 = Reserved			101 = RZQ/5 (48 Ω)		
		110 = Reserved			110 = RZQ/3 (80 Ω)		
		111 = Reserved			111 = RZQ/7 (34 Ω)		
NOTE Standard values are defined in the DDR4 LRDIMM Reference Design Annexes for JEDEC standard module reference designs.							

9.3.22 Byte 150 (0x096) (Load Reduced): DRAM ODT (RTT_WR and RTT_NOM) for $1866 < \text{data rate} \leq 2400$

This byte, shown in Table 143, defines the ODT termination strength for the DRAMs.

Table 143 — Byte 150 (0x096) (Load Reduced): DRAM ODT (RTT_WR and RTT_NOM) for $1866 < \text{data rate} \leq 2400$

DRAM ODT Strength							
Reserved		RTT_WR			RTT_NOM		
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Reserved; must be coded as 00		000 = Dynamic ODT Off			000 = Disabled		
		001 = RZQ/2 (120Ω)			001 = RZQ/4 (60 Ω)		
		010 = RZQ (240 Ω)			010 = RZQ/2 (120 Ω)		
		011 = Hi-Impedance			011 = RZQ/6 (40 Ω)		
		100 = RZQ/3 (80 Ω)			100 = RZQ (240 Ω)		
		101 = Reserved			101 = RZQ/5 (48 Ω)		
		110 = Reserved			110 = RZQ/3 (80 Ω)		
		111 = Reserved			111 = RZQ/7 (34 Ω)		
NOTE Standard values are defined in the DDR4 LRDIMM Reference Design Annexes for JEDEC standard module reference designs.							

9 SPD Bytes Specific to DDR4 Module Families (Cont'd)

9.3 Module Specific Bytes for Load Reduced Memory Module Types (Bytes 128~191, 0x080~0x0BF) (Cont'd)

**9.3.23 Byte 151 (0x097) (Load Reduced): DRAM ODT (RTT_WR and RTT_NOM)
for 2400 < data rate ≤ 3200**

This byte, shown in Table 144, defines the ODT termination strength for the DRAMs.

**Table 144 — Byte 151 (0x097) (Load Reduced): DRAM ODT (RTT_WR and RTT_NOM)
for 2400 < data rate ≤ 3200**

DRAM ODT Strength							
Reserved		RTT_WR			RTT_NOM		
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Reserved; must be coded as 00		000 = Dynamic ODT Off			000 = Disabled		
		001 = RZQ/2 (120Ω)			001 = RZQ/4 (60 Ω)		
		010 = RZQ (240 Ω)			010 = RZQ/2 (120 Ω)		
		011 = Hi-Impedance			011 = RZQ/6 (40 Ω)		
		100 = RZQ/3 (80 Ω)			100 = RZQ (240 Ω)		
		101 = Reserved			101 = RZQ/5 (48 Ω)		
		110 = Reserved			110 = RZQ/3 (80 Ω)		
		111 = Reserved			111 = RZQ/7 (34 Ω)		
NOTE Standard values are defined in the DDR4 LRDIMM Reference Design Annexes for JEDEC standard module reference designs.							

9.3.24 Byte 152 (0x098) (Load Reduced): DRAM ODT (RTT_PARK) for data rate ≤ 1866

This byte, shown in Table 145, defines the ODT termination strength for the DRAMs.

Table 145 — Byte 152 (0x098) (Load Reduced): DRAM ODT (RTT_PARK) for data rate ≤ 1866

DRAM ODT Strength							
Reserved		RTT_PARK, Package Ranks 2 & 3			RTT_PARK, Package Ranks 0 & 1		
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Reserved; must be coded as 00		000 = Disabled			000 = Disabled		
		001 = RZQ/4 (60 Ω)			001 = RZQ/4 (60 Ω)		
		010 = RZQ/2 (120 Ω)			010 = RZQ/2 (120 Ω)		
		011 = RZQ/6 (40 Ω)			011 = RZQ/6 (40 Ω)		
		100 = RZQ (240 Ω)			100 = RZQ (240 Ω)		
		101 = RZQ/5 (48 Ω)			101 = RZQ/5 (48 Ω)		
		110 = RZQ/3 (80 Ω)			110 = RZQ/3 (80 Ω)		
		111 = RZQ/7 (34 Ω)			111 = RZQ/7 (34 Ω)		
NOTE Standard values are defined in the DDR4 LRDIMM Reference Design Annexes for JEDEC standard module reference designs.							

9.3.25 Byte 153 (0x099) (Load Reduced): DRAM ODT (RTT_PARK) for 1866 < data rate ≤ 2400

This byte, shown in Table 146, defines the ODT termination strength for the DRAMs.

Table 146 — Byte 153 (0x099) (Load Reduced): DRAM ODT (RTT_PARK) for 1866 < data rate ≤ 2400

DRAM ODT Strength							
Reserved		RTT_PARK, Package Ranks 2 & 3			RTT_PARK, Package Ranks 0 & 1		
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Reserved; must be coded as 00		000 = Disabled			000 = Disabled		
		001 = RZQ/4 (60 Ω)			001 = RZQ/4 (60 Ω)		
		010 = RZQ/2 (120 Ω)			010 = RZQ/2 (120 Ω)		
		011 = RZQ/6 (40 Ω)			011 = RZQ/6 (40 Ω)		
		100 = RZQ (240 Ω)			100 = RZQ (240 Ω)		
		101 = RZQ/5 (48 Ω)			101 = RZQ/5 (48 Ω)		
		110 = RZQ/3 (80 Ω)			110 = RZQ/3 (80 Ω)		
		111 = RZQ/7 (34 Ω)			111 = RZQ/7 (34 Ω)		
NOTE Standard values are defined in the DDR4 LRDIMM Reference Design Annexes for JEDEC standard module reference designs.							

9.3.26 Byte 154 (0x09A) (Load Reduced): DRAM ODT (RTT_PARK) for 2400 < data rate ≤ 3200

This byte, shown in Table 147, defines the ODT termination strength for the DRAMs.

Table 147 — Byte 154 (0x09A) (Load Reduced): DRAM ODT (RTT_PARK) for 2400 < data rate ≤ 3200

DRAM ODT Strength							
Reserved		RTT_PARK, Package Ranks 2 & 3			RTT_PARK, Package Ranks 0 & 1		
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Reserved; must be coded as 00		000 = Disabled			000 = Disabled		
		001 = RZQ/4 (60 Ω)			001 = RZQ/4 (60 Ω)		
		010 = RZQ/2 (120 Ω)			010 = RZQ/2 (120 Ω)		
		011 = RZQ/6 (40 Ω)			011 = RZQ/6 (40 Ω)		
		100 = RZQ (240 Ω)			100 = RZQ (240 Ω)		
		101 = RZQ/5 (48 Ω)			101 = RZQ/5 (48 Ω)		
		110 = RZQ/3 (80 Ω)			110 = RZQ/3 (80 Ω)		
		111 = RZQ/7 (34 Ω)			111 = RZQ/7 (34 Ω)		
NOTE Standard values are defined in the DDR4 LRDIMM Reference Design Annexes for JEDEC standard module reference designs.							

9.3.27 Byte 155 (0x09B) (Load Reduced): Data Buffer VrefDQ for DRAM Interface Range

This byte, shown in Table 148, defines the DRAM interface VrefDQ Range for the Data Buffer component.

Table 148 — Byte 155 (0x09B) (Load Reduced): Data Buffer VrefDQ for DRAM Interface Range

Reserved	Data Buffer VrefDQ for DRAM Interface	VrefDQ Range for DRAM interface			
	Data Buffer	Rank 3	Rank 2	Rank 1	Rank 0
Bit 7~5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Reserved, must be programmed to 000.	0 = Range 1 1 = Range 2	0 = Range 1 1 = Range 2	0 = Range 1 1 = Range 2	0 = Range 1 1 = Range 2	0 = Range 1 1 = Range 2
NOTE Standard values are defined in the DDR4 LRDIMM Reference Design Annexes for JEDEC standard module reference designs and DDR4DBx specifications.					

9 SPD Bytes Specific to DDR4 Module Families (Cont'd)**9.3 Module Specific Bytes for Load Reduced Memory Module Types (Bytes 128~191, 0x080~0x0BF) (Cont'd)****9.3.28 Byte 156(0x09C) (Load Reduced): Data Buffer DQ Decision Feedback Equalization (DFE)**

This byte, shown in Table 149, defines the DFE capabilities of the LRDIMM data buffer.

Table 149 — Byte 156(0x09C) (Load Reduced): Data Buffer DQ Decision Feedback Equalization (DFE)

Data Buffer DQ DFE		
Reserved	Data Buffer DFE	Data Buffer Gain Adjustment
Bits 7~2	Bit 1	Bit 0
Reserved; must be programmed to 000000	0 = not supported 1 = supported	0 = not supported 1 = supported
NOTE See Data Buffer supplier's data sheet for details.		

9.3.29 Bytes 157~191 (0x09D~0x0BF) (Load Reduced):

Reserved – must be coded as 0x00

**9.3.30 Bytes 254 (0x0FE) (Load Reduced): Cyclical Redundancy Code (CRC) for SPD Block 1, LSB
Bytes 255 (0x0FF) (Load Reduced): Cyclical Redundancy Code (CRC) for SPD Block 1, MSB**

This two-byte field contains the calculated CRC for bytes 128~253 (0x080~0x0FD) in the SPD. See bytes 126~127, Section 8.1.53, for coding example.

10 SPD Bytes Specific to Hybrid Memory Architecture Specific Parameters**10.1 Non-Volatile (NVDIMM-N) Hybrid Memory Parameters: Bytes 192~253 (0x0C0~0x1FD)**

This section defines the encoding of SPD bytes 192~255 when Module Type Key Byte 3 contains the following:

- 0x9M, NVDIMM
- where M refers to the base memory architecture

NVDIMM types defined at this time are:

NVDIMM-N	Persistent DRAM using NAND flash
NVDIMM-F	NAND Flash accessed as a block oriented device
NVDIMM-P	Combined persistent DRAM and block accessed NAND Flash

Table 150 provides the SPD address map for the hybrid module specific section, bytes 192~255, of the SPD for Non-Volatile (NVDIMM) Module Types. Note that most NVDIMM types will also have extended function parameters in block 2, bytes 256~319, specific to each type.

Table 150 — Non-Volatile (NVDIMM-N) Hybrid Memory Parameters

Byte Number	Function Described	Notes
192	0x0C0	Module Product Identifier, LSB
193	0x0C1	Module Product Identifier, MSB
194	0x0C2	Non-Volatile Memory Subsystem Controller Manufacturer's ID Code, LSB
195	0x0C3	Non-Volatile Memory Subsystem Controller Manufacturer's ID Code, MSB
196	0x0C4	Non-Volatile Memory Subsystem Controller Product Identifier, LSB
197	0x0C5	Non-Volatile Memory Subsystem Controller Product Identifier, MSB
198	0x0C6	Non-Volatile Memory Subsystem Controller Revision Code

10 SPD Bytes Specific to Hybrid Memory Architecture Specific Parameters (Cont'd)
10.1 Non-Volatile (NVDIMM-N) Hybrid Memory Parameters: Bytes 192~253 (0x0C0~0x1FD) (Cont'd)

Table 150 — Non-Volatile (NVDIMM-N) Hybrid Memory Parameters (Cont'd)

Byte Number		Function Described	Notes
199	0x0C7	Reference Raw Card Used	
200	0x0C8	Module Characteristics	
201~202	0x0C9~0x0CA	Hybrid Module Media Types	
203	0x0CB	Maximum Non-Volatile Memory Initialization Time	
204~205	0x0CC~0x0CD	Function 0 Format Interface Descriptor	
206~207	0x0CE~0x0CF	Function 1 Format Interface Descriptor	
208~209	0x0D0~0x0D1	Function 2 Format Interface Descriptor	
210~211	0x0D2~0x0D3	Function 3 Format Interface Descriptor	
212~213	0x0D4~0x0D5	Function 4 Format Interface Descriptor	
214~215	0x0D6~0x0D7	Function 5 Format Interface Descriptor	
216~217	0x0D8~0x0D9	Function 6 Format Interface Descriptor	
218~219	0x0DA~0x0DB	Function 7 Format Interface Descriptor	
220~253	0x0DC~0x0FD	Reserved	
254	0x0FE	CRC for SPD Block 1, Least Significant Byte	1
255	0x0FF	CRC for SPD Block 1, Most Significant Byte	1
NOTE 1 1. CRC in bytes 254~255 covers all of SPD Block 1, bytes 128~255 (0x080~0x0FF)			

10.1.1 Byte 192 (0x0C0) (NVDIMM): Module Product Identifier, LSB
Byte 193 (0x0C1) (NVDIMM): Module Product Identifier, MSB

These bytes are defined and controlled by the memory module manufacturer to distinguish a specific product among their product offerings. Combined with the Module Manufacturer's ID Code in bytes 320~321 (0x14D~0x14E), creates a unique 32-bit code for each product type.

10.1.2 Byte 194 (0x0C2) (NVDIMM): Non-Volatile Memory Subsystem Controller Manufacturer's ID Code, LSB
Byte 195 (0x0C3) (NVDIMM): Non-Volatile Memory Subsystem Controller Manufacturer's ID Code, MSB

This two-byte field, shown in Table 151, indicates the manufacturer of the Non-Volatile Memory Subsystem controller, encoded as follows: the first byte is the number of continuation bytes indicated in JEP-106; the second byte is the last non-zero byte of the manufacturer's ID code, again as indicated in JEP-106.

Table 151 — Bytes 194 and 195, Non-Volatile Memory Subsystem Controller Manufacturer's ID Code

Byte 195, Bits 7~0	Byte 194, Bit 7	Byte 194, Bits 6~0
Last non-zero byte, Flash Manufacturer	Odd Parity for Byte 194, bits 6~0	Number of continuation codes, Flash Manufacturer
See JEP-106		See JEP-106

Example: See SPD bytes 320~321, Section 8.5.1, for example manufacturer codes.

10.1.3 Byte 196 (0x0C4) (NVDIMM): Non-Volatile Memory Subsystem Controller Identifier, LSB
Byte 197 (0x0C5) (NVDIMM): Non-Volatile Memory Subsystem Controller Identifier, MSB

These bytes are defined and controlled by the non-volatile memory subsystem controller manufacturer to distinguish a specific product among their product offerings. In combination with the Non-Volatile Subsystem controller Manufacturer's ID Code in bytes 194~195 (0x0C2~0x0C3), they create a unique 32-bit code for each product type.

10 SPD Bytes Specific to Hybrid Memory Architecture Specific Parameters (Cont'd)**10.1 Non-Volatile (NVDIMM-N) Hybrid Memory Parameters: Bytes 192~253 (0x0C0~0x1FD) (Cont'd)****10.1.4 Byte 198 (0x0C6) (NVDIMM): Non-Volatile Memory Subsystem Controller Revision Code**

This byte refers to the non-volatile subsystem controller revision code. While the SPD specification will not attempt to define the format for this information, the individual non-volatile subsystem controller manufacturer may keep track of the revision code and its appropriate decode represented in this byte.

10.1.5 Byte 199 (0x0C7) (NVDIMM): Reference Raw Card Used

This byte, shown in Tables 152 and 153, indicates which JEDEC reference design raw card was used as the basis for the module assembly, if any. Bits 4~0 describe the raw card and bits 6~5 describe the revision level of that raw card. Special raw card indicator, ZZ, is used when no JEDEC standard raw card was used as the basis for the design. Pre-production modules should be encoded as revision 0 in bits 6~5.

Table 152 — Byte 199 (0x0C7) (NVDIMM): Reference Raw Cards A through AL

Bit 7	Bits 6~5	Bits 4~0
Reference Raw Card Extension	Reference Raw Card Revision	Reference Raw Card
0 = Reference raw cards A through AL	00 = revision 0 01 = revision 1 10 = revision 2 11 = revision 3 See byte 200 for extensions beyond revision 3; these bits must be coded as 11 for all revisions greater than 3	When bit 7 = 0, 00000 = Reference raw card A 00001 = Reference raw card B 00010 = Reference raw card C 00011 = Reference raw card D 00100 = Reference raw card E 00101 = Reference raw card F 00110 = Reference raw card G 00111 = Reference raw card H 01000 = Reference raw card J 01001 = Reference raw card K 01010 = Reference raw card L 01011 = Reference raw card M 01100 = Reference raw card N 01101 = Reference raw card P 01110 = Reference raw card R 01111 = Reference raw card T 10000 = Reference raw card U 10001 = Reference raw card V 10010 = Reference raw card W 10011 = Reference raw card Y 10100 = Reference raw card AA 10101 = Reference raw card AB 10110 = Reference raw card AC 10111 = Reference raw card AD 11000 = Reference raw card AE 11001 = Reference raw card AF 11010 = Reference raw card AG 11011 = Reference raw card AH 11100 = Reference raw card AJ 11101 = Reference raw card AK 11110 = Reference raw card AL 11111 = ZZ (no JEDEC reference raw card design used)

10.1 Non-Volatile (NVDIMM-N) Hybrid Memory Parameters: Bytes 192~253 (0x0C0~0x1FD) (Cont'd)

10.1.5 Byte 199 (0x0C7) (NVDIMM): Reference Raw Card Used (Cont'd)

Table 153 — Byte 199 (0x0C7) (NVDIMM): Reference Raw Cards AM through CB

Bit 7	Bits 6~5	Bits 4~0
Reference Raw Card Extension	Reference Raw Card Revision	Reference Raw Card
1 = Reference raw cards AM through CB	00 = revision 0 01 = revision 1 10 = revision 2 11 = revision 3 See byte 200 for extensions beyond revision 3; these bits must be coded as 11 for all revisions greater than 3	When bit 7 = 1, 00000 = Reference raw card AM 00001 = Reference raw card AN 00010 = Reference raw card AP 00011 = Reference raw card AR 00100 = Reference raw card AT 00101 = Reference raw card AU 00110 = Reference raw card AV 00111 = Reference raw card AW 01000 = Reference raw card AY 01001 = Reference raw card BA 01010 = Reference raw card BB 01011 = Reference raw card BC 01100 = Reference raw card BD 01101 = Reference raw card BE 01110 = Reference raw card BF 01111 = Reference raw card BG 10000 = Reference raw card BH 10001 = Reference raw card BJ 10010 = Reference raw card BK 10011 = Reference raw card BL 10100 = Reference raw card BM 10101 = Reference raw card BN 10110 = Reference raw card BP 10111 = Reference raw card BR 11000 = Reference raw card BT 11001 = Reference raw card BU 11010 = Reference raw card BV 11011 = Reference raw card BW 11100 = Reference raw card BY 11101 = Reference raw card CA 11110 = Reference raw card CB 11111 = ZZ (no JEDEC reference raw card design used)

10.1.6 Byte 200 (0x0C8) (NVDIMM): Module Characteristics

This byte, shown in Table 154, contains additional information about the module characteristics. Bits 7~5 allow for extended raw card revision levels.

Table 154 — Byte 200 (0x0C8) (NVDIMM): Module Characteristics

Bits 7~5	Bits 4~0
Raw Card Extension	Reserved
000 = raw card revisions 0 to 3; see byte 199 001 = raw card revision 4 010 = raw card revision 5 011 = raw card revision 6 100 = raw card revision 7 101 = raw card revision 8 110 = raw card revision 9 111 = raw card revision 10	Reserved; must be coded as 00000

10.1.7 Bytes 201~202 (0x0C9~0x0CA) (NVDIMM): Hybrid Module Media Types

These bytes, shown in Table 155, define a media bit mask for all media types on the module. A setting of 1 in each bit position indicates the presence of that memory type on the module. The SPD is not considered as a media type in this context.

10.1 Non-Volatile (NVDIMM-N) Hybrid Memory Parameters: Bytes 192~253 (0x0C0~0x1FD) (Cont'd)**10.1.7 Bytes 201~202 (0x0C9~0x0CA) (NVDIMM): Hybrid Module Media Types (Cont'd)****Table 155 — Bytes 201~202 (0x0C9~0x0CA) (NVDIMM): Hybrid Module Media Types**

Byte 201		Byte 202	
Bit	Media Type	Bit	Media Type
0	Unknown/undefined	0	Reserved
1	SDRAM	1	Reserved
2	NAND Flash	2	Reserved
3	Reserved	3	Reserved
4	Reserved	4	Reserved
5	Reserved	5	Reserved
6	Reserved	6	Reserved
7	Reserved	7	Reserved

Examples: See Table 156.

Table 156 — Examples for Hybrid Module Media Types

Module Type	Media Types	Byte 201	Byte 202
NVDIMM-N	SDRAM, NAND Flash	0000 0110	0000 0000
NVDIMM-F	NAND Flash	0000 0100	0000 0000
NVDIMM-P	SDRAM, NAND Flash	0000 0110	0000 0000

10.1.8 Byte 203 (0x0CB) (NVDIMM): Maximum Non-Volatile Memory Initialization Time

This byte, shown in Table 157, defines the maximum time required for all subsystem(s) to complete initialization from release of RESET or initialization programming of all subsystems present on the module, whichever is later, in seconds. If the initialization time is not known, this byte must be coded as 255 (0xFF).

Table 157 — Byte 203 (0x0CB) (NVDIMM): Maximum Non-Volatile Memory Initialization Time

Minimum Non-Volatile Memory Initialization Time Seconds
Values defined from 0 to 254

Examples: See Table 158.

Table 158 — Examples for Maximum Non-Volatile Memory Initialization Time

Byte 203	Hex	Use
0	0x00	No initialization time required
1	0x01	1 second
2	0x02	2 seconds
...		...
255	0xFF	Initialization time not known or specified

10.1.9 Bytes 204~219 (0x0CC~0x0DB) (NVDIMM): Function Interface Descriptors

These sets of 2 bytes, shown in Table 159, define the functions available on this module plus the location in the SPD Extended Function Parameters block of the information specific to each function.

Table 159 — Bytes 204~219 (0x0CC~0x0DB) (NVDIMM): Function Interface Descriptors

Descriptor	LSB		MSB	
Function 0 Interface Descriptor	204	0x0CC	205	0x0CD
Function 1 Interface Descriptor	206	0x0CE	207	0x0CF
Function 2 Interface Descriptor	208	0x0D0	209	0x0D1
Function 3 Interface Descriptor	210	0x0D2	211	0x0D3
Function 4 Interface Descriptor	212	0x0D4	213	0x0D5
Function 5 Interface Descriptor	214	0x0D6	215	0x0D7
Function 6 Interface Descriptor	216	0x0D8	217	0x0D9
Function 7 Interface Descriptor	218	0x0DA	219	0x0DB

Each descriptor encodes the information depicted in Table 160:

Table 160 — Function Interface Descriptor

MSB			LSB	
Bit 15	Bit 14	Bits 13~10	Bits 9~5	Bits 4~0
Implemented	Reserved	Block Offset	Function Class	Function Interface
Bit 15	0 = Not implemented 1 = Implemented			
Bit 14	Reserved; must be coded as 0			
Bits 13~10	Offset of additional information in Extended Function Parameter Block. Calculate as: 256 + (Block Offset * 4). Field shall have value of 0000 if there is no additional information in the Extended Function Parameter Block. Block Offset values shall be contiguous from one Function Interface Descriptor to the next. If a Function Interface Descriptor N has information in the Extended Function Parameter Block, size of Extended Function Parameter Block for Function Interface Descriptor N is ((Block Offset Function Interface Descriptor N+1 – Block Offset Function Interface Descriptor N) * 4) where Block Offset Function Interface Descriptor N+1 is next Function Interface Descriptor that has a non-zero Block Offset value.			
Bits 9~5	See table for Function Classes and Function Interfaces			
Bits 4~0	See table for Function Classes and Function Interfaces			
NOTE 1 When bit 15 = 0, all other bits in the descriptor must also be coded as 0				
NOTE 2 Block Offset granularity of 4 bytes may require rounding up to the next available byte in the Extended Function Parameter Block; unused bytes must be coded as 0x00				

The Function Class and Function Interface are related as shown in Table 161:

Table 161 — Function Classes and Function Interfaces

Function Class Bits 9~5	Function	Definition	Function Interface Bits 4~0
00 000	Undefined function	Undefined	0 0000 = Proprietary interface All other codes reserved
00 001	Byte addressable energy backed	A function containing byte addressable persistent memory whose persistence is achieved through the use of DRAM, nonvolatile memory (e.g., Flash) and an energy source. Only the DRAM portion is addressable by the system.	0 0000 = Proprietary interface 0 0001 = Byte Addressable Energy Backed Function Interface 1 All other codes reserved

10.1 Non-Volatile (NVDIMM-N) Hybrid Memory Parameters: Bytes 192~253 (0x0C0~0x1FD) (Cont'd)**10.1.9 Bytes 204~219 (0x0CC~0x0DB) (NVDIMM): Function Interface Descriptors (Cont'd)****Table 161 — Function Classes and Function Interfaces (Cont'd)**

00 010	Block addressed	A function containing persistent memory that requires block-oriented access.	0 0000 = Proprietary interface 0 0001 = Block Addressed Function Interface 1 All other codes reserved
00 011	Byte addressable, no energy backed	A function containing byte addressable persistent memory. All of the persistent memory is addressable by the system. No external energy source is required.	0 0000 = Proprietary interface 0 0001 = Byte Addressable No Energy Backed Function Interface 1 All other codes reserved

Example 1: See Table 162.

NVDIMM with two functions:

Function Interface Descriptor 0 = A byte addressable energy backed function class using the byte addressable energy backed function with standard interface 1. This function class defines 14 bytes in the SPD Extended Function Parameters block.

Function Interface Descriptor 1 = A block addressed function class using the block addressed function interface 1. This function class requires 23 bytes in the SPD Extended Function Parameters block.

Table 162 — Example 1: NVDIMM with Two Functions

Function 0 Interface Descriptor		
Byte 205	Byte 204	Meaning
1000 0000	0010 0001	Bit 15 = 1: Implemented Bits 9~5 = 00001: Function Class is Byte addressable energy backed Bits 4~0 = 00001: Function Interface is Byte addressable energy backed Interface 1 Bits 13~10 = 0000: Offset of Extended Function Parameters = $256 + 0 * 4 = 256$ (0x100) Implied by the example (i.e., 14 bytes used): Extended Function block from 256 through 269 (0x100~0x10D) is defined Extended Function block bytes 270, 271 not used (coded as 0x00)
Function 1 Interface Descriptor		
Byte 207	Byte 206	Meaning
1001 0000	0100 0001	Bit 15 = 1: Implemented Bits 9~5 = 00010: Function Class is Block addressed Bits 4~0 = 00001: Function Interface is Block Addressed Function Interface 1 Bits 13~10 = 0100: Offset of Extended Function Parameters = $256 + 4 * 4 = 272$ (0x110) Implied by the example (i.e., 23 bytes used): Extended Function block from 272 through 294 (0x110~0x126) is defined Extended Function block byte 295 not used (coded as 0x00)
Functions 2~7 Interface Descriptors		
Byte n+1	Byte n	Meaning
0000 0000	0000 0000	Bit 15 = 0: Not Implemented; all other bits must be coded as 0

Example 2: See Table 163.

NVDIMM with three functions:

Function Interface Descriptor 0 = a byte addressable no energy backed function using a proprietary interface. This function class defines 14 bytes in the SPD Extended Function Parameters block.

Function Interface Descriptor 1 = a block addressed function (2) using a proprietary interface. This function class does not have any bytes in the SPD Extended Function Parameter block.

Function Interface Descriptor 2 = an undefined function using a proprietary interface. This function class defines 23 bytes in the SPD Extended Function Parameters block.

Table 163 — Example 1: NVDIMM with Three Functions

Function 0 Interface Descriptor		
Byte 205	Byte 204	Meaning
1000 0000	0110 0000	Bit 15 = 1: Implemented Bits 9~5 = 00011: Function Class is Byte addressable no energy backed Bits 4~0 = 00000: Function Interface is Proprietary interface Bits 13~10 = 0000: Offset of Extended Function Parameters = $256 + 0 * 4 = 256$ (0x100) Implied by the example (i.e., 14 bytes used): Extended Function block from 256 through 269 (0x100~0x10D) is defined Extended Function block bytes 270, 271 not used (coded as 0x00)
Function 1 Interface Descriptor		
Byte 207	Byte 206	Meaning
1000 0000	0100 0000	Bit 15 = 1: Implemented Bits 9~5 = 00010: Function Class is Block addressed Bits 4~0 = 00000: Function Interface is Proprietary interface Implied by the example (i.e., no bytes used): No changes to Extended Function block Bits 13~10 coded as 0000
Function 2 Interface Descriptor		
Byte 209	Byte 208	Meaning
1001 0000	0000 0000	Bit 15 = 1: Implemented Bits 9~5 = 00000: Function Class is Undefined Bits 4~0 = 00000: Function Interface is Proprietary interface Offset of Extended Function Parameters = $256 + 4 * 4 = 272$ (0x110) Implied by the example (i.e., 23 bytes used): Extended Function block from 272 through 294 (0x110~0x126) is used Extended Function block byte 295 not used (coded as 0x00)
Functions 3~7 Interface Descriptors		
Byte n+1	Byte n	Meaning
0000 0000	0000 0000	Bit 15 = 0: Not Implemented; all other bits must be coded as 0

10.1.10 Bytes 220~253 (0x0DC~0x0FD) (NVDIMM):

Reserved, must be coded as 0x00

10.1.11 Bytes 254 (0x0FE) (NVDIMM): Cyclical Redundancy Code (CRC) for SPD Block 1, LSB Bytes 255 (0x0FF) (NVDIMM): Cyclical Redundancy Code (CRC) for SPD Block 1, MSB

This two-byte field contains the calculated CRC for bytes 128~253 (0x080~0x0FD) in the SPD. See bytes 126~127 for coding example.

11 Extended Function and Interface Protocol Parameter Sets for Hybrid Memory Modules (EXFUNC): Bytes 256~319 (0x100~0x13F)

This section contains zero or more blocks of extended function and interface protocol parameter sets for Hybrid memory modules. The blocks of parameters are contiguous in storage, and the index into this block is stored in the Offset descriptor field for each function type. Granularity of extended parameter blocks is a multiple 4 bytes; unused bytes in a block must be coded as 0x00.

The definition of each parameter block is shown here based on relative addressing within this SPD parameter block, i.e., the beginning address of the parameter block is calculated as:

$$256 + (4 * \text{Extended Parameter Block Offset for that function})$$

11 Extended Function and Interface Protocol Parameter Sets for Hybrid Memory Modules (EXFUNC): Bytes 256~319 (0x100~0x13F) (Cont'd)**Example:**

Function 0 offset = 0, length = 14: byte n = 256, n+1 = 257, ... bytes 268, 269 = 0x00

Function 1 offset = 16, length = 23: byte n = 270, n+1 = 271, ... byte 293 = 0x00

Function 2 offset = 40, length = 9: byte n = 296, n+1 = 297, ... bytes 305~307 = 0x00

Functions 3~7 = not implemented

11.1 Extended Function Parameter Block for Byte Addressable Energy Backed NVDIMM (BYTE-E)

This section defines the encoding of Extended Function Parameter SPD bytes when a Function Interface Descriptor Function Class bits 9~5 contain 00001:

Byte n (BYTE-E)

Byte n + 1 (BYTE-E)

Byte n + 2 (BYTE-E)

11.2 Extended Function Parameter Block for Block Addressed NVDIMM (BLOCK)

This section defines the encoding of Extended Function Parameter SPD bytes when a Function Interface Descriptor Function Class bits 9~5 contain 00010:

11.2.1 Byte n (BLOCK): Module Characteristics

This byte, shown in Table 164, contains additional information about the module characteristics including the interleave capability and energy backup of the NVDIMM. The interleave stride indicates either the NVDIMM cannot be interleaved and must be placed in a contiguous address space or it can be interleaved with other DIMMs on a stride greater than or equal to the encoded value.

Table 164 — Byte n: Module Characteristics

Bit 7	Bit 6	Bits 5~0
Energy Backed	Interleave Capable	Interleave Stride $\geq 2^n$
0 = DIMM does not have energy backup 1 = DIMM has energy backup	0 = DIMM cannot be interleaved 1 = DIMM can be interleaved on a stride greater than or equal to 2^{Stride}	000000 = 2^0 : Stride ≥ 1 byte 000001 = 2^1 : Stride ≥ 2 bytes ... 001111 = 2^{15} : Stride ≥ 32 KB 010000 = 2^{16} : Stride ≥ 64 KB 010001 = 2^{17} : Stride ≥ 128 KB ... 111111 = 2^{63} : Stride $\geq 2^{63}$ bytes

Examples: See Table 165.

Table 165 — Examples for Byte n: Module Characteristics

Code	Hex	Meaning
11 010000	0xD0	Energy backup is available DIMM can be interleaved $\geq 2^{16}$ (64 KB) boundaries
00 000000	0x00	Energy backup is not available DIMM interleaving not supported

11.2.2 Byte n + 1 ~ n + 2 (BLOCK): Module Storage Capacity

These 2 bytes define the offered capacity of the block storage on the NVDIMM. The capacity is encoded as shown in Table 166:

Table 166 — Bytes n+2, n+1: Module Storage Capacity

Byte n+2 (MSB)		Byte n+1 (LSB)	
Bits 15~14		Bits 13~4	Bits 3~0
Capacity Base		Capacity Digits	Capacity, Tenths
00 = Mega Bytes 01 = Giga Bytes 10 = Tera Bytes 11 = Peta Bytes		Values from 0 to 999 All other values reserved	0000 = 0 0001 = 0.1 0010 = 0.2 0011 = 0.3 0100 = 0.4 0101 = 0.5 0110 = 0.6 0111 = 0.7 1000 = 0.8 1001 = 0.9 All other codes reserved

Examples: See Table 167.

Table 167 — Examples for Bytes n+2, n+1: Module Storage Capacity

Code	Hex	Meaning
01 00 1000 0000 0000	0x2800	Module storage capacity = 128 GB
10 00 0000 0001 0000	0x8010	Module storage capacity = 1 TB
10 00 0000 0001 0101	0x8015	Module storage capacity = 1.5 TB
00 11 0010 0000 0000	0x3200	Module storage capacity = 800 MB

11.3 Extended Function Parameter Block for Byte Addressable No Energy Backed NVDIMM (BYTE-NOE)

This section defines the encoding of Extended Function Parameter SPD bytes when a Function Interface Descriptor Function Class bits 9~5 contain 00011:

Byte n (BYTE-NOE)

Byte n + 1 (BYTE-NOE)

Byte n + 2 (BYTE-NOE)

11.3.1 Bytes 318 (0x13E) (EXFUNC): Cyclical Redundancy Code (CRC) for SPD Block 2 lower half, LSB Bytes 319 (0x13F) (EXFUNC): Cyclical Redundancy Code (CRC) for SPD Block 2 lower half, MSB

This two-byte field contains the calculated CRC for bytes 256~318 (0x100~0x13D) in the SPD. See bytes 126~127, Section 8.1.53, for coding example.